



(12) **United States Patent**
Conrad et al.

(10) **Patent No.:** **US 12,410,010 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **STORAGE AND RETRIEVAL SYSTEM**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 332 days.

(21) Appl. No.: **18/298,038**

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(65) **Prior Publication Data**

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Related U.S. Application Data

(63) Continuation of application No. 17/652,012, filed on Feb. 22, 2022, now Pat. No. 11,623,822, which is a (Continued)

(51) **Int. Cl.**
B65G 1/04 (2006.01)
B65G 1/00 (2006.01)
B65G 1/137 (2006.01)

(52) **U.S. Cl.**
CPC **B65G 1/0492** (2013.01); **B65G 1/00** (2013.01); **B65G 1/0485** (2013.01); **B65G 1/0435** (2013.01); **B65G 1/1376** (2013.01)

(58) **Field of Classification Search**
CPC .. **B65G 1/0492**; **B65G 1/0485**; **B65G 1/0435**; **B65G 1/00**; **B65G 1/1376**
(Continued)

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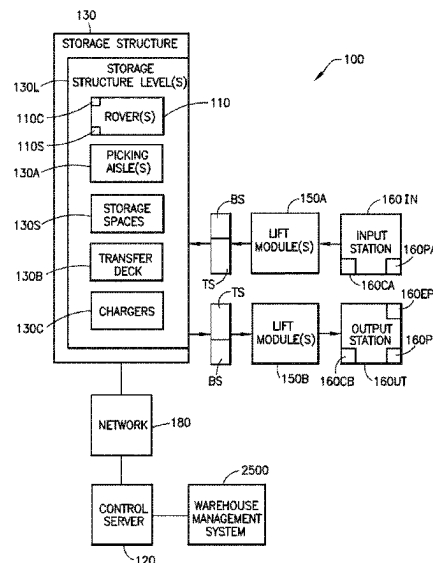
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(57) **ABSTRACT**

An automated storage and retrieval system including at least one autonomous transport vehicle, a transfer deck that defines a transport surface for the vehicle, at least one reciprocating lift, a first and second pickface interface station connected to the deck and spaced apart from each other, each station forming a pickface transfer interfacing between the vehicle on the deck and the lift at each station so that a pickface is transferred between the lift and the vehicle at each station, wherein the vehicle is configured to pick a first pickface at the first station, traverse the deck and buffer one first pickface, or at least a portion thereof, at the second station so that the second station has multiple pickfaces buffered on a common support in an order sequence of pickfaces according to a predetermined case out order sequence of mixed case pickfaces.

18 Claims, 41 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/108,265, filed on Dec. 1, 2020, now Pat. No. 11,254,502, which is a continuation of application No. 16/222,283, filed on Dec. 17, 2018, now Pat. No. 10,850,921, which is a continuation of application No. 15/848,809, filed on Dec. 20, 2017, now Pat. No. 10,155,623, which is a continuation of application No. 14/997,920, filed on Jan. 18, 2016, now Pat. No. 9,856,083.

- (60) Provisional application No. 62/104,520, filed on Jan. 16, 2015.

(58) Field of Classification Search

USPC 414/277, 279, 281–285
See application file for complete search history.

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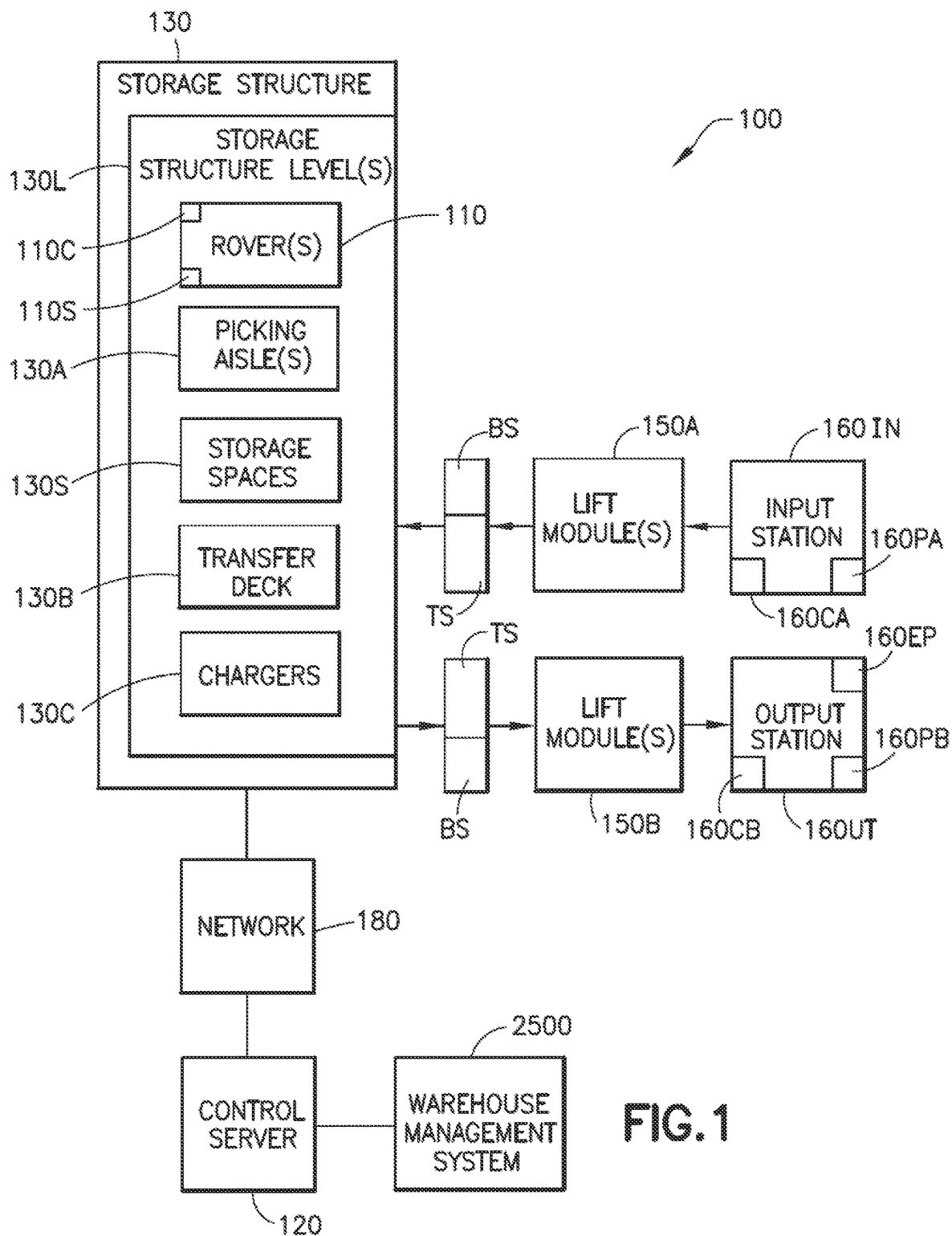
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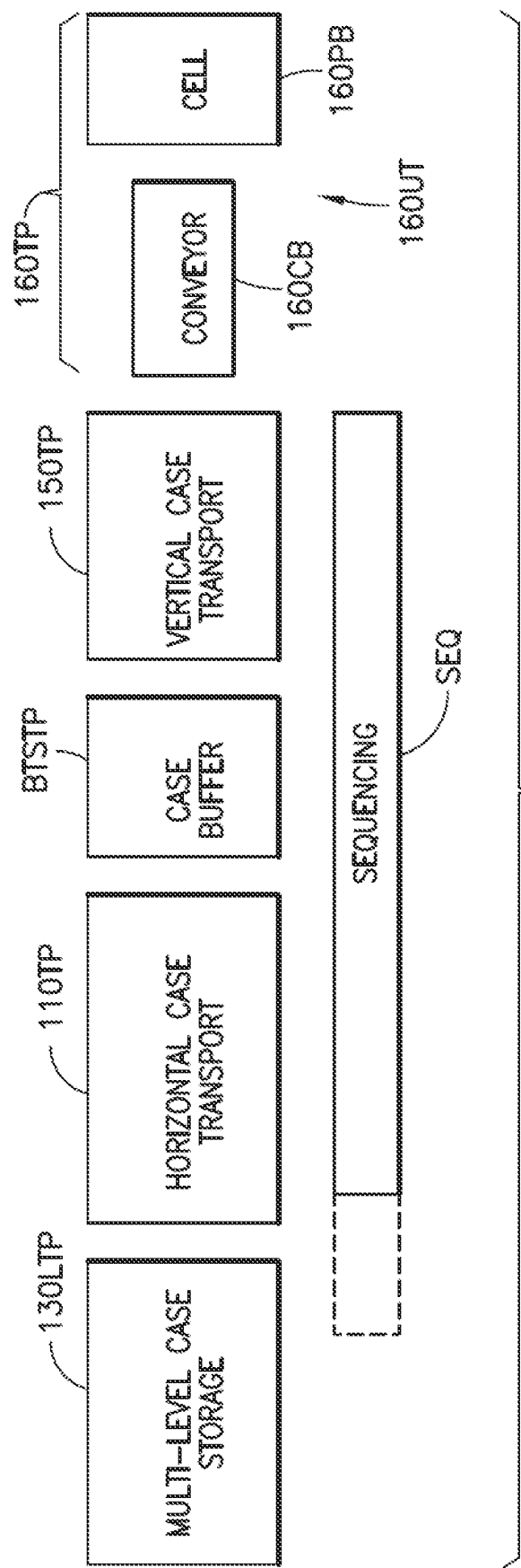
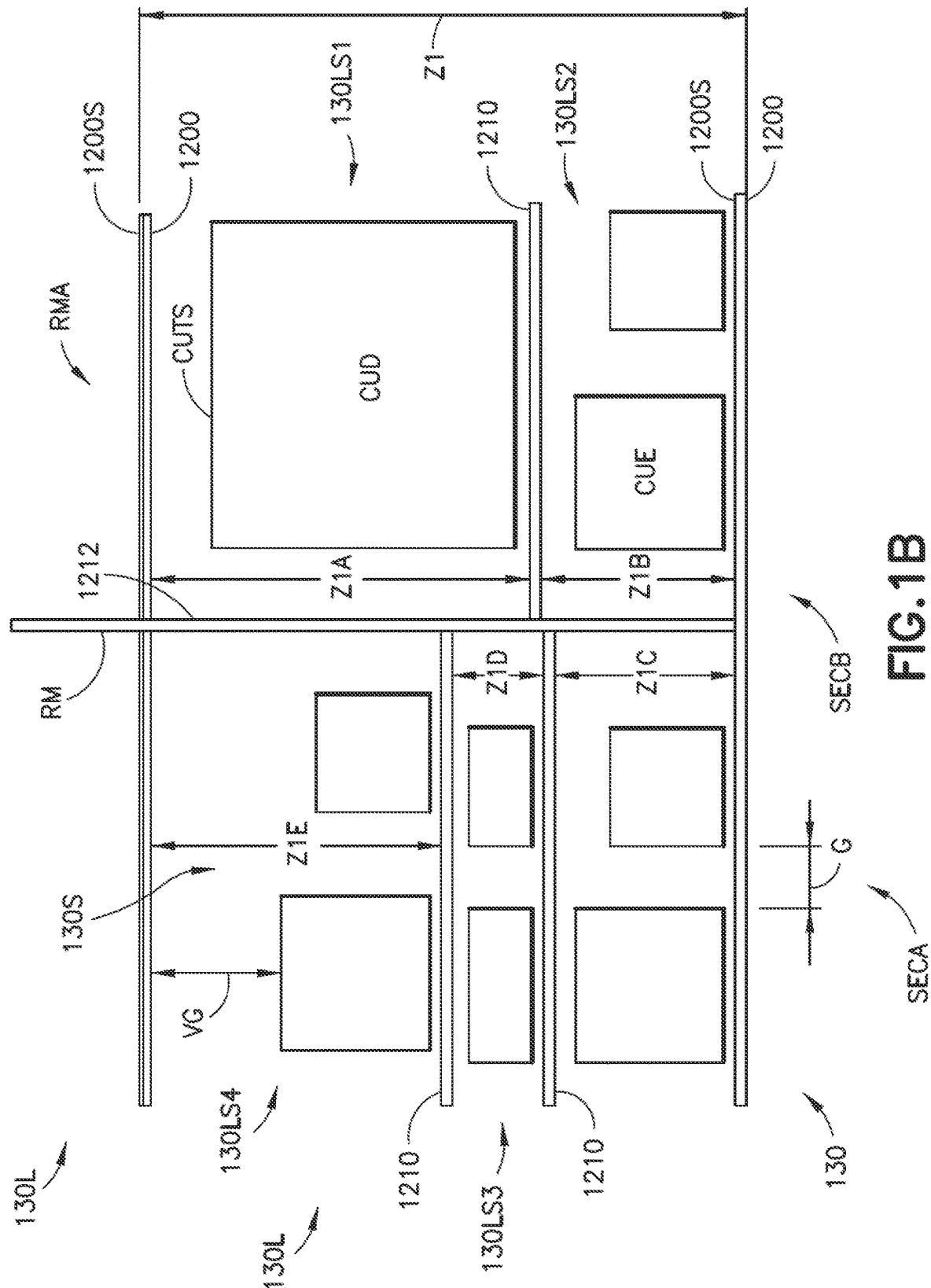


FIG.1A



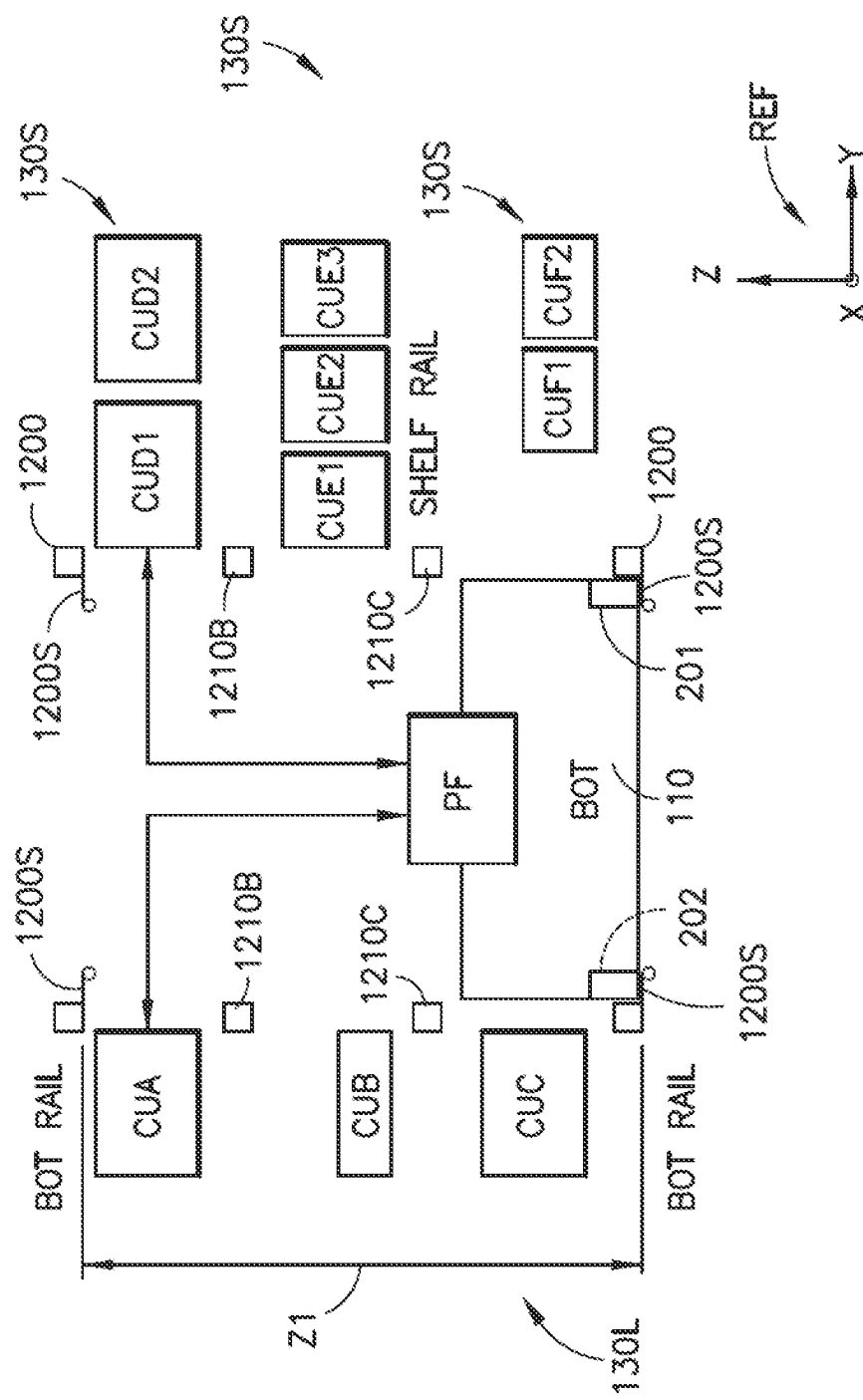


FIG. 1C

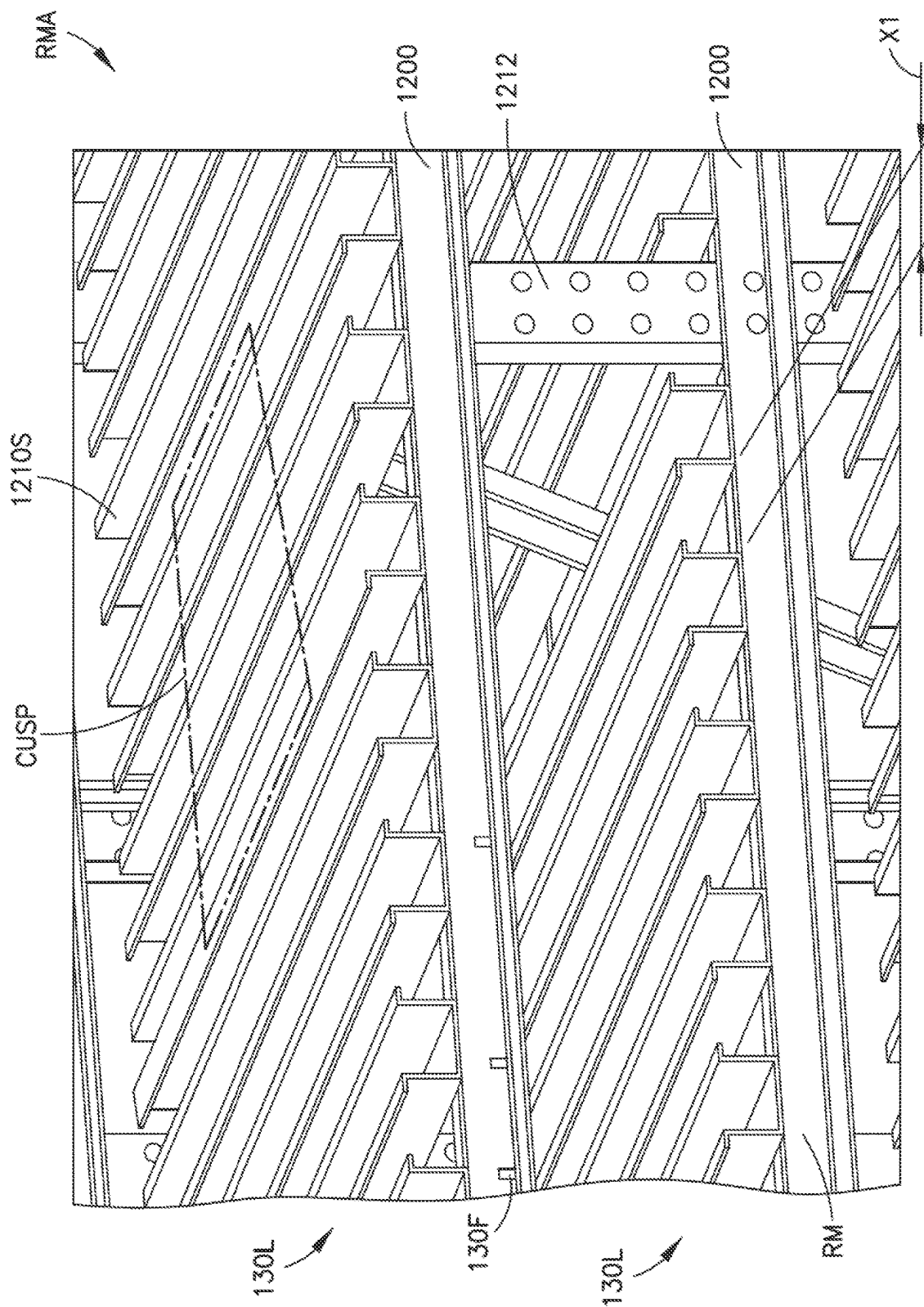


FIG. 1D

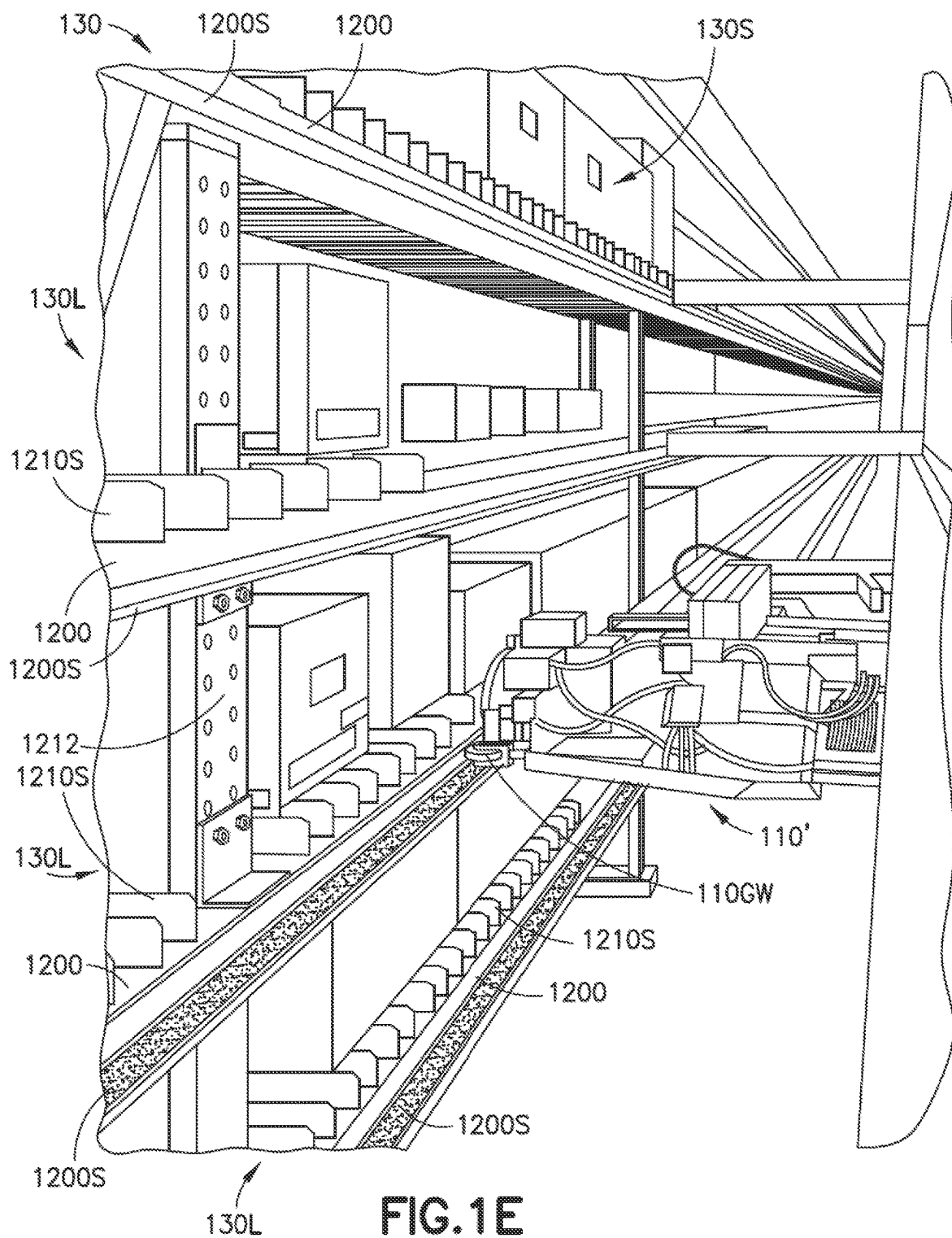
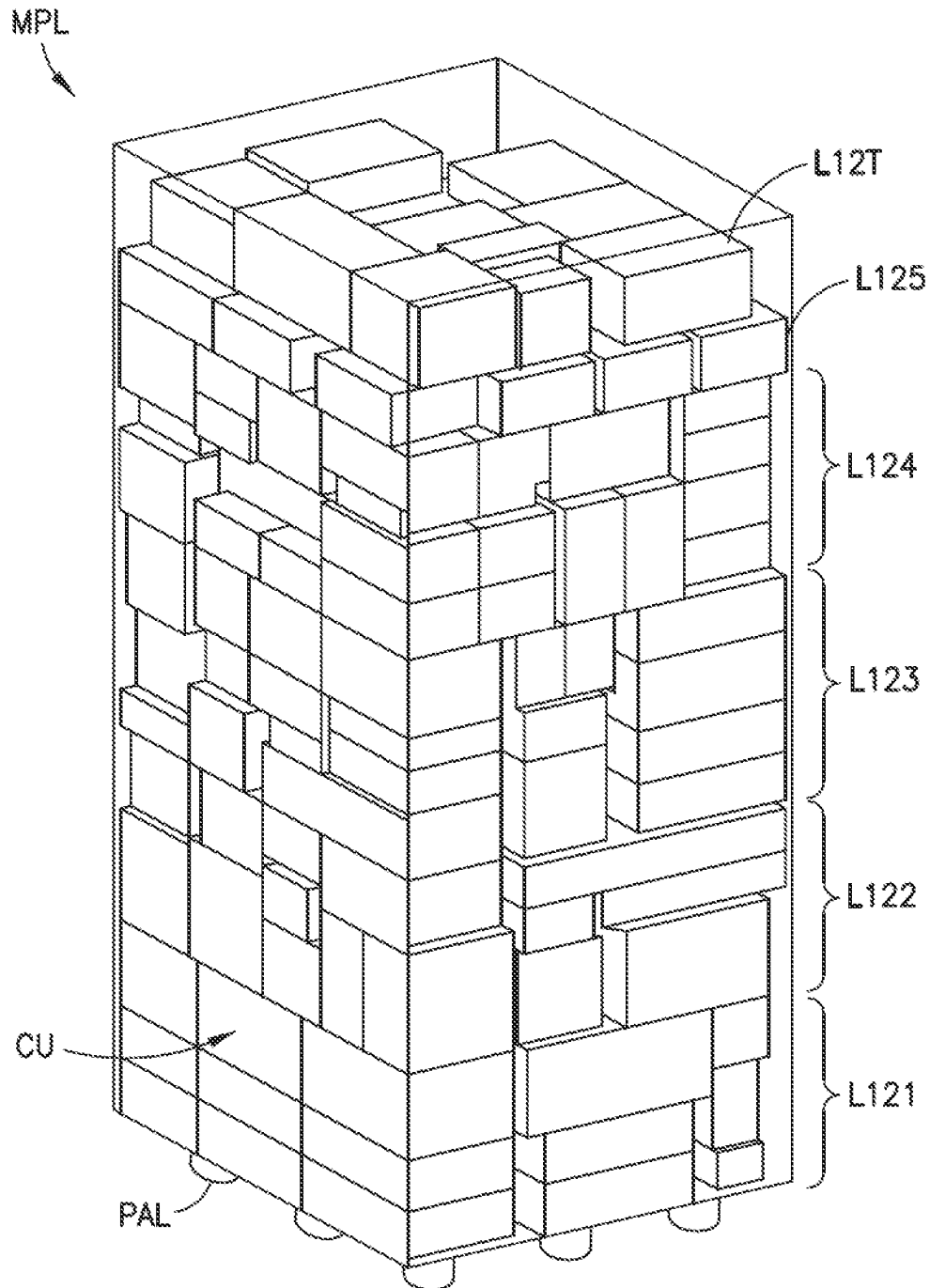


FIG. 1E

**FIG. 1F**

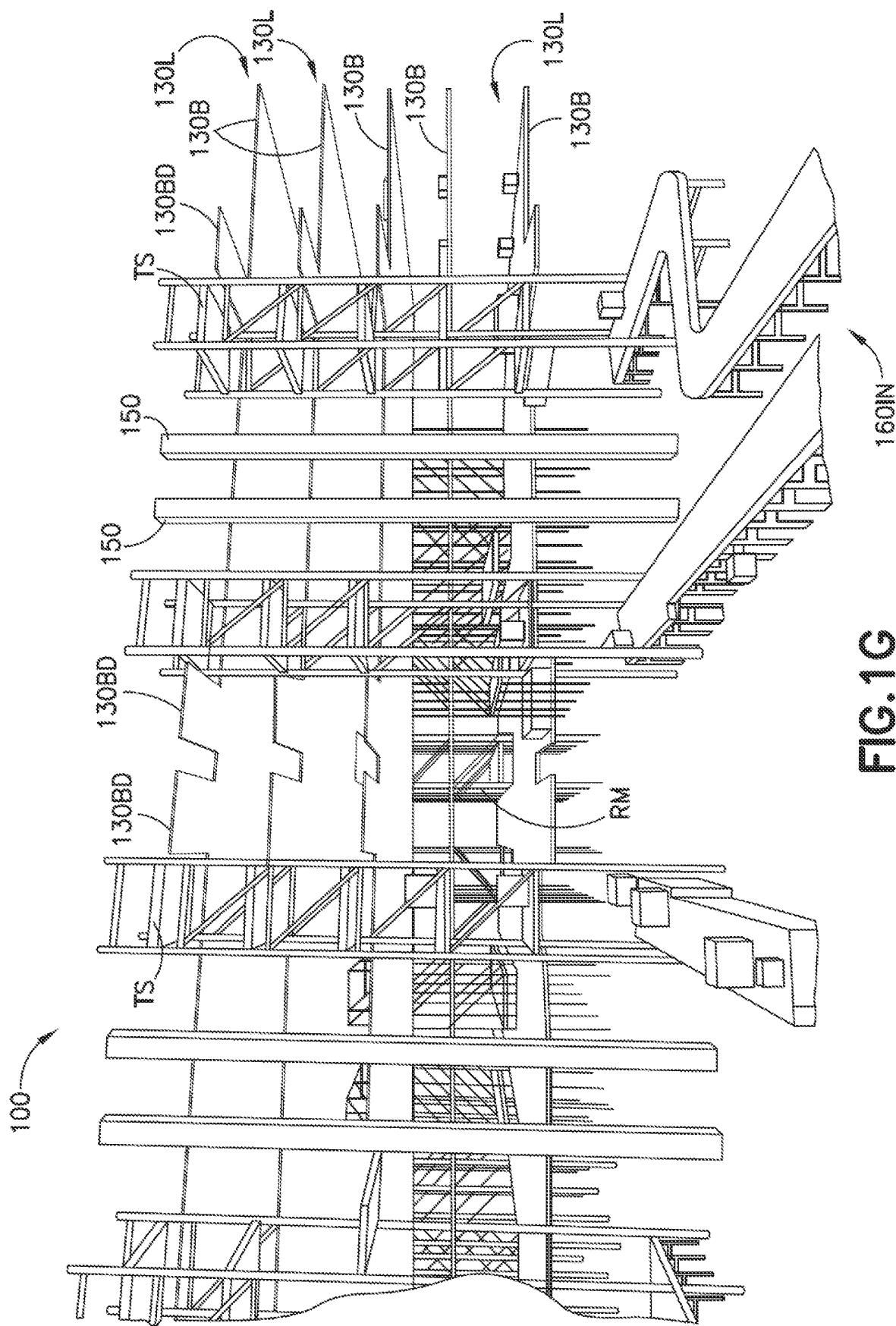
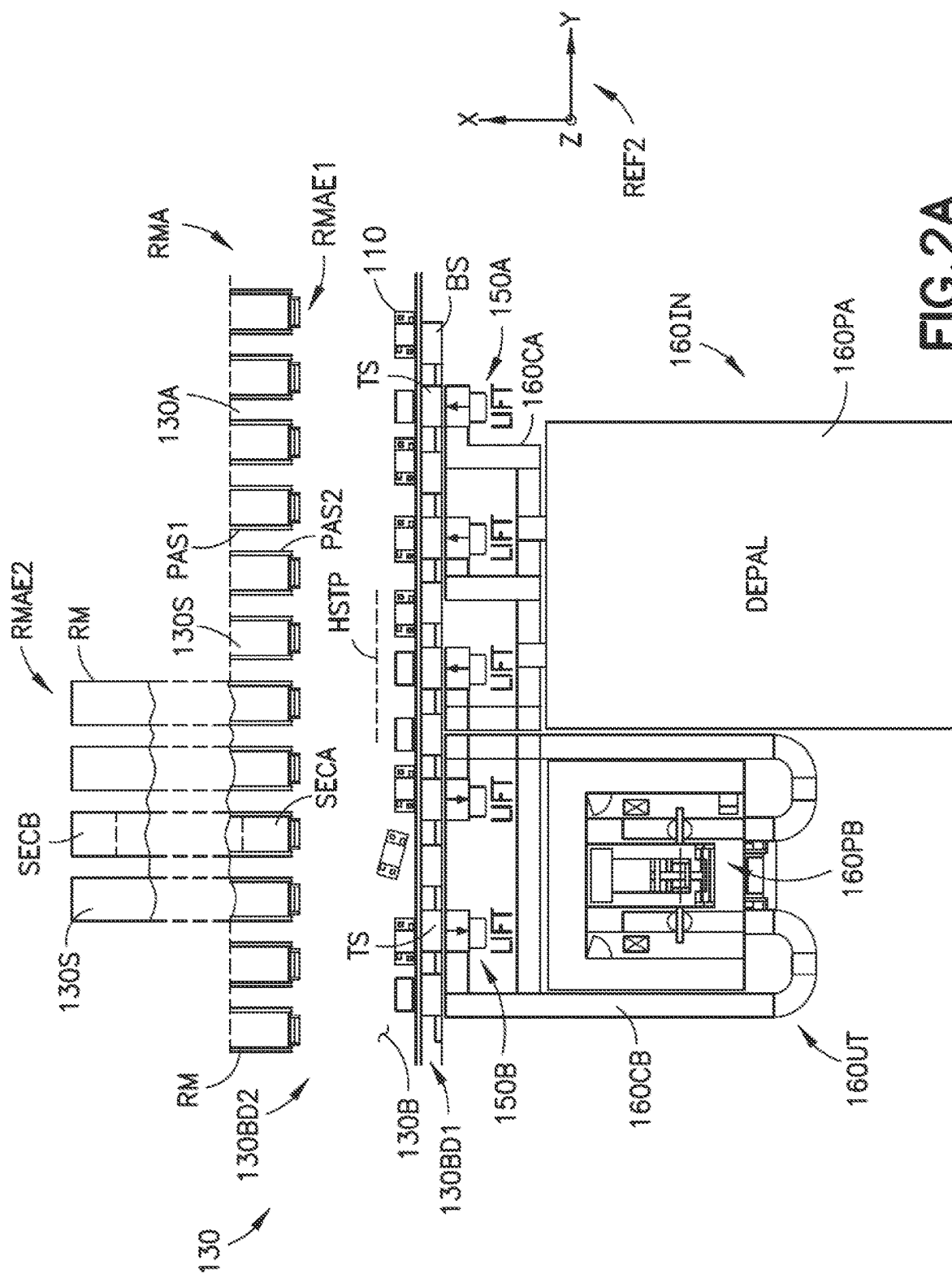


FIG.1G



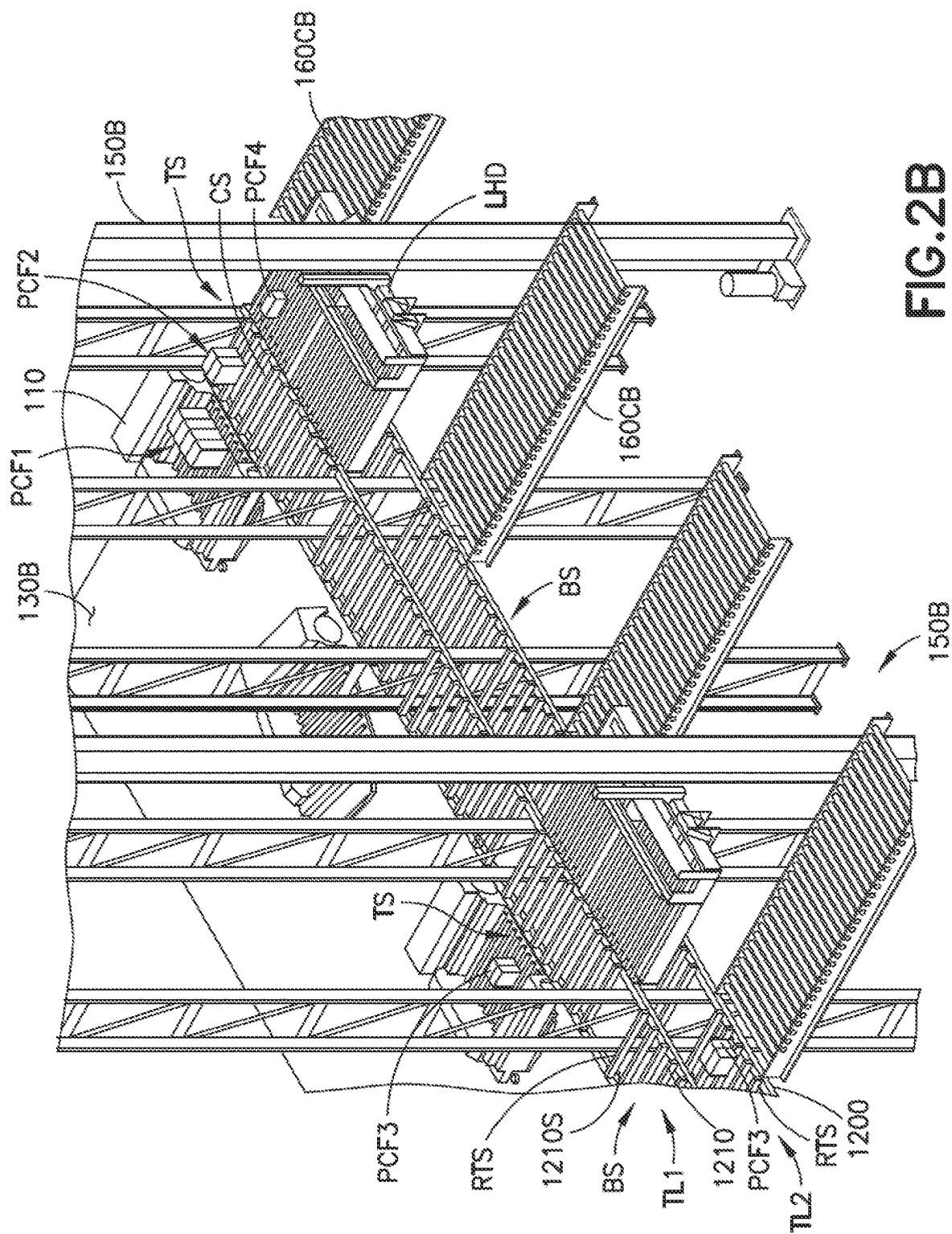


FIG. 2B

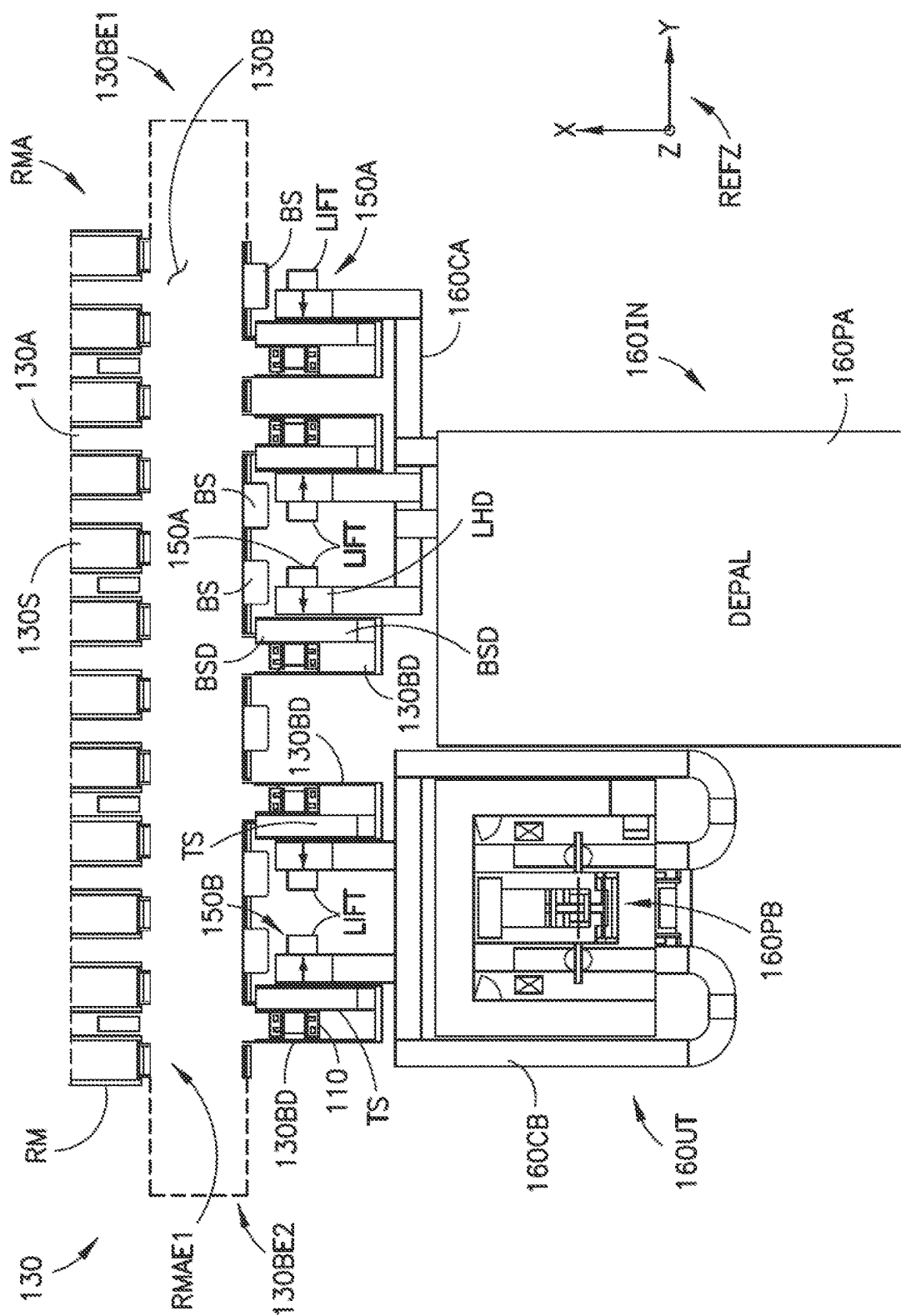
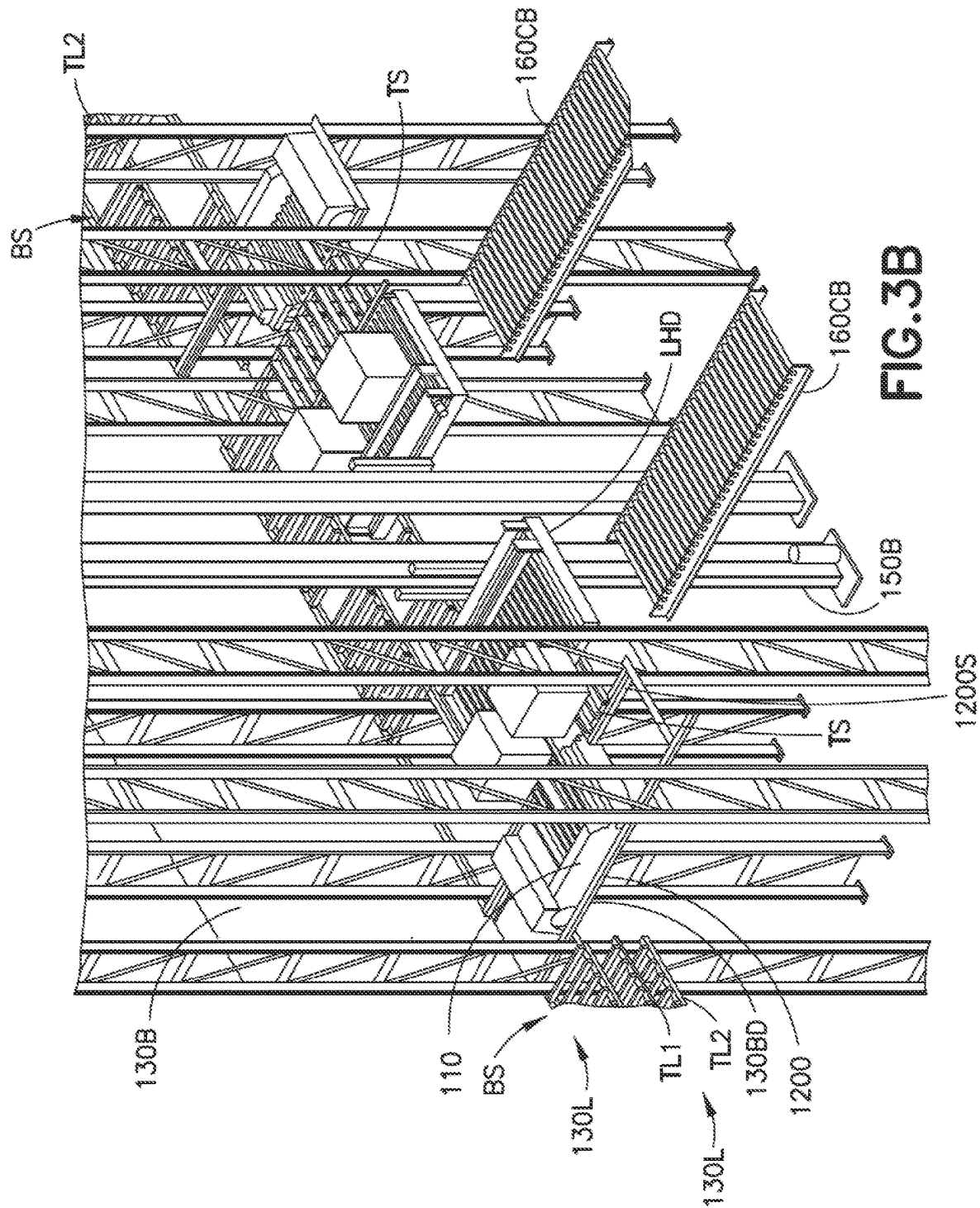


FIG. 3A



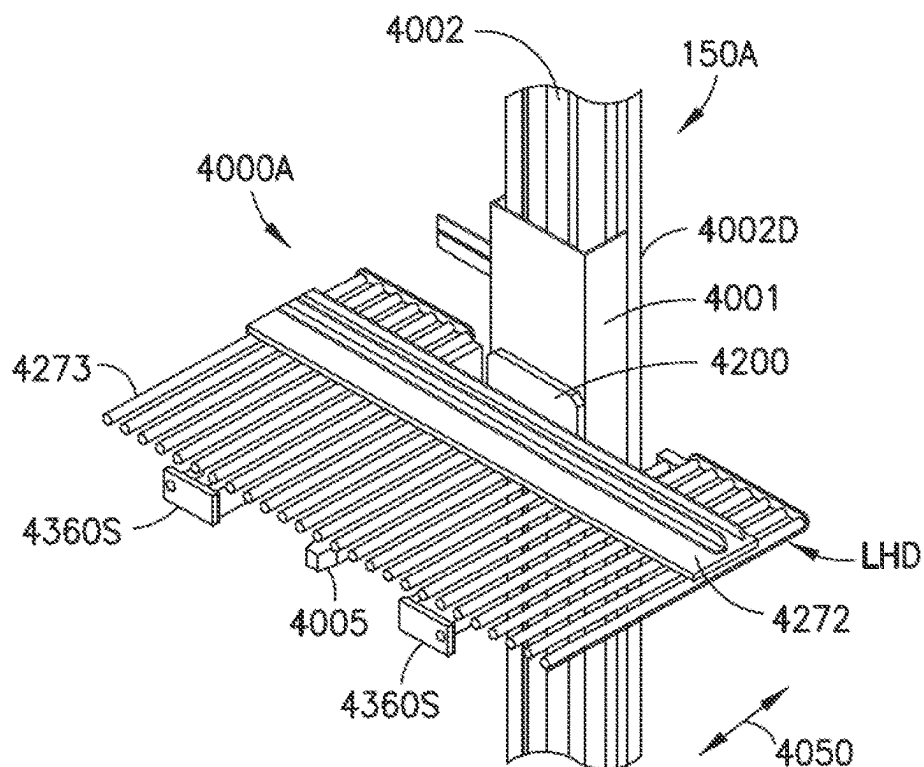


FIG. 4A

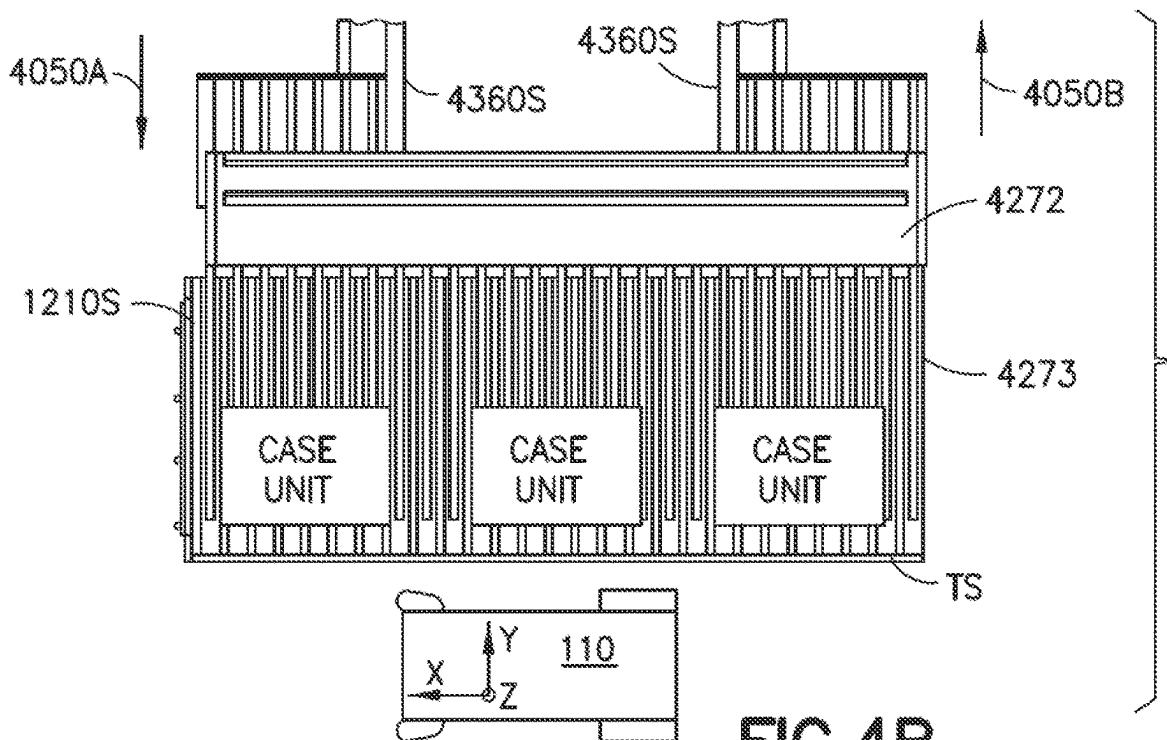


FIG. 4B

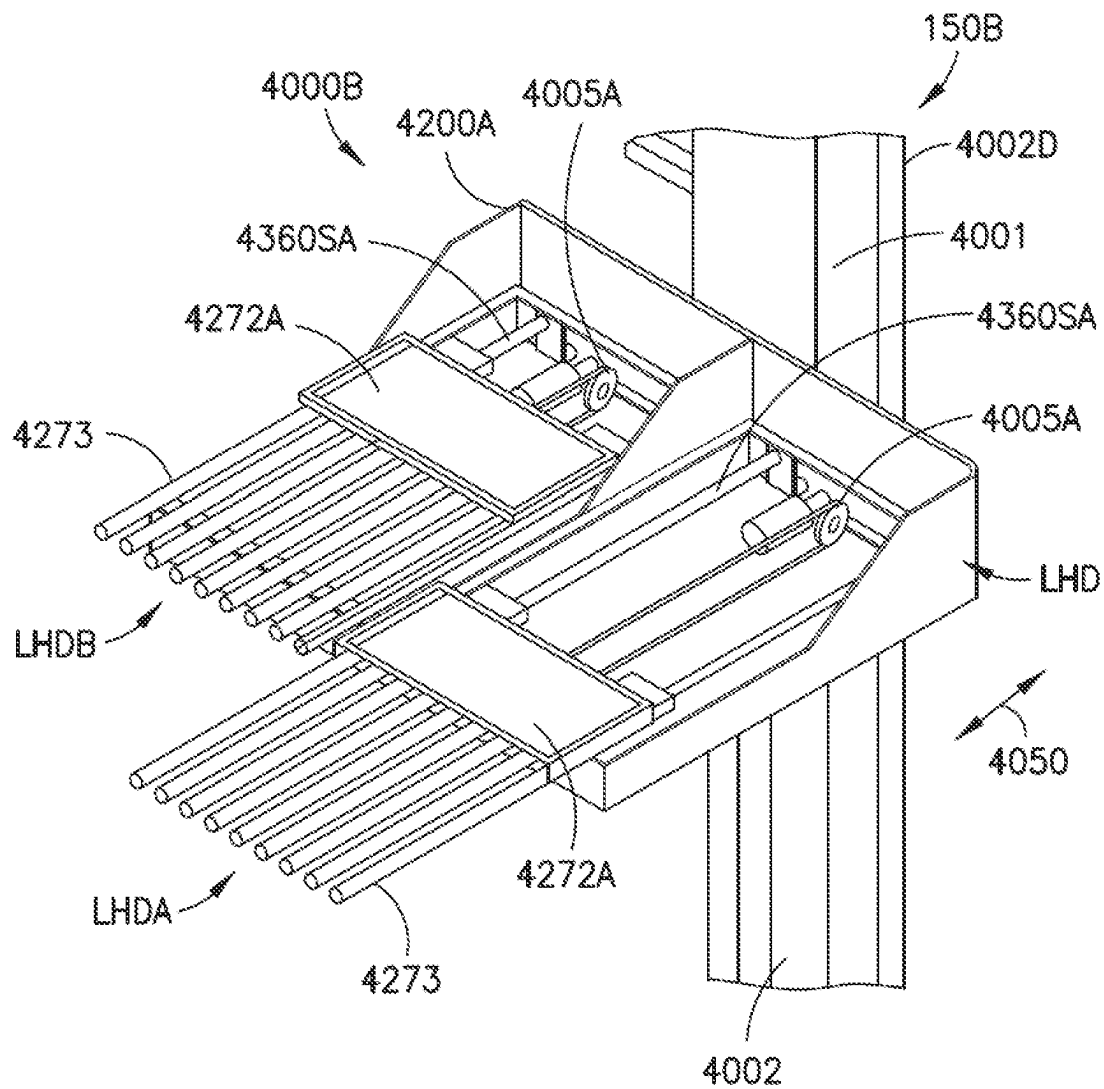


FIG.5

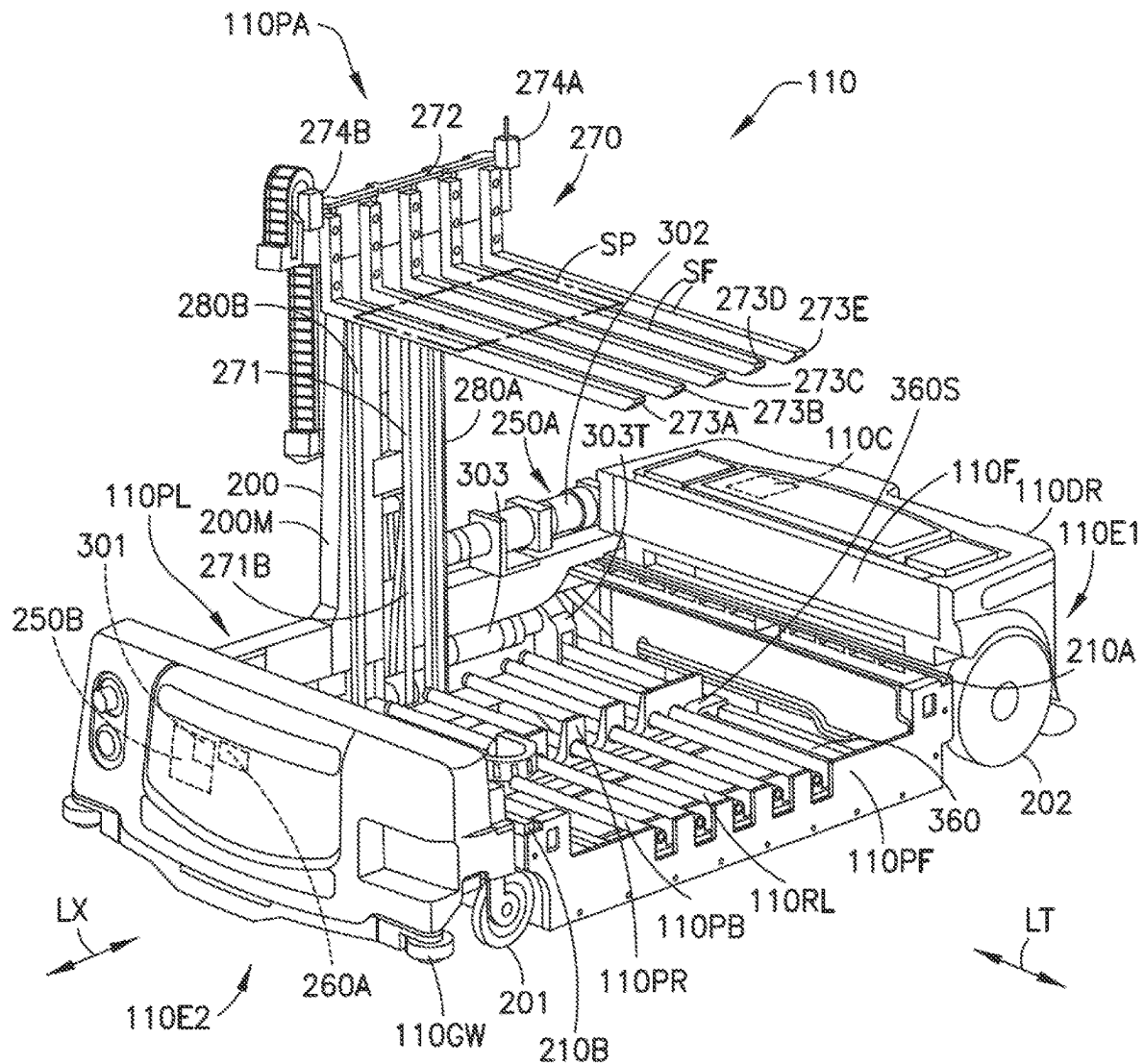
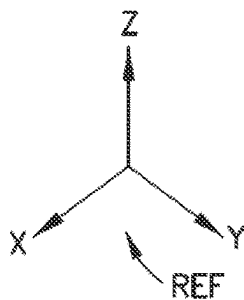
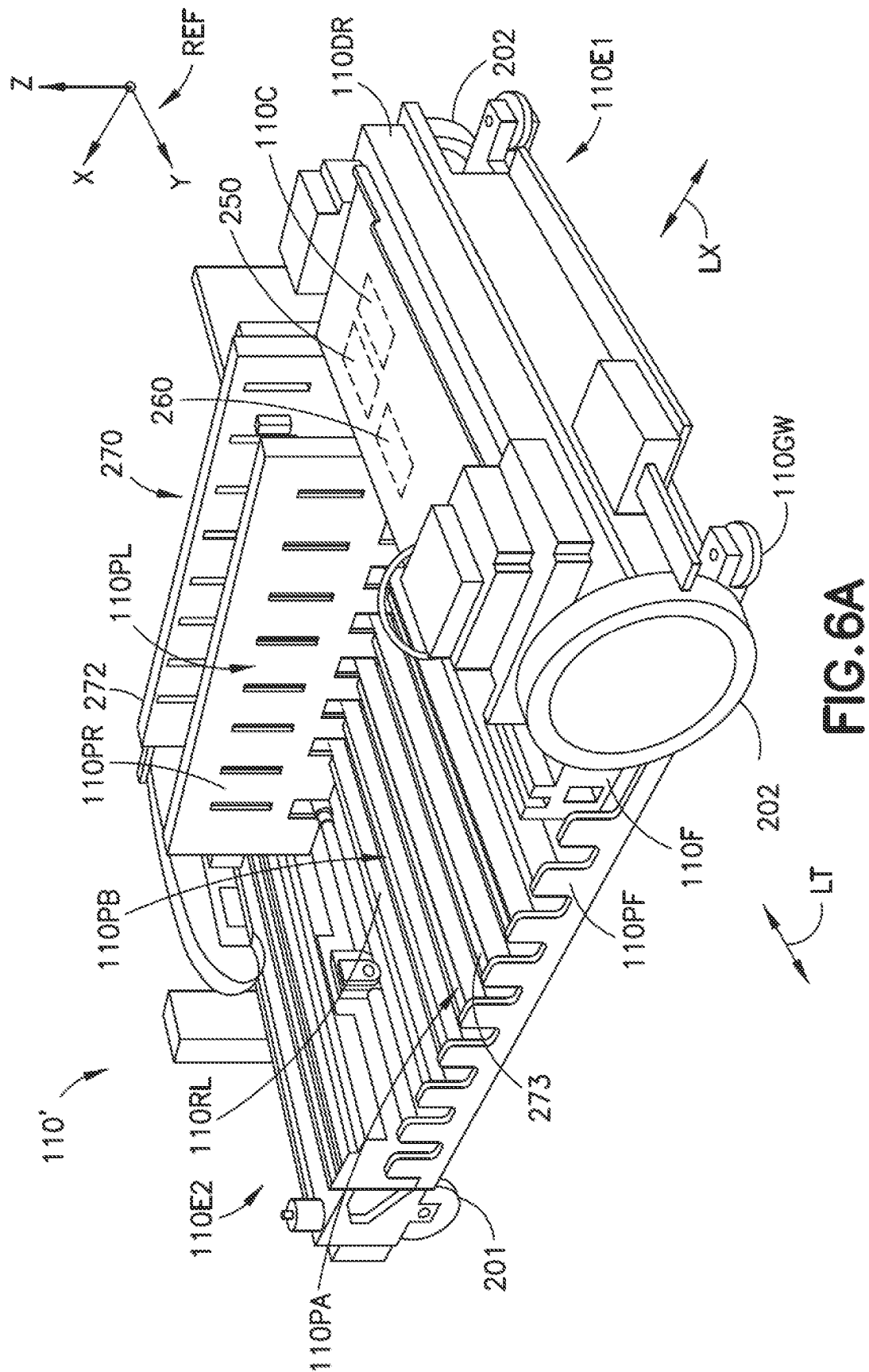


FIG. 6





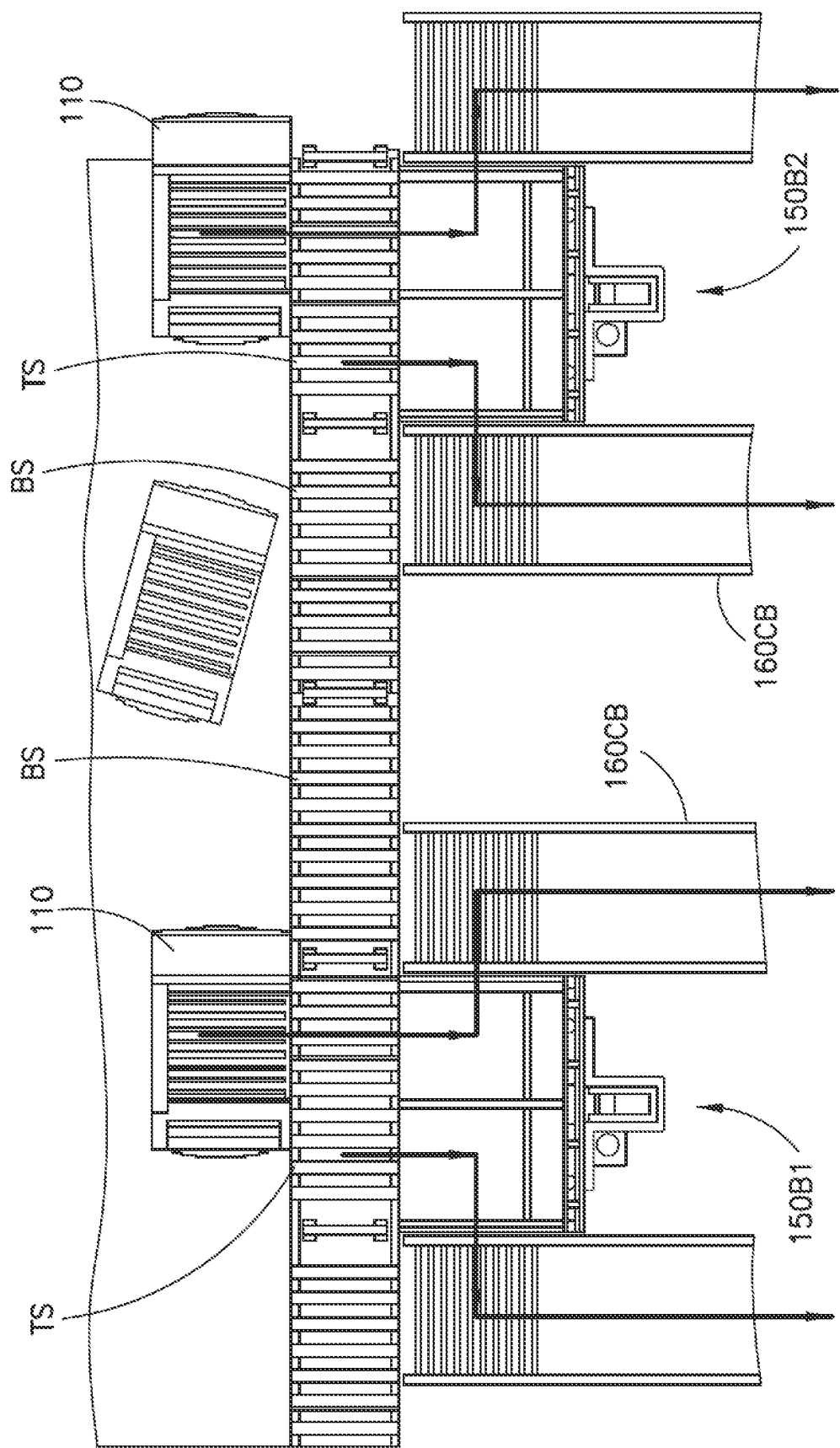


FIG. 7

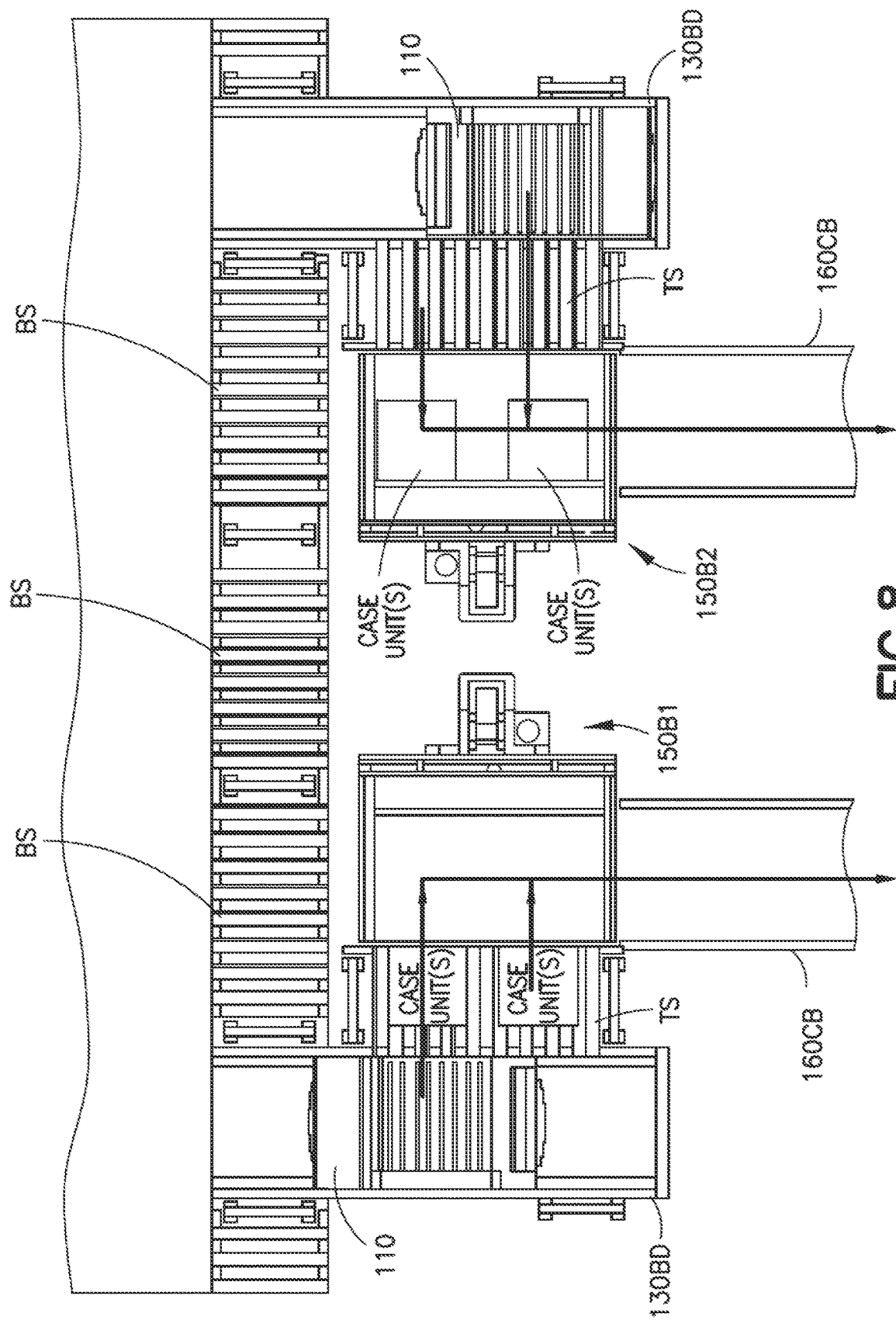


FIG. 8

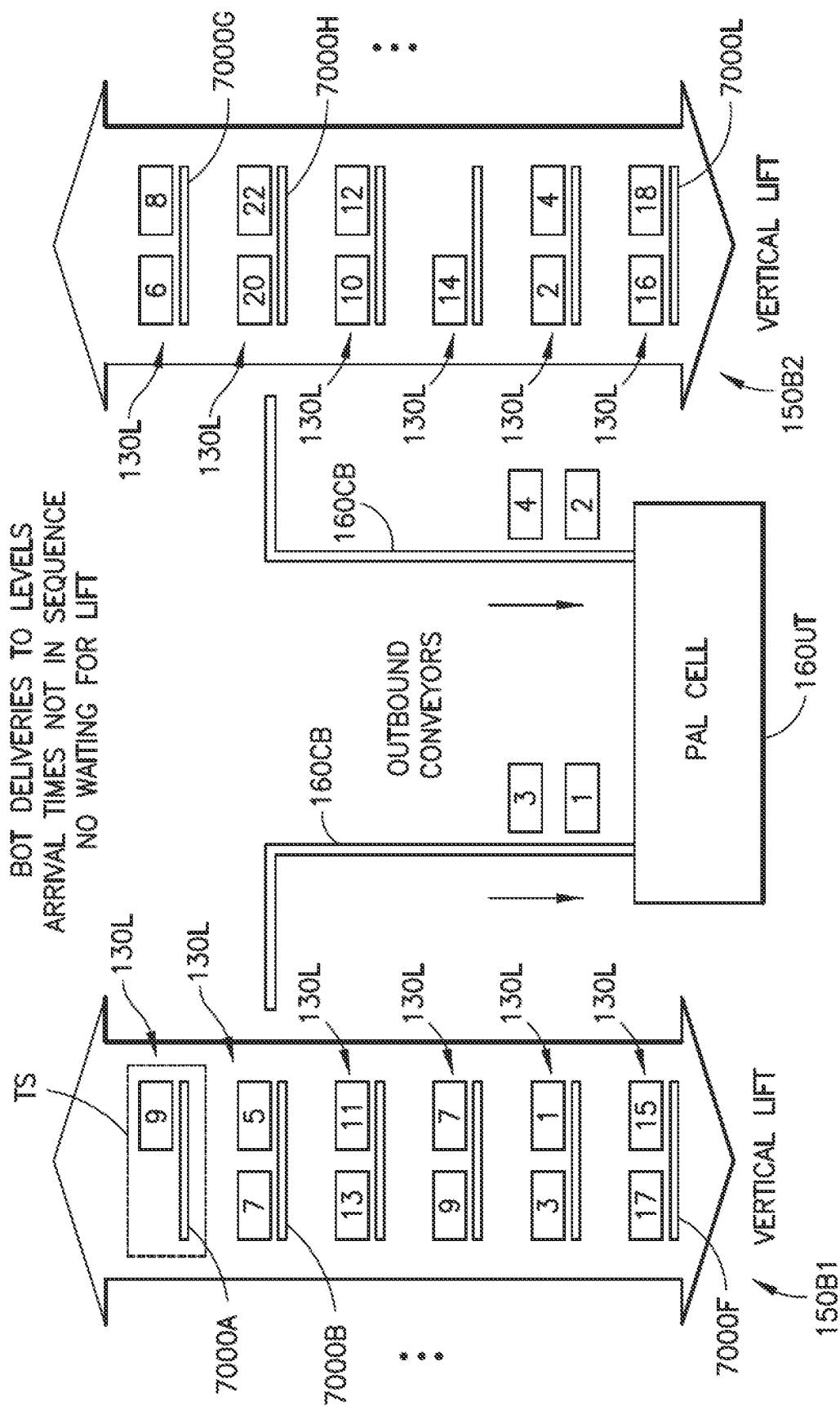


FIG.9

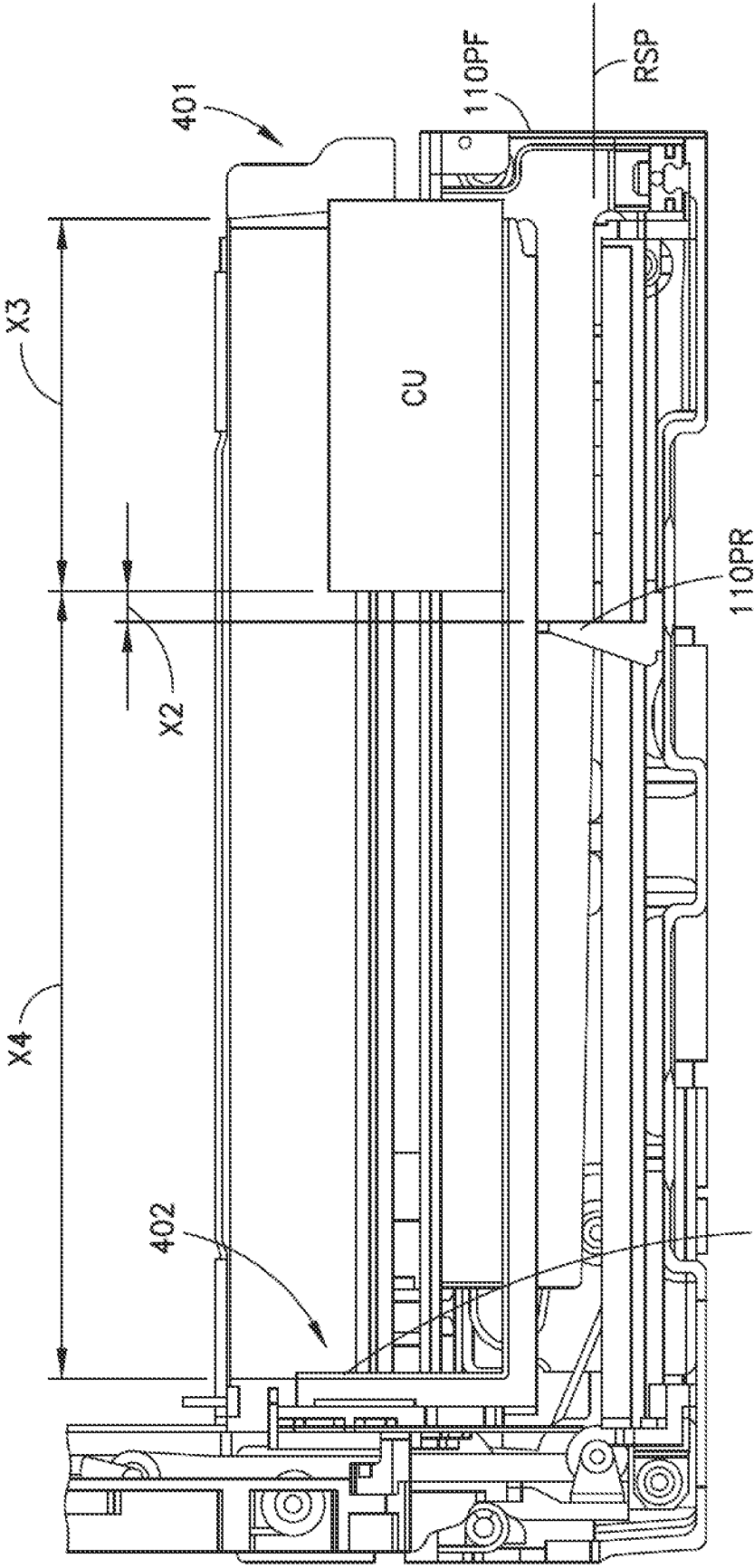
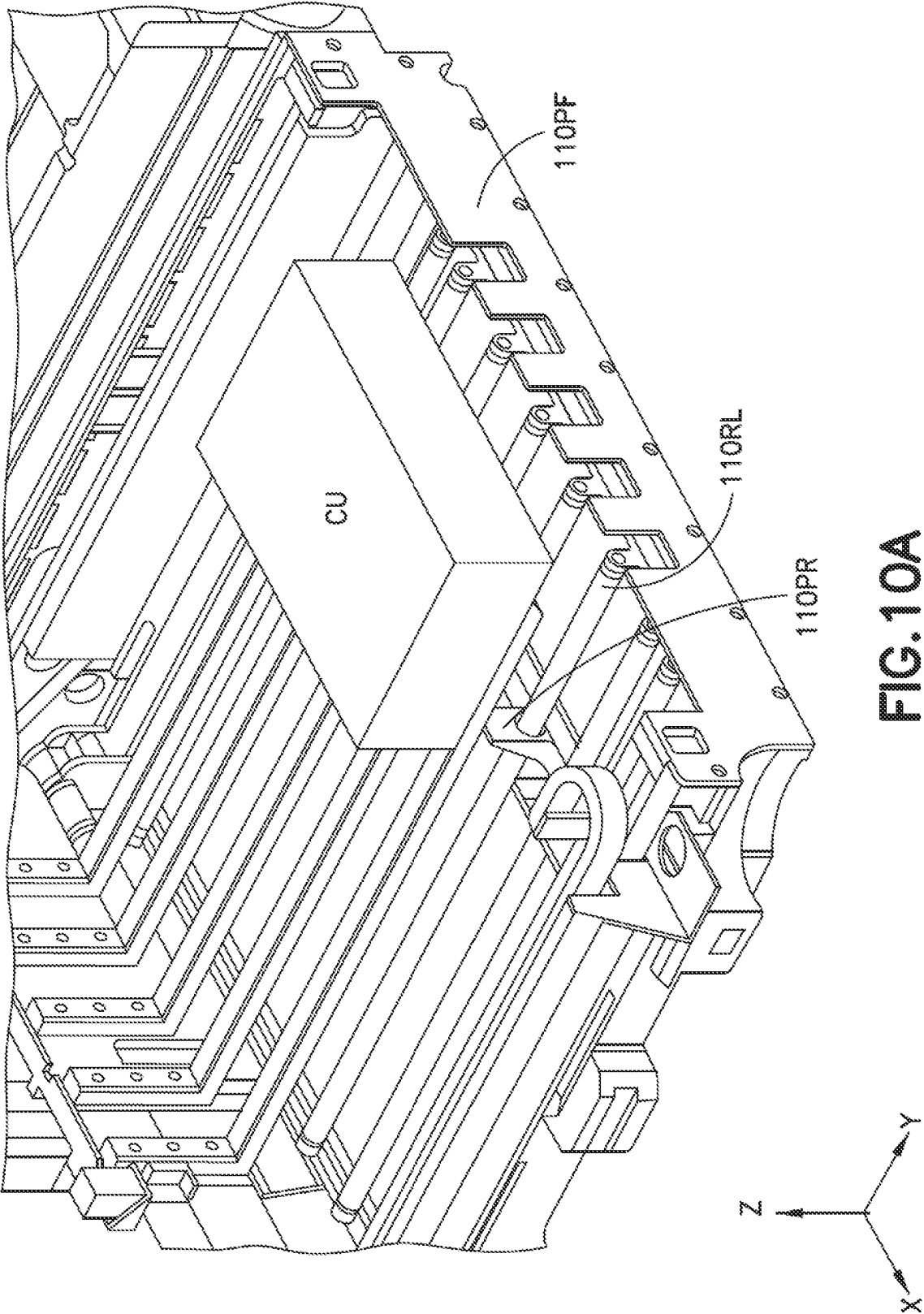


FIG.10



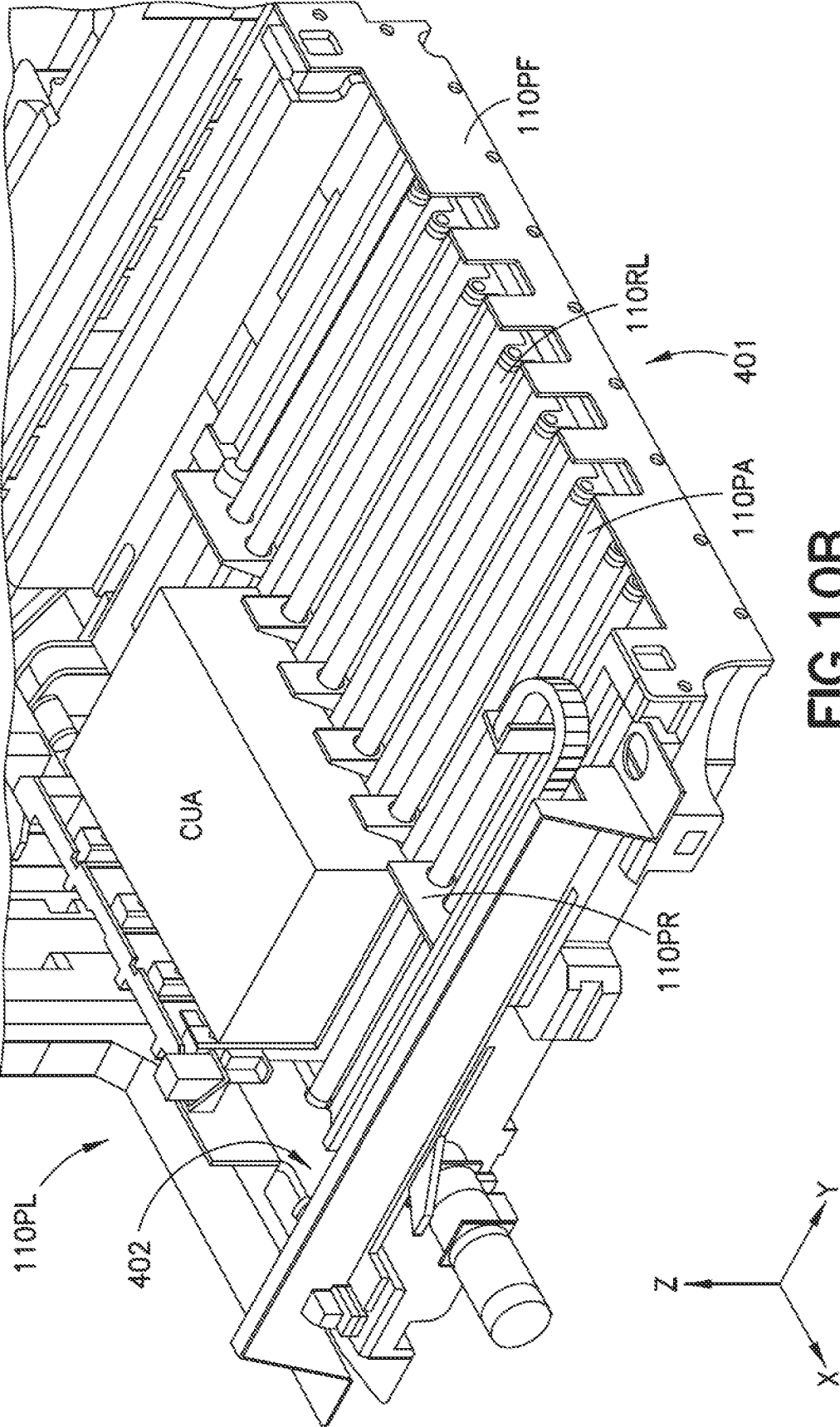


FIG.10B

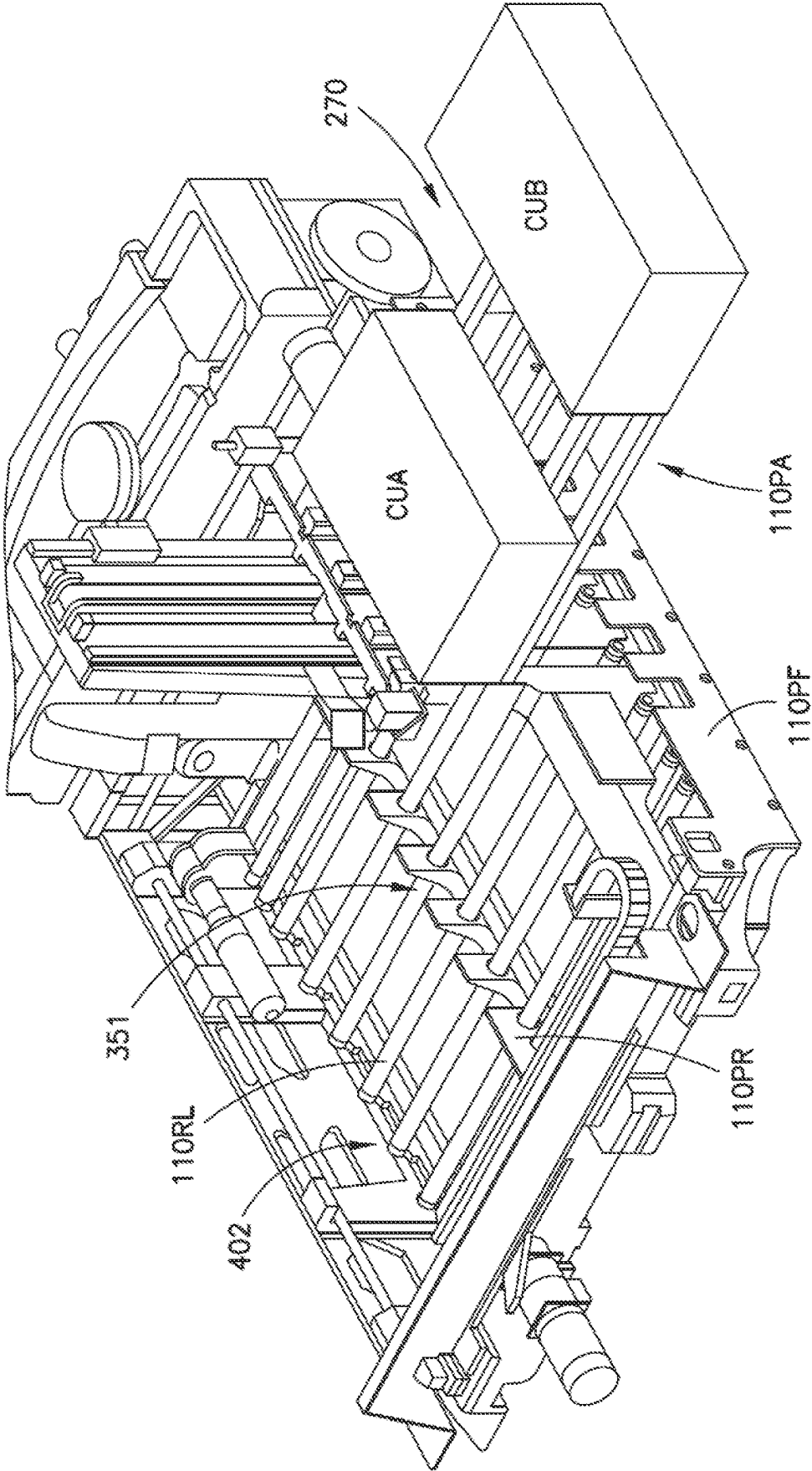
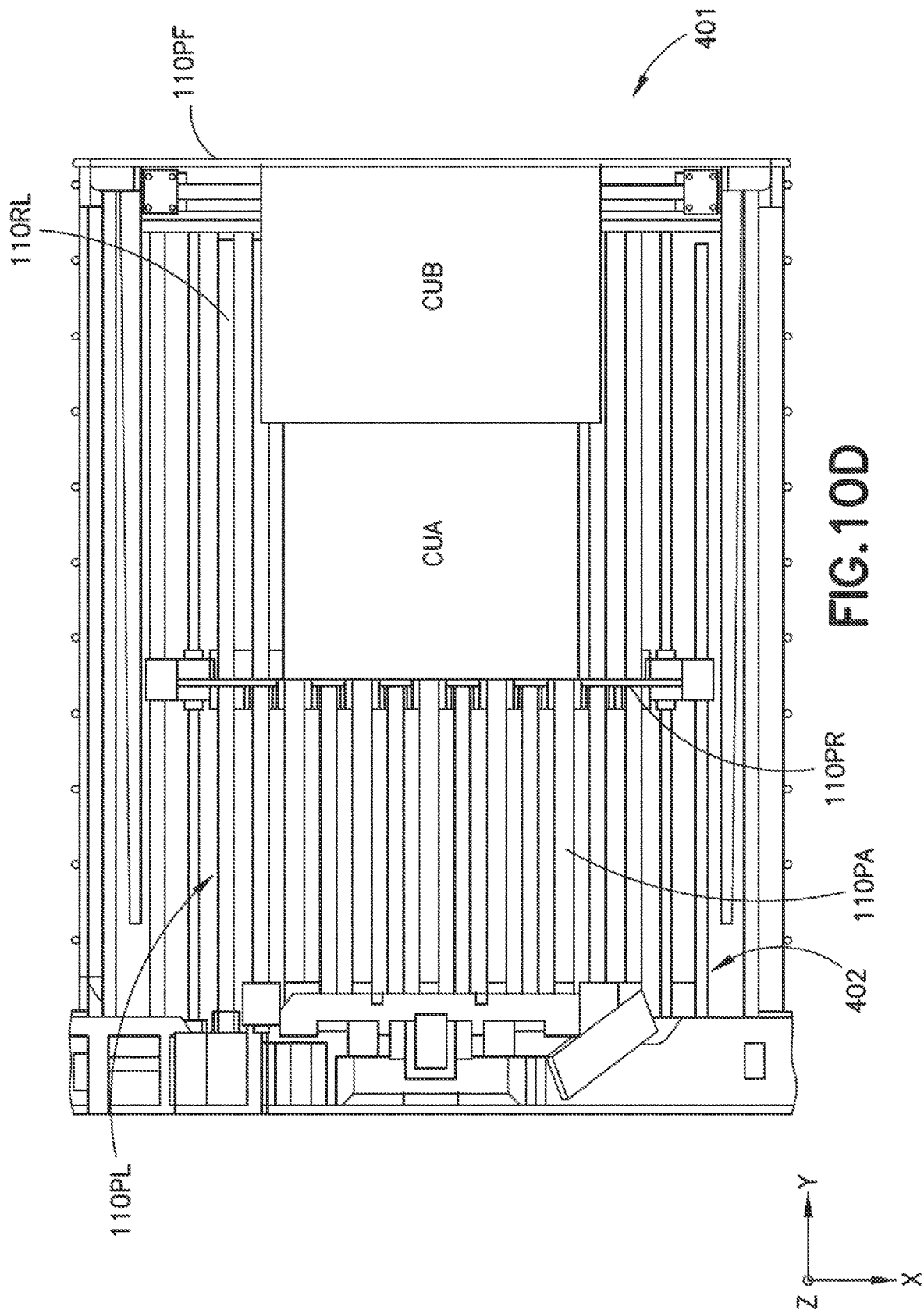


FIG.10C



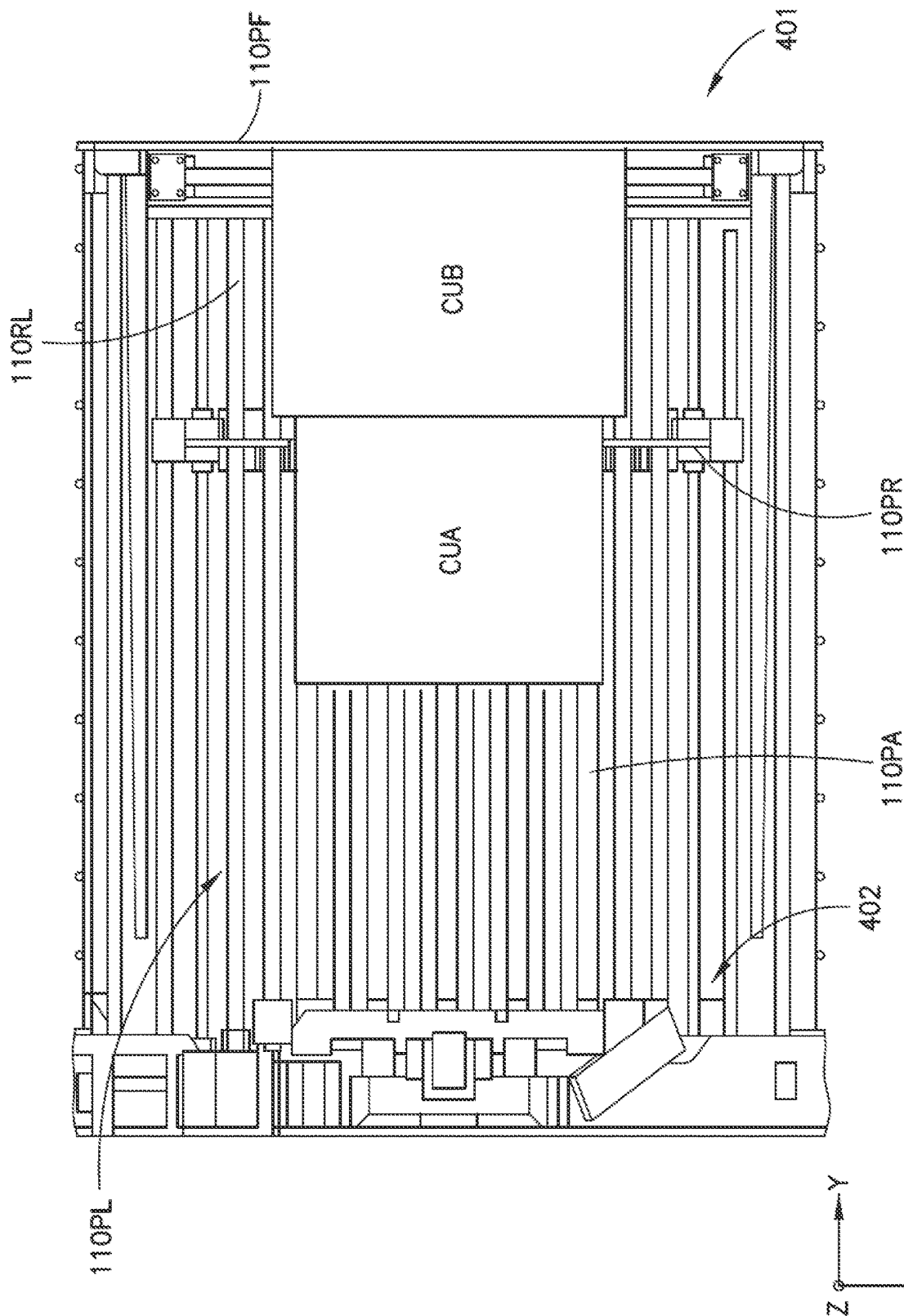


FIG. 10E

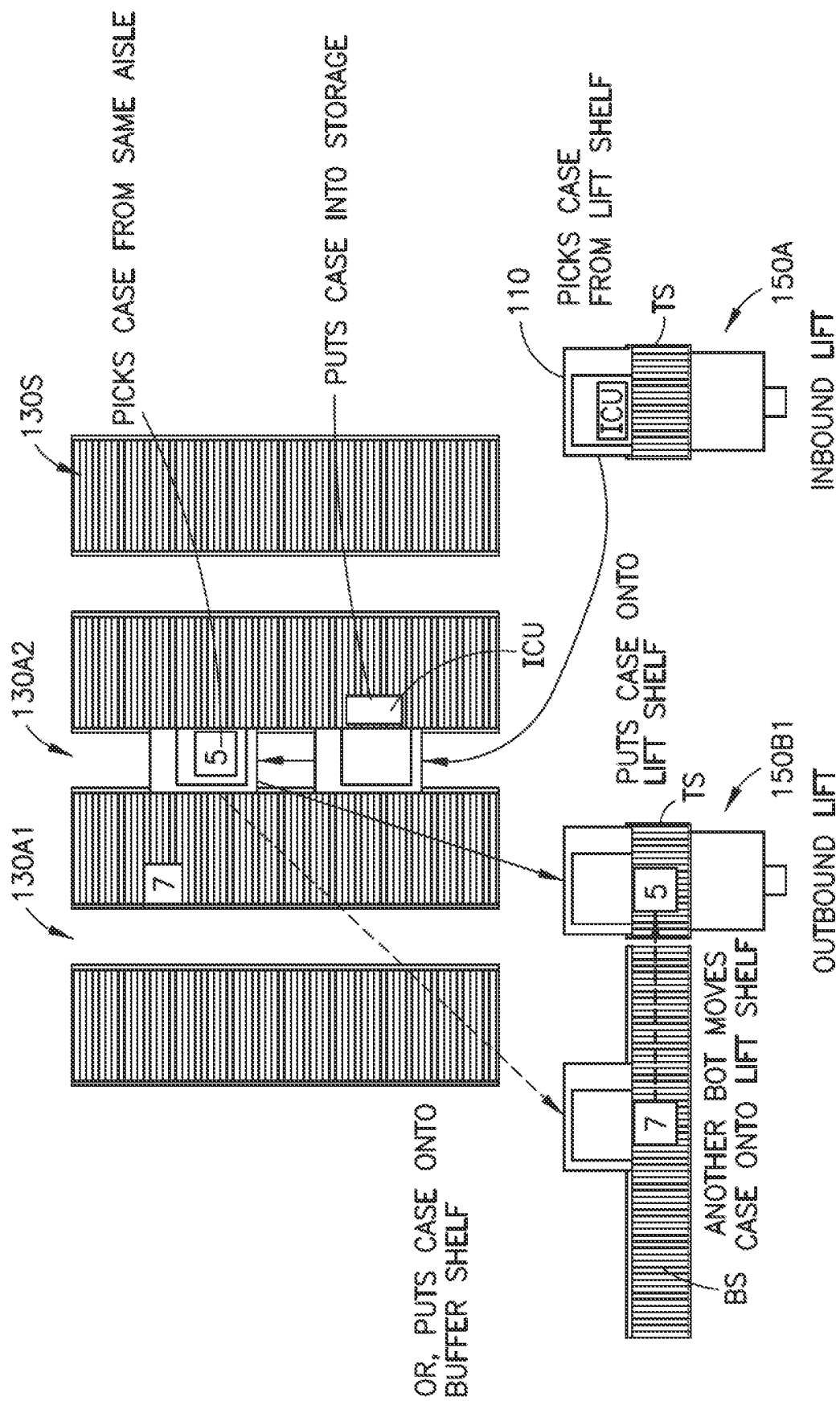


FIG.11

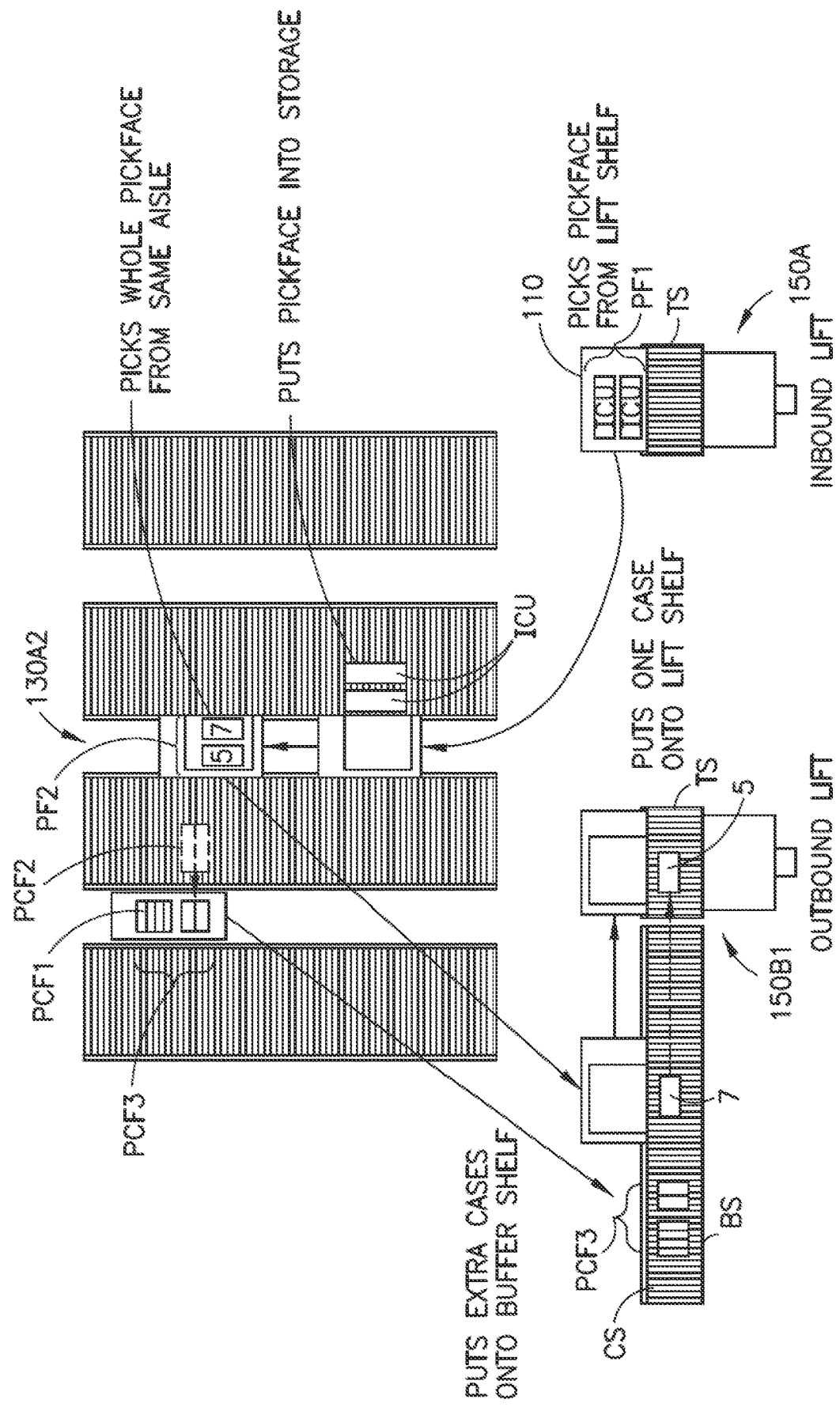


FIG.12

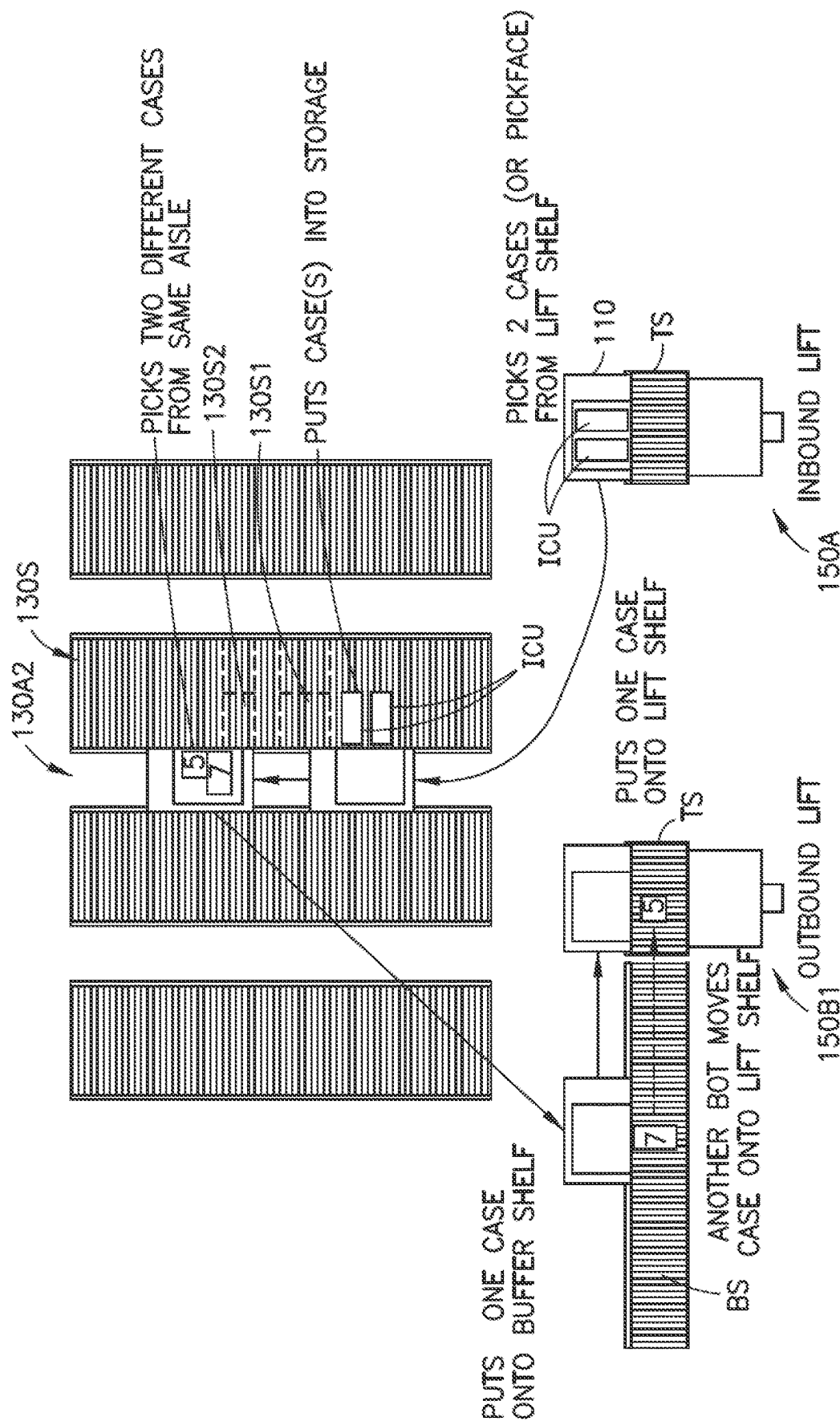
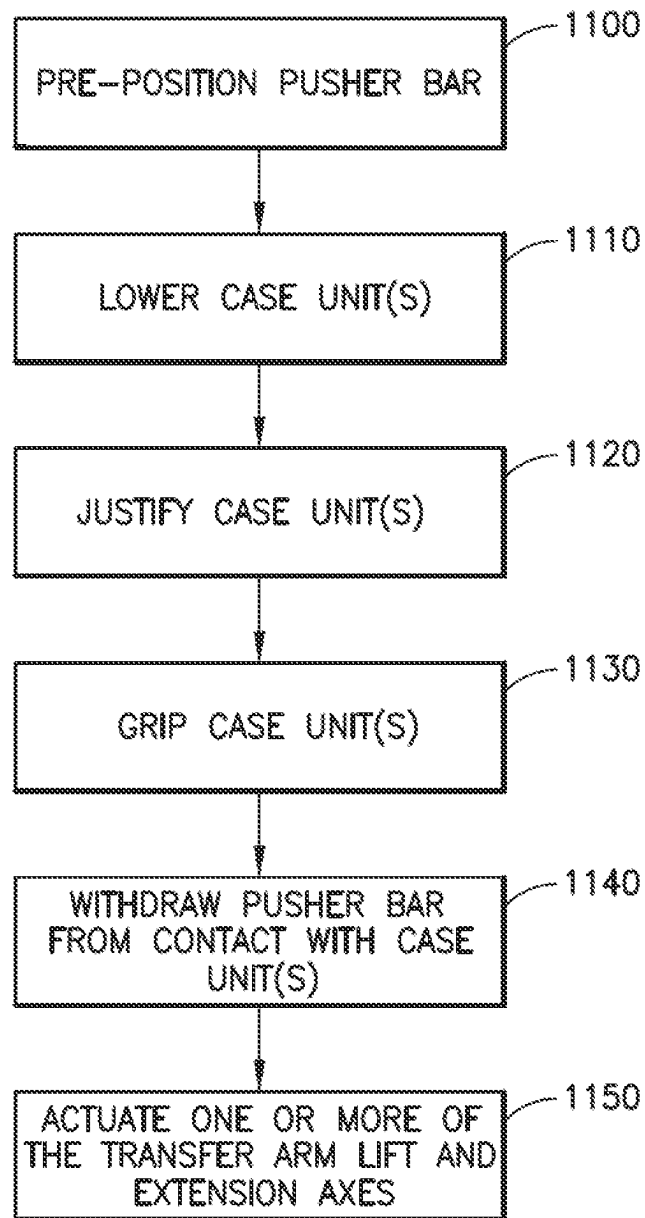


FIG.13

**FIG.14**

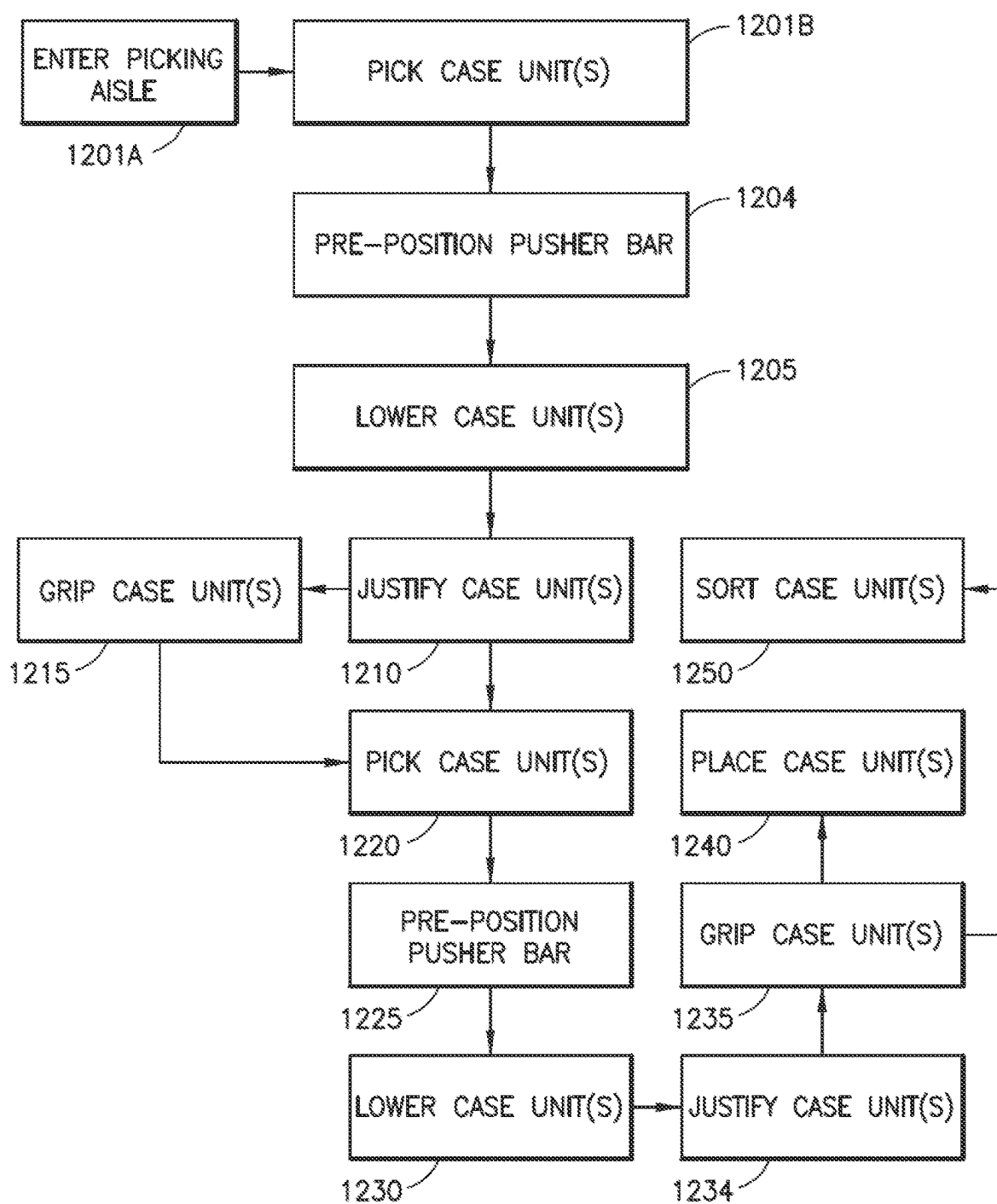
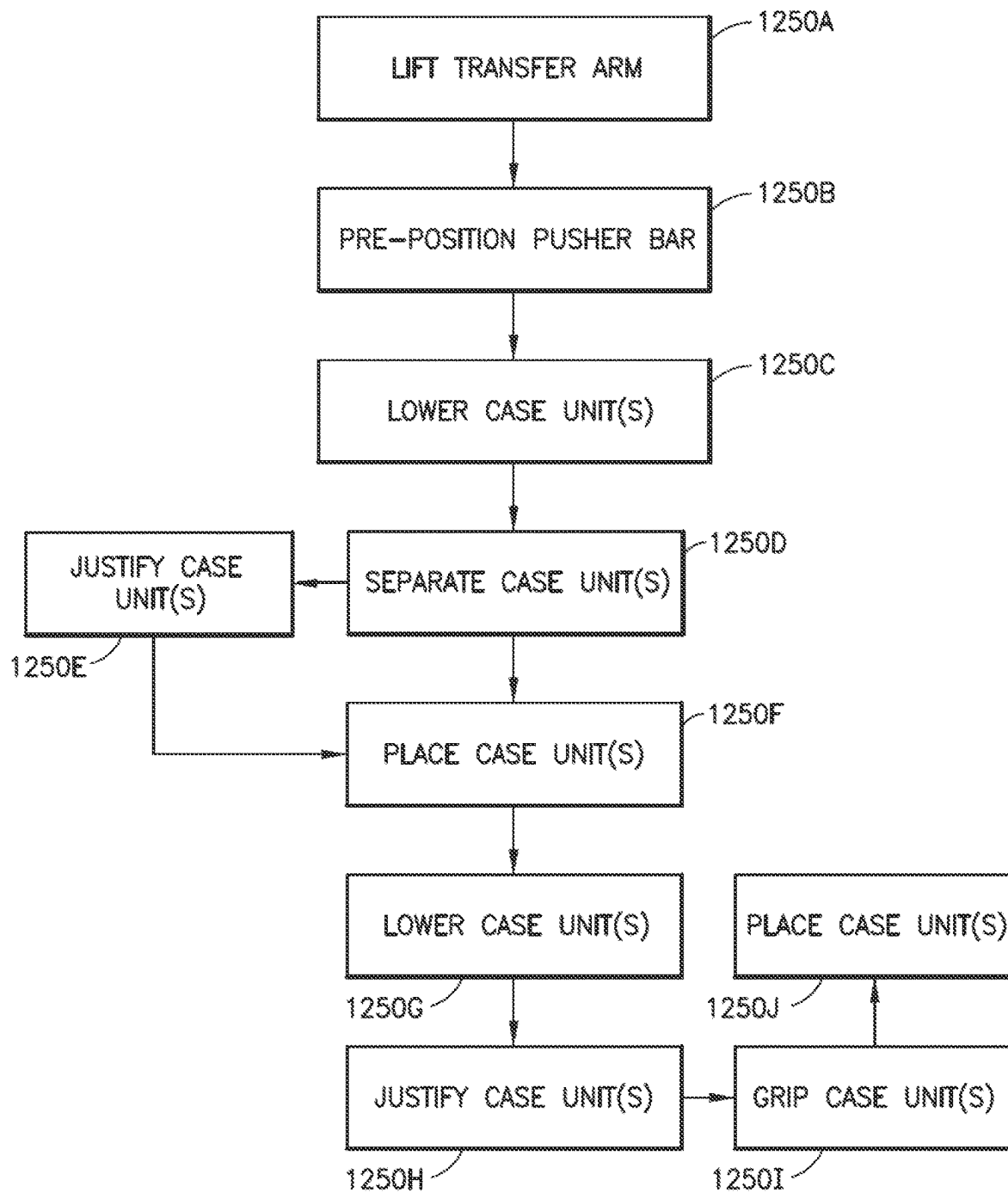
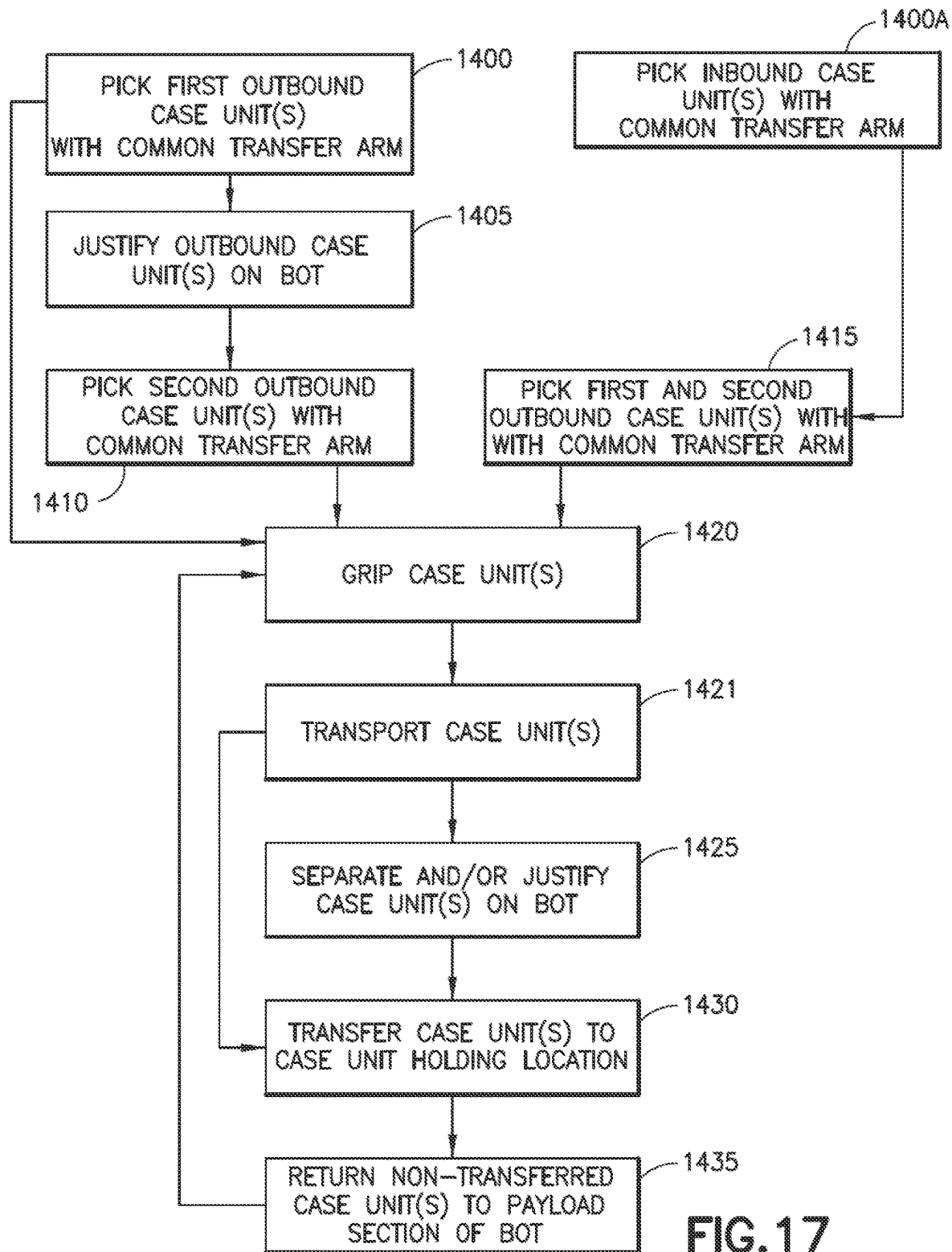
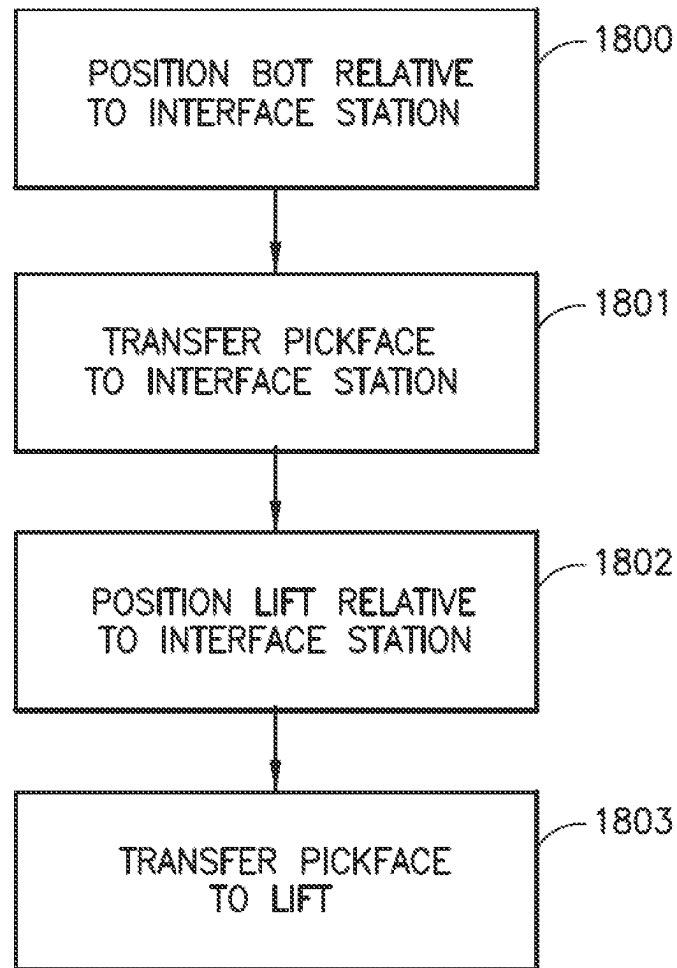
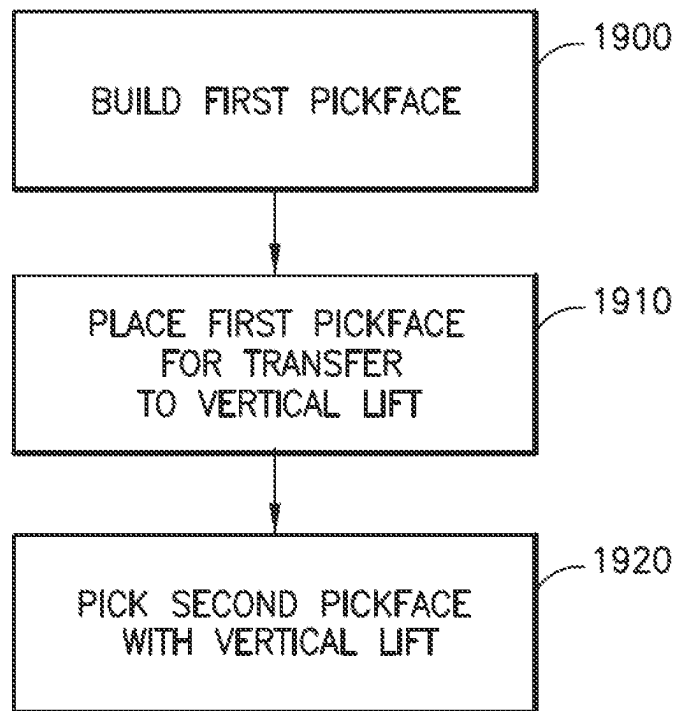
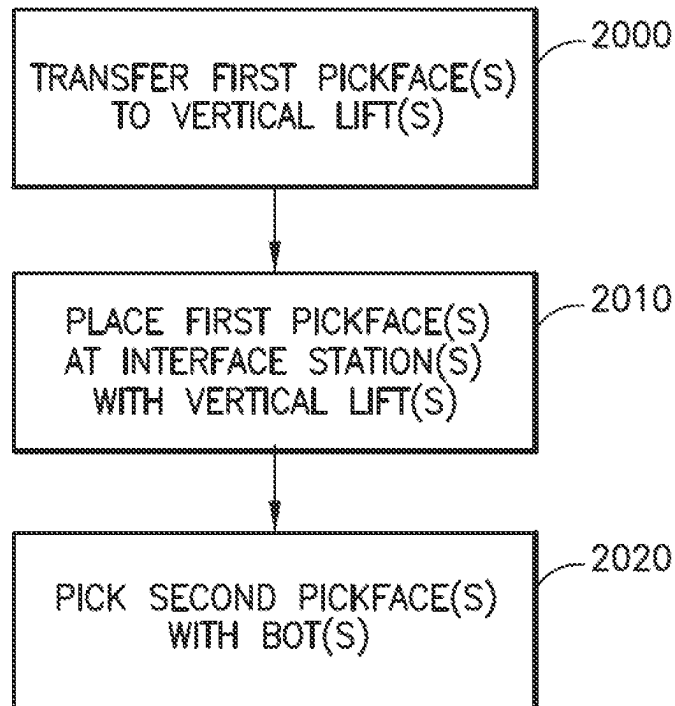


FIG. 15

**FIG.16**



**FIG.18**

**FIG.19****FIG.20**

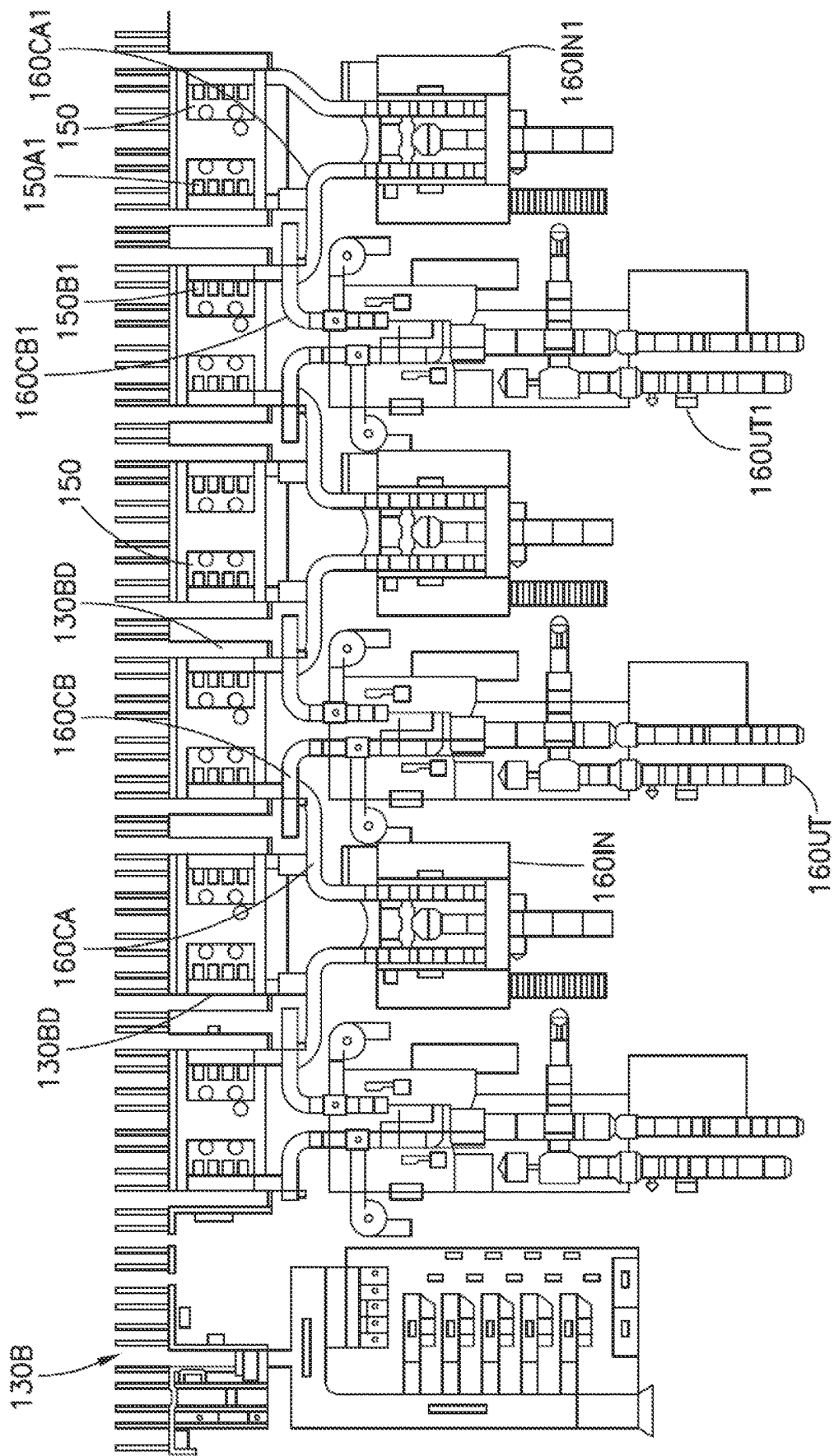


FIG. 21

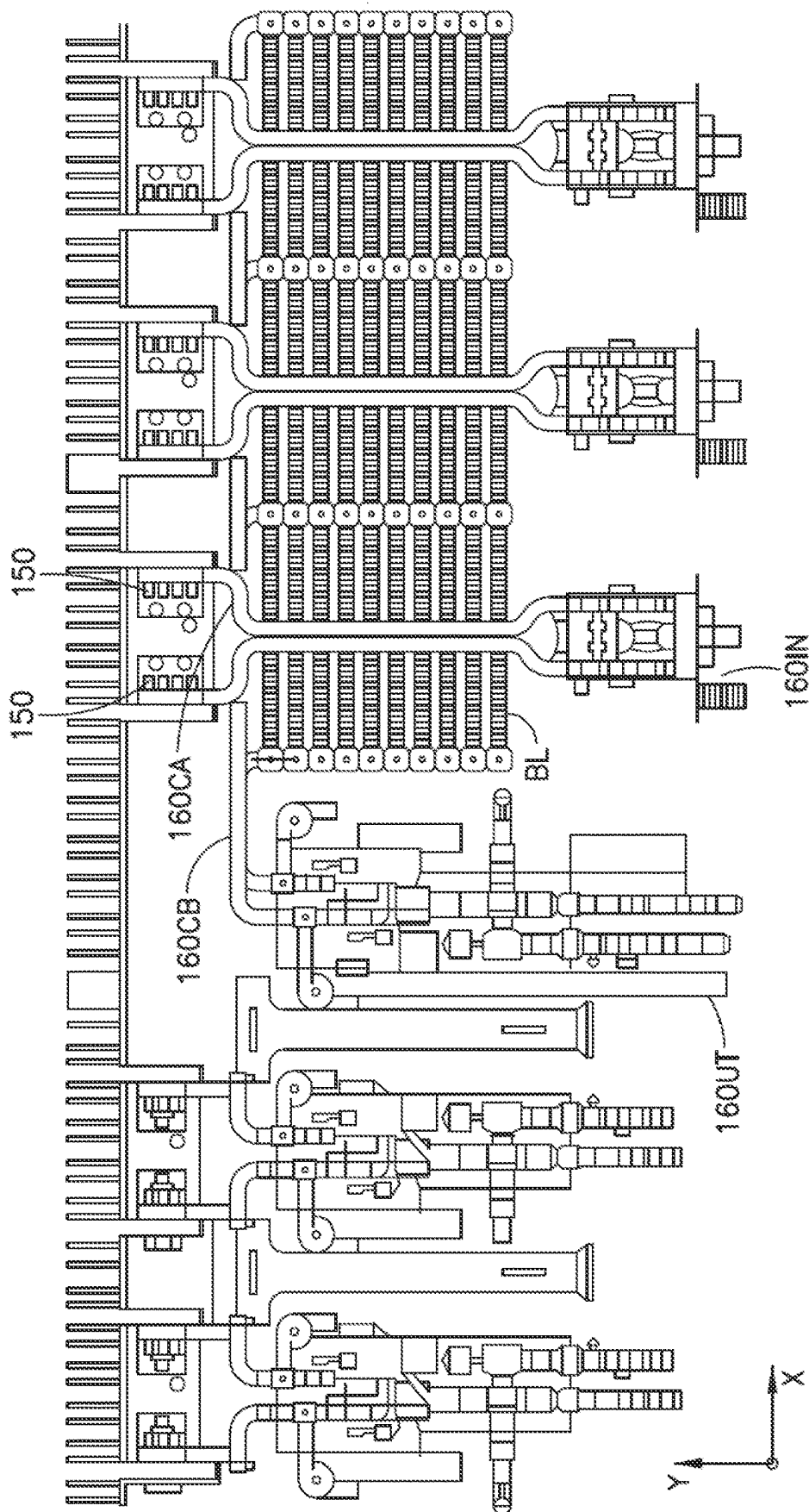


FIG. 22A

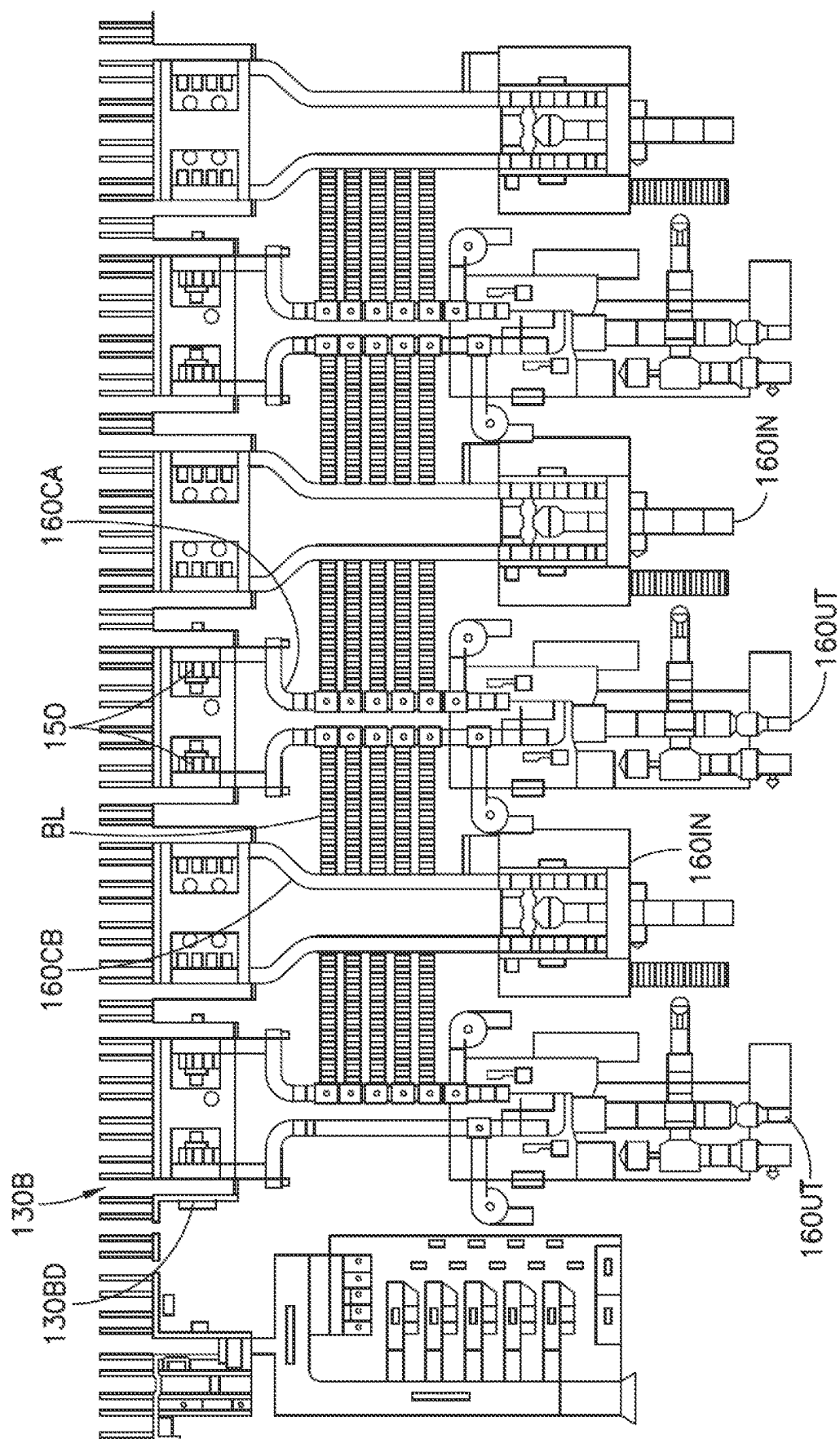
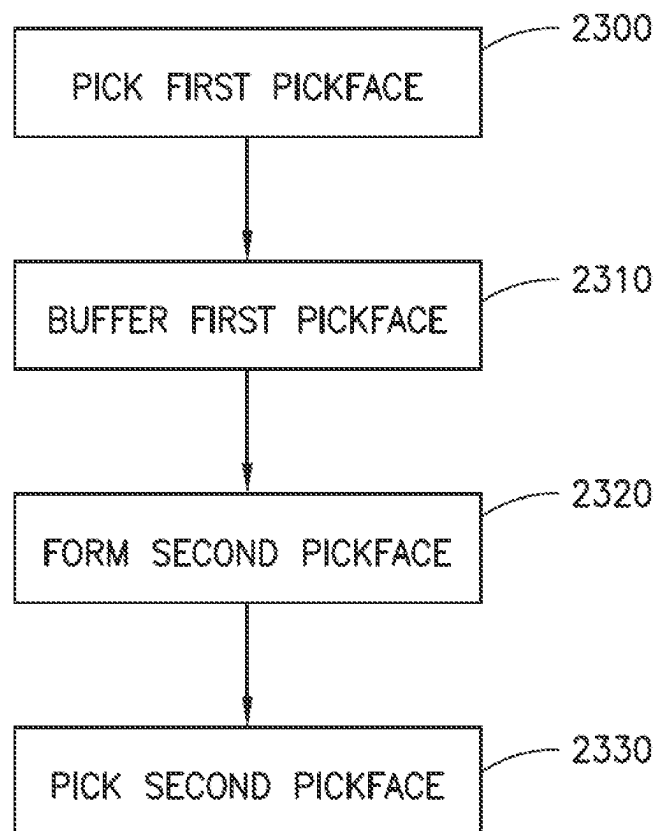


FIG. 22B

**FIG.23**

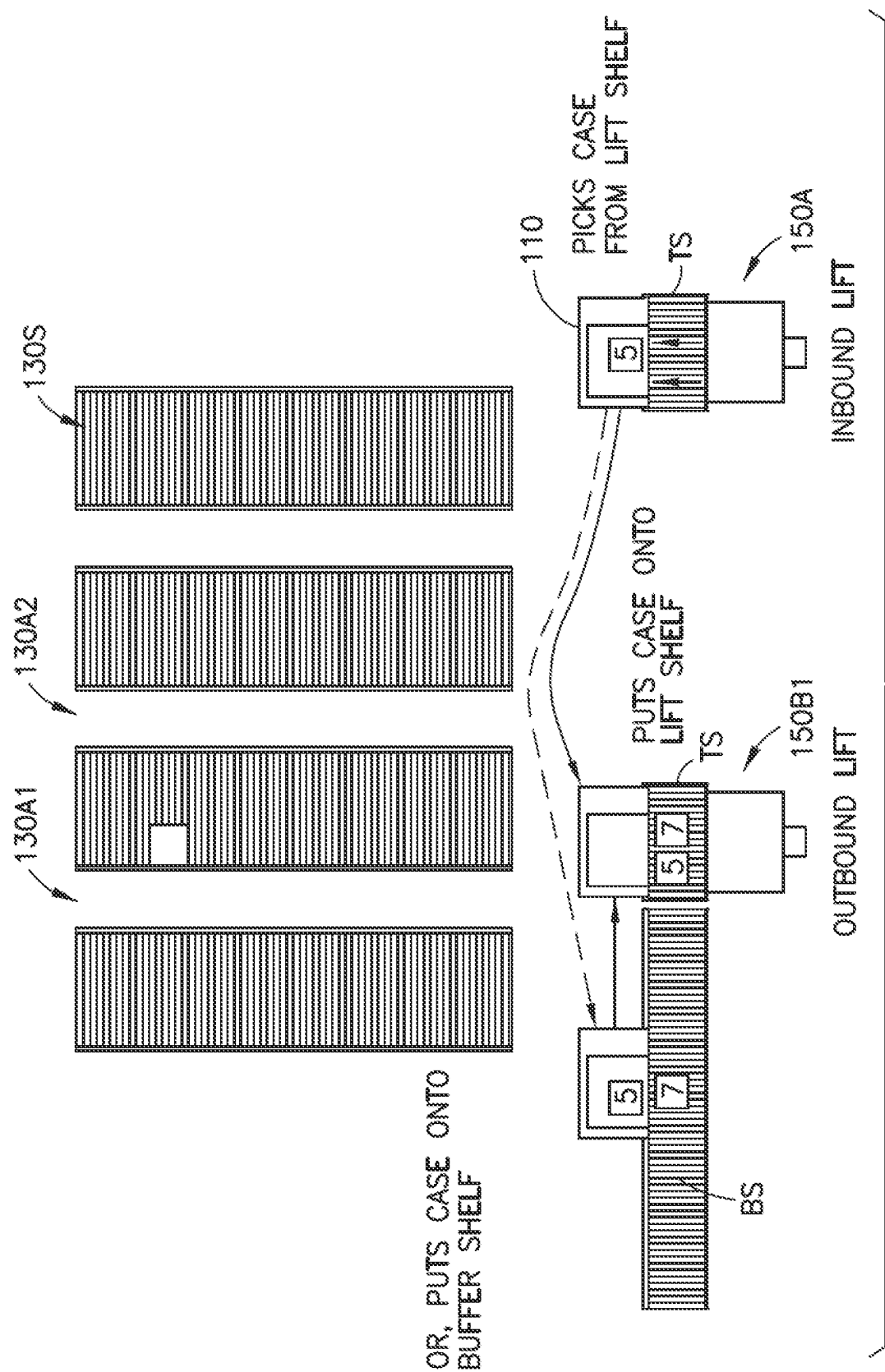
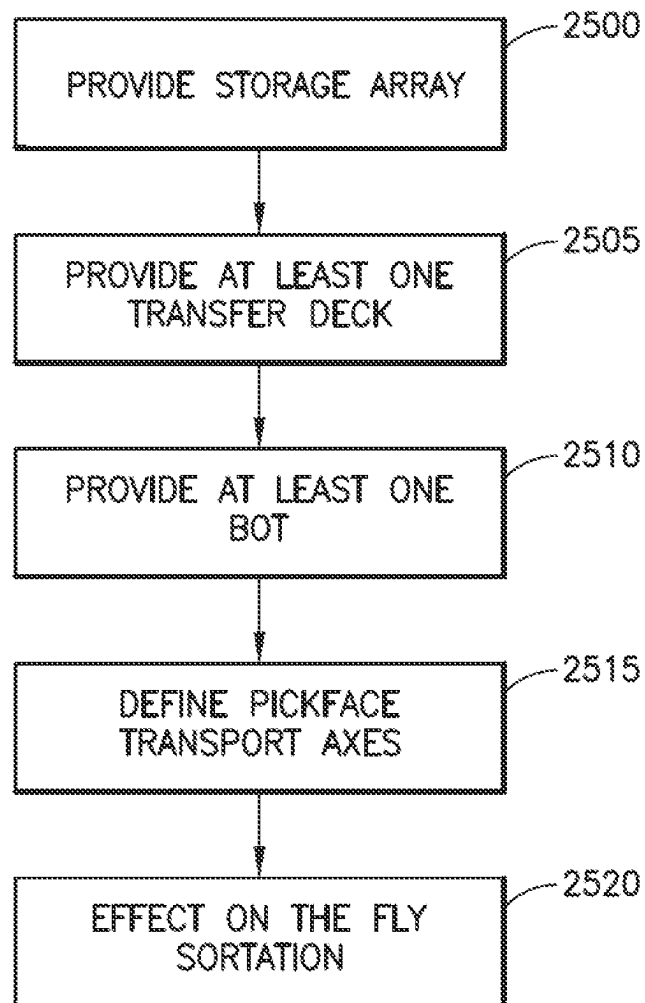
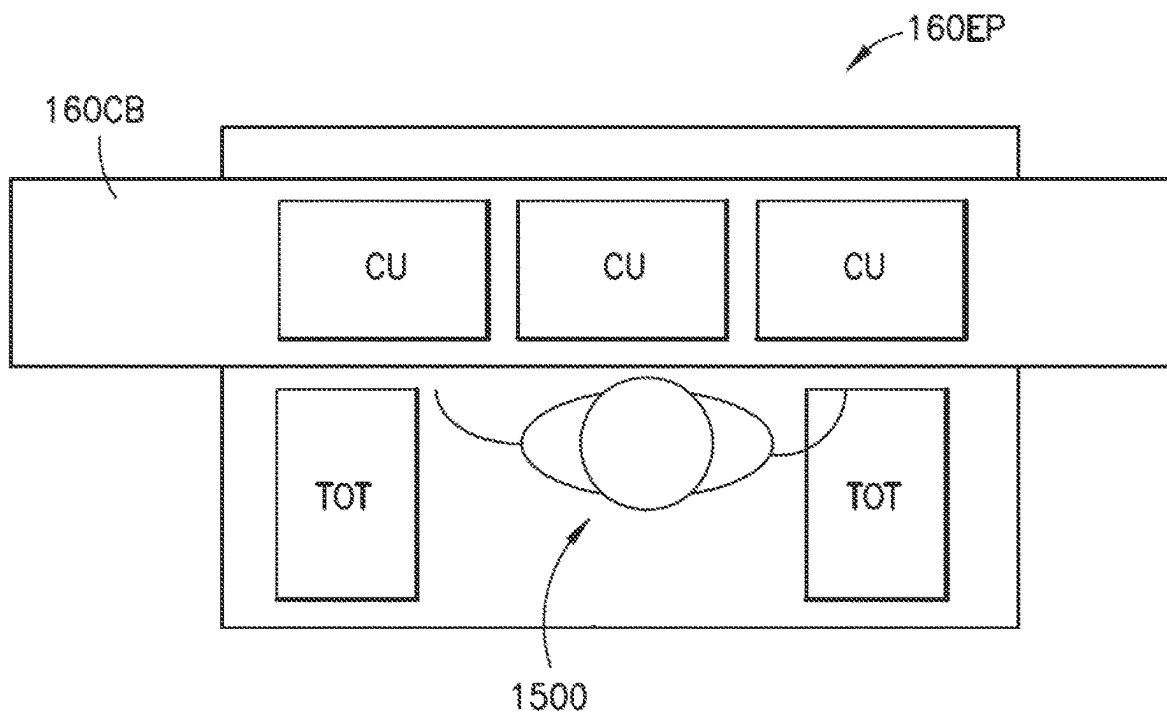
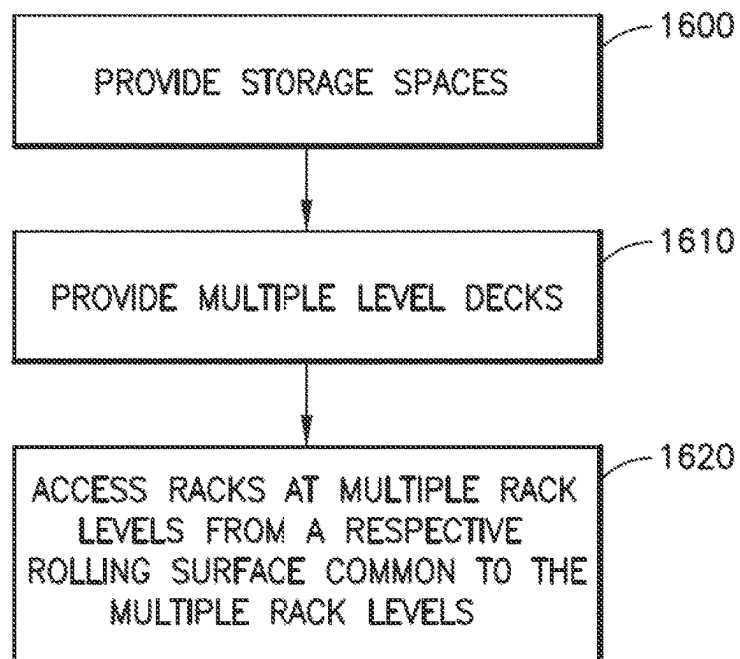


FIG. 24

**FIG.25**

**FIG. 26****FIG. 27**

1

STORAGE AND RETRIEVAL SYSTEM**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. Non-Provisional patent application Ser. No. 17/652,012, filed Feb. 22, 2022, (Now U.S. Pat. No. 11,623,822), which is a continuation of U.S. Non-Provisional patent application Ser. No. 17/108,265, filed Dec. 1, 2020, (Now U.S. Pat. No. 11,254,502), which is a continuation of U.S. Non-Provisional patent application Ser. No. 16/222,283, filed Dec. 17, 2018, (Now U.S. Pat. No. 10,850,921), which is a continuation of U.S. Non-Provisional patent application Ser. No. 15/848,809, filed Dec. 20, 2017, (Now U.S. Pat. No. 10,155,623), which is a continuation of U.S. Non-Provisional patent application Ser. No. 14/997,920, filed on Jan. 18, 2016, (Now U.S. Pat. No. 9,856,083), which claims priority from and the benefit of U.S. Provisional Patent Application No. 62/104,520, filed on Jan. 16, 2015, the disclosures of which are incorporated herein by reference in their entireties.

This application is also related to U.S. patent application Ser. No. 14/966,978, filed on Dec. 11, 2015; U.S. patent application Ser. No. 14/997,892, filed on Jan. 18, 2016; U.S. patent application Ser. No. 14/997,902, filed on Jan. 18, 2016; U.S. patent application Ser. No. 14/997,925, filed on Jan. 18, 2016; and U.S. Provisional Patent Application No. 62/107,135, filed on Jan. 23, 2015, the disclosures of which are incorporated herein by reference in their entireties.

BACKGROUND**1. Field**

The exemplary embodiments generally relate to material handling systems and, more particularly, to transport and storage of items within the material handling system.

2. Brief Description of Related Developments

Multilevel storage and retrieval systems may be used in warehouses for the storage and retrieval of goods. Generally the transportation of goods into and out of the storage structure is done with lifts for transfer to a vehicle on a storage level, vehicles travelling up ramps to a predetermined storage level, or with vehicles that include lifts traveling along guide ways. Goods stored within the storage and retrieval system are generally stored in storage spaces on each storage level such that a transport vehicle disposed on that level has access to one level of storage spaces. Generally, the lifts that transfer items to and from the storage spaces carry the vehicles between different storage levels, are incorporated into the vehicles (such as with a gantry crane) or have a paternoster configuration where the lift payload shelves continually circulate around a frame at a predetermined rate.

The case units output from the multilevel storage and retrieval systems are transferred to a packing station where the case units are placed on pallets for shipping. Generally the pallets include case units of similar size and shape so that stable case levels, sometimes with paperboard sheets disposed between the levels, are formed on the pallets. In some instances each level of tier of the pallet is separately formed and then placed on the pallet to form stacked tiers. Mixed pallets are also possible. Generally when forming a pallet layer cases are placed in a buffer station or other location at the palletizing station so that the dimensions of the case are

2

measured. A computer or other processor determines an arrangement of the cases based on the dimensions and instructs a robot to pick the cases for placement in the pallet layer.

It would be advantageous to sort case units for placement on a pallet during transport of the case units out of the storage and retrieval system storage structure to increase throughput of the storage and retrieval system.

BRIEF DESCRIPTION OF THE DRAWINGS

The foregoing aspects and other features of the disclosed embodiment are explained in the following description, taken in connection with the accompanying drawings, wherein:

FIGS. 1 and 1A are schematic illustrations of an automated storage and retrieval system in accordance with aspects of the disclosed embodiment;

FIGS. 1B, 1C, 1D and 1E are schematic illustrations of portions of the automated storage and retrieval system in accordance with aspects of the disclosed embodiment;

FIG. 1F is a schematic illustration of a mixed pallet load formed by the automated storage and retrieval system in accordance with aspects of the disclosed embodiment;

FIG. 1G is a schematic illustration of a portion of the automated storage and retrieval system in accordance with aspects of the disclosed embodiment;

FIGS. 2A and 2B are schematic illustrations of portions of the storage and retrieval system in accordance with aspects of the disclosed embodiment;

FIGS. 3A and 3B are schematic illustrations of portions of the storage and retrieval system in accordance with aspects of the disclosed embodiment;

FIGS. 4A, 4B and 5 are schematic illustrations of portions of the storage and retrieval system in accordance with aspects of the disclosed embodiment;

FIG. 6 is a schematic illustration of a transport vehicle in accordance with aspects of the disclosed embodiment;

FIG. 6A is a schematic illustration of a transport vehicle in accordance with aspects of the disclosed embodiment;

FIGS. 7 and 8 are schematic illustrations of portions of the transport vehicle in accordance with aspects of the disclosed embodiment;

FIG. 9 is a schematic illustration of a portion of the storage and retrieval system in accordance with aspects of the disclosed embodiment;

FIGS. 10, 10A-10E are schematic illustrations of portions of the transport vehicle in accordance with aspects of the disclosed embodiment;

FIGS. 11-13 are schematic illustrations of portions of the storage and retrieval system in accordance with aspects of the disclosed embodiment;

FIGS. 14-20 are exemplary flow diagrams in accordance with aspects of the disclosed embodiment;

FIGS. 21, 22A and 22B are schematic illustrations of portions of the automated storage and retrieval system in accordance with aspects of the disclosed embodiment;

FIG. 23 is an exemplary flow diagram in accordance with aspects of the disclosed embodiment;

FIG. 24 is a schematic illustration of a portion of the storage and retrieval system in accordance with aspects of the disclosed embodiment;

FIG. 25 exemplary flow diagram in accordance with aspects of the disclosed embodiment;

FIG. 26 is a schematic illustration of an operator station of the storage and retrieval system in accordance with aspects of the disclosed embodiment; and

FIG. 27 is an exemplary flow diagram in accordance with aspects of the disclosed embodiment.

DETAILED DESCRIPTION

FIG. 1 is a schematic illustration of an automated storage and retrieval system 100 in accordance with aspects of the disclosed embodiment. Although the aspects of the disclosed embodiment will be described with reference to the drawings, it should be understood that the aspects of the disclosed embodiment can be embodied in many forms. In addition, any suitable size, shape or type of elements or materials could be used.

In accordance with aspects of the disclosed embodiment the automated storage and retrieval system 100 may operate in a retail distribution center or warehouse to, for example, fulfill orders received from retail stores for case units such as those described in U.S. patent application Ser. No. 13/326,674 filed on Dec. 15, 2011, the disclosure of which is incorporated by reference herein in its entirety. For example, the case units are cases or units of goods not stored in trays, on totes or on pallets (e.g. uncontained). In other examples, the case units are cases or units of goods that are contained in any suitable manner such as in trays, on totes or on pallets. In still other examples, the case units are a combination of uncontained and contained items. It is noted that the case units, for example, include cased units of goods (e.g. case of soup cans, boxes of cereal, etc.) or individual goods that are adapted to be taken off of or placed on a pallet. In accordance with the aspects of the disclosed embodiment, shipping cases for case units (e.g. cartons, barrels, boxes, crates, jugs, or any other suitable device for holding case units) may have variable sizes and may be used to hold case units in shipping and may be configured so they are capable of being palletized for shipping. It is noted that when, for example, bundles or pallets of case units arrive at the storage and retrieval system the content of each pallet may be uniform (e.g. each pallet holds a predetermined number of the same item—one pallet holds soup and another pallet holds cereal) and as pallets leave the storage and retrieval system the pallets may contain any suitable number and combination of different case units (e.g. a mixed pallet where each mixed pallet holds different types of case units—a pallet holds a combination of soup and cereal) that are provided to, for example the palletizer in a sorted arrangement for forming the mixed pallet. In the embodiments the storage and retrieval system described herein may be applied to any environment in which case units are stored and retrieved.

Also referring to FIG. 1F, it is noted that when, for example, incoming bundles or pallets (e.g. from manufacturers or suppliers of case units arrive at the storage and retrieval system for replenishment of the automated storage and retrieval system 100, the content of each pallet may be uniform (e.g. each pallet holds a predetermined number of the same item—one pallet holds soup and another pallet holds cereal). As may be realized, the cases of such pallet load may be substantially similar or in other words, homogeneous cases (e.g. similar dimensions), and may have the same SKU (otherwise, as noted before the pallets may be “rainbow” pallets having layers formed of homogeneous cases). As pallets PAL leave the storage and retrieval system 100, with cases filling replenishment orders, the pallets PAL may contain any suitable number and combination of different case units CU (e.g. each pallet may hold different types of case units—a pallet holds a combination of canned soup, cereal, beverage packs, cosmetics and household

cleaners). The cases combined onto a single pallet may have different dimensions and/or different SKU's. In one aspect of the exemplary embodiment, the storage and retrieval system 100 may be configured to generally include an in-feed section, a storage and sortation section (where, in one aspect, storage of items is optional) and an output section as will be described in greater detail below. As may be realized, in one aspect of the disclosed embodiment the system 100 operating for example as a retail distribution center may serve to receive uniform pallet loads of cases, breakdown the pallet goods or disassociate the cases from the uniform pallet loads into independent case units handled individually by the system, retrieve and sort the different cases sought by each order into corresponding groups, and transport and assemble the corresponding groups of cases into what may be referred to as mixed case pallet loads MPL. As may also be realized, as illustrated in FIG. 26, in one aspect of the disclosed embodiment the system 100 operating for example as a retail distribution center may serve to receive uniform pallet loads of cases, breakdown the pallet goods or disassociate the cases from the uniform pallet loads into independent case units handled individually by the system, retrieve and sort the different cases sought by each order into corresponding groups, and transport and sequence the corresponding groups of cases (in the manner described herein) at an operator station 160EP where items are picked from the different case units CU, and/or the different case units CU themselves, are placed in one or more bag(s), tote(s) or other suitable container(s) TOT by an operator 1500, or any suitable automation, in a predetermined order sequence of picked items according to, for example, an order, fulfilling one or more customer orders, in which the case units CU are sequenced at the operator station 160EP in accordance with the predetermined order sequence, noting that the sequencing of the case units CU as described herein effects the sequencing of the case units CU at the operator station 160EP. The in-feed section may generally be capable of resolving the uniform pallet loads to individual cases, and transporting the cases via suitable transport, for input to the storage and sortation section. In other aspects the output section assembles the appropriate group of ordered case units, that may be different in SKU, dimensions, etc. into bags, totes or other suitable containers to according to the predetermined order sequence of picked items at the operator station 160EP (such as to fill a customer order).

The storage and sortation section includes, as will be described in greater detail below, a multilevel automated storage array that has a transport system that in turn receives or feeds individual cases into the multilevel storage array for storage in a storage area. The storage and sortation section also defines outbound transport of case units from the multilevel storage array such that desired case units are individually retrieved in accordance with commands generated in accordance to orders entered into a warehouse management system, such as warehouse management system 2500, for transport to the output section. In other aspects, the storage and sortation section receives individual cases, sorts the individual cases (utilizing, for example, the buffer and interface stations described herein) and transfers the individual cases to the output section in accordance to orders entered into the warehouse management system. The sorting and grouping of cases according to order (e.g. an order out sequence) may be performed in whole or in part by either the storage and retrieval section or the output section, or both, the boundary between being one of convenience for the description and the sorting and grouping being capable of being performed any number of ways. The intended result

is that the output section assembles the appropriate group of ordered cases, that may be different in SKU, dimensions, etc. into mixed case pallet loads in the manner described in, for example, U.S. patent application Ser. No. 13/654,293 filed on Oct. 17, 2012 (now U.S. Pat. No. 8,965,559) the disclosure of which is incorporated herein by reference in its entirety.

In the exemplary embodiment, the output section generates the pallet load in what may be referred to as a structured architecture of mixed case stacks. The structured architecture of the pallet load described herein is representative and in other aspects the pallet load may have any other suitable configuration. For example, the structured architecture may be any suitable predetermined configuration such as a truck bay load or other suitable container or load container envelope holding a structural load. The structured architecture of the pallet load may be characterized as having several flat case layers **L121-L125**, **L12T**, at least one of which is formed of non-intersecting, free-standing and stable stacks of multiple mixed cases. The mixed case stacks of the given layer have substantially the same height, to form as may be realized substantially flat top and bottom surfaces of the given layer, and may be sufficient in number to cover the pallet area, or a desired portion of the pallet area. Overlaying layer(s) may be orientated so that corresponding cases of the layer(s) bridge between the stacks of the supporting layer. Thus, stabilizing the stacks and correspondingly the interfacing layer(s) of the pallet load. In defining the pallet load into a structured layer architecture, the coupled 3-D pallet load solution is resolved into two parts that may be saved separately, a vertical (1-D) part resolving the load into layers, and a horizontal (2-D) part of efficiently distributing stacks of equal height to fill out the pallet height of each layer. As will be described below, the storage and retrieval system outputs case units to the output section so that the two parts of the 3-D pallet load solution are resolved. The predetermined structure of the mixed pallet load defines an order of case units, whether the case units are a singular case unit pickface or a combined case unit pickface provided by the sortation and output sections to a load construction system (which may be automated or manual loading).

In accordance with aspects of the disclosed embodiment, referring again to FIG. 1, the automated storage and retrieval system **100** includes input stations **160IN** (which include depalletizers **160PA** and/or conveyors **160CA** for transporting items to lift modules for entry into storage) and output stations **160UT** (which include palletizers **160PB**, operator stations **160EP** and/or conveyors **160CB** for transporting case units from lift modules for removal from storage), input and output vertical lift modules **150A**, **150B** (generally referred to as lift modules **150**—it is noted that while input and output lift modules are shown, a single lift module may be used to both input and remove case units from the storage structure), a storage structure **130**, and a number of autonomous rovers or transport vehicles **110** (referred to herein as “bots”). It is noted that the depalletizers **160PA** may be configured to remove case units from pallets so that the input station **160IN** can transport the items to the lift modules **150** for input into the storage structure **130**. The palletizers **160PB** may be configured to place items removed from the storage structure **130** on pallets **PAL** (FIG. 1F) for shipping. As used herein the lift modules **150**, storage structure **130** and bots **110** may be collectively referred to herein as the multilevel automated storage array (e.g. storage and sorting section) noted above so as to define (e.g. relative to e.g. a bot **110** frame of reference **REF**—FIG. 6—or any other suitable storage and retrieval system frame of reference) transport/

throughput axes (in e.g. three dimensions) that serve the three dimensional multilevel automated storage array where each throughput axis has an integral “on the fly sortation” (e.g. sortation of case units during transport of the case units) so that case unit sorting and throughput occurs substantially simultaneously without dedicated sorters as will be described in further detail below. Sortation along each throughput axis is selectable so that a load out (e.g. transfer of outgoing case units to form a pallet load) is effected along all throughput axes or any combination of throughput axes/axis with substantially no throughput cost relative to a throughput of case units without any sortation (e.g. a “sortationless” throughput). As an example of case unit throughput as it relates to sortation, referring also to FIG. 1A, the storage and retrieval system **100** includes several areas or regions of throughput. For example, there is multi-level case unit storage throughput **130LTP** (e.g. placement of case units into storage), horizontal case unit transport throughput **110TP** (e.g. a transfer of case unit(s) from storage along the picking aisles and transfer decks), case buffering throughput **BTSTP** (e.g. buffering of case units to facilitate transfer of the case units between storage and vertical transport), vertical transport throughput **150TP** (e.g. transfer of case units by the vertical lifts), and throughput at the output stations **160TP** which includes, e.g., transport by conveyors **160CB** and palletizing by palletizer **160PB**. In one aspect sortation of case units, as described herein, is effected substantially coincident (e.g. “on the fly”) with throughput **130LTP**, **110TP**, **BTSTP**, **150TP** of case units along each throughput axis (e.g. the X, Y, Z axes relative to, for example, a bot **110** and or lift **150** frame of reference) and sortation along each axis is independently selectable so that sortation is effected along one or more X, Y, Z axes.

As may be realized, an on the fly sortation of case units occurs, in one aspect, on the bot **110** without offloading case units/pickfaces carried by the bot **110** when the bot **110** is one of both moving between case unit/pickface holding locations and static/standing (e.g. not traversing a transfer deck, picking aisle, etc.) As will be described below, one or more of high density multi-level shelving aisles, linear buffer stations **BS** along the transfer decks **130B** and linear multi-place transfer stations **TS** effect on the fly substantially coincident sortation with throughput along the X axis. As will also be described below, one or more of the bot **110** transfer arm and end effector **110PA** (which is configured to sort cases/pickfaces through end effector traverse along the Y axis for multi-independent picking/placing of cases/pickfaces where the Y axis is defined by the extension of the transfer arm **110PA** and is in a different direction angled relative to another of the transport axes defined by the bot **110** along the picking aisle **130A**) and independent load handling devices of the lifts **150** (configured for sortation on the lift platforms through extension of the load handling device in along the Y axis) effect on the fly substantially coincident sortation with throughput along the Y axis. As may be realized, the lifts **150** are configured to transport pickfaces between different transfer deck levels and provide on the fly substantially coincident sortation with throughput along the Z axis (which is defined by the lifts **150**) as will be described herein. In one aspect the lift is configured to pick one or more pickfaces from one or more transfer deck levels and transport the one or more pickfaces to a load fill section or cell (such as output station **160UT**) of the storage and retrieval system **100**. The term load fill section or load fill cell (used interchangeably herein, and generally referred to as a load fill) refers to either a pallet load fill section/cell

(such as for the creation of a mixed pallet load MPL) or an itemized load fill section/cell as described with respect to FIG. 26.

Also referring to FIGS. 1G and 2A, the storage structure 130 may include multiple storage rack modules RM, configured in a high density three dimensional rack array RMA, that are accessible by storage or deck levels 130L. As used herein the term “high density three dimensional rack array” refers to the three dimensional rack array RMA having undeterministic open shelving distributed along picking aisles 130A where multiple stacked shelves are accessible from a common picking aisle travel surface or picking aisle level (e.g. case units are placed at each picking aisle level within dynamically allocated storage spaces so that the vertical space/gap VG and horizontal space/gap G between case units is minimized at each picking aisle level, as will be described in greater detail below).

Each storage level 130L includes pickface storage/hand-off spaces 130S (referred to herein as storage spaces 130S) formed by the rack modules RM where the rack modules include shelves that are disposed along storage or picking aisles 130A (that are connected to the transfer deck 130B) which, e.g., extend linearly through the rack module array RMA and provide bot 110 access to the storage spaces 130S and transfer deck(s) 130B. In one aspect, the shelves of the rack modules RM are arranged as multi-level shelves that are distributed along the picking aisles 130A. As may be realized the bots 110 travel on a respective storage level 130L along the picking aisles 130A and the transfer deck 130B for transferring case units between any of the storage spaces 130S of the storage structure 130 (e.g. on the level which the bot 110 is located) and any of the lift modules 150 (e.g. each of the bots 110 has access to each storage space 130S on a respective level and each lift module 150 on a respective storage level 130L). The transfer decks 130B are arranged at different levels (corresponding to each level 130L of the storage and retrieval system) that may be stacked one over the other or horizontally offset, such as having one transfer deck 130B at one end or side RMAE1 of the storage rack array RMA or at several ends or sides RMAE1, RMAE2 of the storage rack array RMA as described in, for example, U.S. patent application Ser. No. 13/326,674 filed on Dec. 15, 2011 the disclosure of which is incorporated herein by reference in its entirety.

The transfer decks 130B are substantially open and configured for the undeterministic traversal of bots 110 along multiple travel lanes (e.g. along the X throughput axis with respect to the bot frame of reference REF illustrated in FIG. 6) across and along the transfer decks 130B. As may be realized, the transfer deck(s) 130B at each storage level 130L communicate with each of the picking aisles 130A on the respective storage level 130L. Bots 110 bi-directionally traverse between the transfer deck(s) 130B and picking aisles 130A on each respective storage level 130L so as to travel along the picking aisles (e.g. along the X throughput axis with respect to the bot frame of reference REF illustrated in FIG. 6) and access the storage spaces 130S disposed in the rack shelves alongside each of the picking aisles 130A (e.g. bots 110 may access, along the Y throughput axis, storage spaces 130S distributed on both sides of each aisle such that the bot 110 may have a different facing when traversing each picking aisle 130A, for example, referring to FIG. 6, drive wheels 202 leading a direction of travel or drive wheels trailing a direction of travel). As may be realized, throughput outbound from the storage array in the horizontal plane corresponding to a predetermined storage or deck level 130L is effected by and manifest in the

combined or integrated throughput along both the X and Y throughput axes. As noted above, the transfer deck(s) 130B also provides bot 110 access to each of the lifts 150 on the respective storage level 130L where the lifts 150 feed and remove case units (e.g. along the Z throughput axis) to and/or from each storage level 130L and where the bots 110 effect case unit transfer between the lifts 150 and the storage spaces 130S.

As described above, referring also to FIG. 2A, in one aspect the storage structure 130 includes multiple storage rack modules RM, configured in a three dimensional array RMA where the racks are arranged in aisles 130A, the aisles 130A being configured for bot 110 travel within the aisles 130A. The transfer deck 130B has an undeterministic transport surface on which the bots 110 travel where the undeterministic transport surface 130BS has more than one juxtaposed travel lane (e.g. high speed bot travel paths HSTP) connecting the aisles 130A. As may be realized, the juxtaposed travel lanes are juxtaposed along a common undeterministic transport surface 130BS between opposing sides 130BD1, 130BD2 of the transfer deck 130B. As illustrated in FIG. 2A, in one aspect the aisles 130A are joined to the transfer deck 130B on one side 130BD2 of the transfer deck 130B but in other aspects, the aisles are joined to more than one side 130BD1, 130BD2 of the transfer deck 130B in a manner substantially similar to that described in U.S. patent application Ser. No. 13/326,674 filed on Dec. 15, 2011, the disclosure of which is previously incorporated by reference herein in its entirety. As will be described in greater detail below the other side 130BD1 of the transfer deck 130B includes deck storage racks (e.g. interface stations TS and buffer stations BS) that are distributed along the other side 130BD1 of the transfer deck 130B so that at least one part of the transfer deck is interposed between the deck storage racks (such as, for example, buffer stations BS or transfer stations TS) and the aisles 130A. The deck storage racks are arranged along the other side 130BD1 of the transfer deck 130B so that the deck storage racks communicate with the bots 110 from the transfer deck 130B and with the lift modules 150 (e.g. the deck storage racks are accessed by the bots 110 from the transfer deck 130B and by the lifts 150 for picking and placing pickfaces so that pickfaces are transferred between the bots 110 and the deck storage racks and between the deck storage racks and the lifts 150 and hence between the bots 110 and the lifts 150).

Referring again to FIG. 1, each storage level 130L may also include charging stations 130C for charging an on-board power supply of the bots 110 on that storage level 130L such as described in, for example, U.S. patent application Ser. No. 14/209,086 filed on Mar. 13, 2014 and Ser. No. 13/326,823 filed on Dec. 15, 2011 (now U.S. Pat. No. 9,082,112), the disclosures of which are incorporated herein by reference in their entireties.

The bots 110 may be any suitable independently operable autonomous transport vehicles that carry and transfer case units along the X and Y throughput axes throughout the storage and retrieval system 100. In one aspect the bots 110 are automated, independent (e.g. free riding) autonomous transport vehicles. Suitable examples of bots can be found in, for exemplary purposes only, U.S. patent application Ser. No. 13/326,674 filed on Dec. 15, 2011; U.S. patent application Ser. No. 12/757,312 filed on Apr. 9, 2010 (now U.S. Pat. No. 8,425,173); U.S. patent application Ser. No. 13/326,423 filed on Dec. 15, 2011; U.S. patent application Ser. No. 13/326,447 filed on Dec. 15, 2011 (now U.S. Pat. No. 8,965,619); U.S. patent application Ser. No. 13/326,505 Dec. 15, 2011 (now U.S. Pat. No. 8,696,010); U.S. patent

application Ser. No. 13/327,040 filed on Dec. 15, 2011 (now U.S. Pat. No. 9,187,244); U.S. patent application Ser. No. 13/326,952 filed on Dec. 15, 2011; U.S. patent application Ser. No. 13/326,993 filed on Dec. 15, 2011; U.S. patent application Ser. No. 14/486,008 filed on Sep. 15, 2014; and U.S. Provisional Patent Application No. 62/107,135 filed on Jan. 23, 2015, the disclosures of which are incorporated by reference herein in their entireties. The bots **110** (described in greater detail below) may be configured to place case units, such as the above described retail merchandise, into picking stock in the one or more levels of the storage structure **130** and then selectively retrieve ordered case units. As may be realized, in one aspect, the throughput axes X and Y (e.g. pickface transport axes) of the storage array are defined by the picking aisles **130A**, at least one transfer deck **130B**, the bot **110** and the extendable end effector (as described herein) of the bot **110** (and in other aspects the extendable end effector of the lifts **150** also, at least in part, defines the Y throughput axis). The pickfaces are transported between an inbound section of the storage and retrieval system **100**, where pickfaces inbound to the array are generated (such as, for example, input station **160IN**) and a load fill section of the storage and retrieval system **100** (such as for example, output station **160UT**), where outbound pickfaces from the array are arranged to fill a load in accordance with a predetermined load fill order sequence. In one aspect, the storage rack modules RM and the bots **110** are arranged so that in combination the storage rack modules RM and the bots **110** effect the on the fly sortation of mixed case pickfaces coincident with transport on at least one (or in other aspects on at least one of each of the more than one) of the throughput axes so that two or more pickfaces are picked from one or more of the storage spaces and placed at one or more pickface holding locations (such as, for example, the buffer and transfer stations BS, TS), that are different than the storage spaces **130S**, according to the predetermined load fill order sequence.

The bots **110**, lift modules **150** and other suitable features of the storage and retrieval system **100** are controlled in any suitable manner such as by, for example, one or more central system control computers (e.g. control server) **120** through, for example, any suitable network **180**. In one aspect the network **180** is a wired network, a wireless network or a combination of wireless and wired networks using any suitable type and/or number of communication protocols. In one aspect, the control server **120** includes a collection of substantially concurrently running programs (e.g. system management software) for substantially automatic control of the automated storage and retrieval system **100**. The collection of substantially concurrently running programs, for example, being configured to manage the storage and retrieval system **100** including, for exemplary purposes only, controlling, scheduling, and monitoring the activities of all active system components, managing inventory (e.g. which case units are input and removed, the order in which the cases are removed and where the case units are stored) and pickfaces (e.g. one or more case units that are movable as a unit and handled as a unit by components of the storage and retrieval system), and interfacing with a warehouse management system **2500**. The control server **120** may, in one aspect, be configured to control the features of the storage and retrieval system in the manner described herein. For simplicity and ease of explanation the term “case unit(s)” is generally used herein for referring to both individual case units and pickfaces (a pickface is formed of multiple case units that are moved as a unit).

Referring also to FIGS. **1B** and **1D** the rack module array RMA of the storage structure **130** includes vertical support members **1212** and horizontal support members **1200** that define the high density automated storage array as will be described in greater detail below. Rails **1200S** may be mounted to one or more of the vertical and horizontal support members **1212**, **1200** in, for example, picking aisles **130A** and be configured so that the bots **110** ride along the rails **1200S** through the picking aisles **130A**. At least one side of at least one of the picking aisles **130A** of at least one storage level **130L** may have one or more storage shelves (e.g. formed by rails **1210**, **1200** and slats **1210S**) provided at differing heights so as to form multiple shelf levels **130LS1-130LS4** between the storage or deck levels **130L** defined by the transfer decks **130B** (and the rails **1200S** which form an aisle deck). Accordingly, there are multiple rack shelf levels **130LS1-130LS4**, corresponding to each storage level **130L**, extending along one or more picking aisles **130A** communicating with the transfer deck **130B** of the respective storage level **130L**. As may be realized, the multiple rack shelf levels **130LS1-130LS4** effect each storage level **130L** having stacks of stored case units (or case layers) that are accessible from a common deck **1200S** of a respective storage level **130L** (e.g. the stacks of stored cases are located between storage levels).

As may be realized, bots **110** traversing a picking aisle **130A**, at a corresponding storage level **130L**, have access (e.g. for picking and placing case units) to each storage space **130S** that is available on each shelf level **130LS1-130LS4**, where each shelf level **130LS1-130LS4** is located between adjacent vertically stacked storage levels **130L** on one or more side(s) PAS1, PAS2 (see e.g. FIG. **2A**) of the picking aisle **130A**. As noted above, each of the storage shelf levels **130LS1-130LS4** is accessible by the bot **110** from the rails **1200** (e.g. from a common picking aisle deck **1200S** that corresponds with a transfer deck **130B** on a respective storage level **130L**). As can be seen in FIGS. **1B** and **1D** there are one or more intermediate shelf rails **1210** vertically spaced (e.g. in the Z direction) from one another (and from rails **1200**) to form multiple stacked storage spaces **130S** each being accessible by the bot **110** from the common rails **1200S**. As may be realized, the horizontal support members **1200** also form shelf rails (in addition to shelf rails **1210**) on which case units are placed.

Each stacked shelf level **130LS1-130LS4** (and/or each single shelf level as described below) of a corresponding storage level **130L** defines an open and undeterministic two dimensional storage surface (e.g. having a case unit support plane CUSP as shown in FIG. **1D**) that facilitates a dynamic allocation of pickfaces both longitudinally (e.g. along a length of the aisle or coincident with a path of bot travel defined by the picking aisle) and laterally (e.g. with respect to rack depth, transverse to the aisle or the path of bot travel). Dynamic allocation of the pickfaces and case units that make up the pickfaces is provided, for example, in the manner described in U.S. Pat. No. 8,594,835 issued on Nov. 26, 2013, the disclosure of which is incorporated by reference herein in its entirety. For example, the controller, such as controller **120** monitors the case units stored on the shelves and the empty spaces or storage locations between the case units. The empty storage locations are dynamically allocated such that, for exemplary purposes only, one case having a first size is replaced by three cases each having a second size which when combined fits into the space previously reserved for the first size case, or vice versa. Dynamic allocation substantially continuously resizes the empty storage locations as case units are placed on and

removed from the storage shelves (e.g. the storage locations do not have a predetermined size and/or location on the storage shelves). As such, case unit (or tote) pickfaces of variable lengths and widths are positioned at each two dimensional storage location on the storage shelves (e.g. on each storage shelf level **130LS1-130LS4**) with minimum gaps **G** (e.g. that effect picking/placing of case units free from contact with other case units stored on the shelves, see FIG. 1B) between adjacent stored case units/storage spaces.

As described above, the spacing between the rails **1200**, **1210** (e.g. storage shelves) is a variable spacing so as to minimize (e.g. provide only sufficient clearance for insertion and removal of case units from a respective storage location) the vertical gap **VG** between vertically stacked case units. As will be described below (e.g. with respect to sections SECA, SECB in FIGS. 1B and 2A), in one aspect the vertical spacing between rails **1200**, **1210** varies along a length of a respective picking aisle **130A** while in other aspects the spacing between rails **1200**, **1210** may be substantially continuous along a picking aisle **130A**. As may be realized and as described in greater detail below, the spacing between the rails **1200**, **1210** on one side **PAS1** (FIG. 2A) of a picking aisle **130A** may be different than the spacing between rails **1200**, **1210** on an opposite side **PAS2** (FIG. 2A) of the same picking aisle **130A**. As may be realized, any suitable number of shelves **1210** may be provided between the decks **1200S** of adjacent vertically stacked storage levels **130L** where the shelves have the same or differing pitches between the shelves (see e.g. FIG. 1C where case units **CUD1**, **CUD2**, **CUE1-CUE3**, **CUF1**, **CUF2** are located in a vertical stack on one side of the picking aisle and case units **CUA**, **CUB**, **CUC** are located in a vertical stack on an opposite side of the picking aisle on storage shelves having a substantially similar pitch). In one aspect of the disclosed embodiment, referring to FIG. 1B, a vertical pitch between rack shelf levels **130LS1-130LS4** (that corresponds to each storage level **130L**) is varied so that a height **Z1A-Z1E** between the shelves is different, rather than equal to, for example, minimize a vertical gap **VG** between an upper or top surface **CUTS** of a case unit **CU** and a bottom of the storage shelf **1200**, **1210** located directly above the case unit. As can be seen in FIG. 1B, minimizing the gaps **G**, **VG** in both the horizontal and vertical directions results in a densely packed case unit arrangement within the storage shelves so as to form the high density three dimensional rack array **RMA** where, for example, the high density multi-level shelving aisles increases throughput along the **X** throughput axis and enables an ordered/sorted (e.g. according to the predetermined load out sequence) multi-pick of two or more case units from a common picking aisle in one common pass of the picking aisle as will be described below. For example, still referring to FIG. 1B, one section **SECB** of the storage level **130L** includes two storage shelves **1200**, **1210** where one shelf has a pitch of **Z1A** and the other shelf has a pitch of **Z1B** where **Z1A** and **Z1B** are different from each other. This differing pitch allows for the placement of case units **CUD**, **CUE** having differing heights in a stack one above the other on a common storage level **130L**. In other aspects pitches **Z1A**, **Z1B** may be substantially the same. In this aspect the storage level **130L** includes another storage section **SECA** that has three storage shelves where one shelf has a pitch of **Z1E**, one storage shelf has a pitch of **Z1D** and the other storage shelf has a pitch of **Z1C** where **Z1E**, **Z1D** and **Z1C** are different from each other. In other aspects at least two of the pitches **Z1E**, **Z1D** and **Z1C** are substantially the same. In one aspect the pitch between the shelves is arranged so that larger and/or heavier case units **CUC**, **CUE**

are arranged closer to the deck **1200S** than smaller and/or lighter case units **CUD**, **CUA**, **CUB**. In other aspects the pitch between the shelves is arranged so that the case units are arranged in any suitable positions that may or may not be related to case unit size and weight.

In other aspects, the vertical pitch between at least some of the rack shelves is the same so that the height **Z1A-Z1E** between at least some shelves is equal while the vertical pitch between other shelves is different. In still other aspects, the pitch of rack shelf levels **130LS1-130LS4** on one storage level is a constant pitch (e.g. the rack shelf levels are substantially equally spaced in the **Z** direction) while the pitch of rack shelf levels **130LS1-130LS4** on a different storage level is a different constant pitch.

In one aspect, the storage space(s) **130S** defined by the storage shelf levels **130LS1-130LS4** between the storage or deck levels **130L** accommodates case units of different heights, lengths, widths and/or weights at the different shelf levels **130LS1-130LS4** as described in, for example, U.S. Non-provisional application Ser. No. 14/966,978, filed on Dec. 11, 2015 and U.S. Provisional Patent Application 62/091,162 filed on Dec. 12, 2014, the disclosures of which are incorporated by reference herein in their entireties. For example, still referring to FIG. 1B the storage level **130L** includes storage sections having at least one intermediate shelf **1210**. In the example shown, one storage section includes one intermediate shelf **1210** while another storage section includes two intermediate shelves **1210** for forming shelf levels **130LS1-130LS4**. In one aspect the pitch **Z1** between storage levels **130L** may be any suitable pitch such as, for example, about 32 inches to about 34 inches while in other aspects the pitch may be more than about 34 inches and/or less than about 32 inches. Any suitable number of shelves may be provided between the decks **1200S** of adjacent vertically stacked storage levels **130L** where the shelves have the same or differing pitches between the shelves (see e.g. FIG. 1C where case units **CUD1**, **CUD2**, **CUE1-CUE3**, **CUF1**, **CUF2** are located in a vertical stack on one side of the picking aisle and case units **CUA**, **CUB**, **CUC** are located in a vertical stack on an opposite side of the picking aisle on storage shelves having a substantially similar pitch).

In one aspect of the disclosed embodiment the storage or deck levels **130L** (e.g. the surface on which the bots **110** travel) are arranged at any suitable predetermined pitch **Z1** that is not, for example, an integer multiple of the intermediate shelf pitch(es) **Z1A-Z1E**. In other aspects the pitch **Z1** may be an integer multiple of the intermediate shelf pitch, such as for example, the shelf pitch may be substantially equal to the pitch **Z1** so that the corresponding storage space has a height substantially equal to the pitch **Z1**. As may be realized, the shelf pitch **Z1A-Z1E** is substantially decoupled from the storage level **130L** pitch **Z1** and corresponds to general case unit heights as illustrated in FIG. 1B. In one aspect of the disclosed embodiment case units of different heights are dynamically allocated or otherwise distributed along each aisle within a storage space **130S** having a shelf height commensurate with the case unit height. The remaining space between the storage levels **130L**, both along the length of the aisle coincident with the stored case unit (e.g. in the **X** direction with respect to the rack frame of reference **REF2** where the **X** direction is the same in the bot frame of reference **REF** as the bot travel through a picking aisle **130A**) and alongside the stored case unit, being freely usable for dynamic allocation for cases of a corresponding height. As may be realized, the dynamic allocation of case units having different heights onto shelves having different

13

pitches provides for stored case layers of different heights, between storage levels **130L** on both sides of each picking aisle **130A**, with each case unit being dynamically distributed along a common picking aisle **130A** so that each case unit within each stored case layer being independently accessible (e.g. for picking/placing) by the bot in the common aisle. This high density placement/allocation of case units and the arrangement of the storage shelves provides maximum efficiency of storage space/volume use between the storage levels **130L**, and hence of maximum efficiency of the rack module array RMA, with optimized distribution of case unit SKU's, as each aisle length may include multiple case units of different heights, yet each rack shelf at each shelf level may be filled by dynamic allocation/distribution (e.g. to fill the three dimensional rack module array RMA space in length, width and height, to provide a high density storage array).

In one aspect, referring to FIGS. **1E** and **6A** each of the storage levels **130L** includes a single level of storage shelves to store a single level of case units (e.g. each storage level includes a single case unit support plane CUSP) and the bots **110** are configured to transfer case units to and from the storage shelves of the respective storage level **130L**. For example, the bot **110'** illustrated in FIG. **6A** is substantially similar to bot **110** described herein however, the bot **110'** is not provided with sufficient Z-travel of the transfer arm **110PA** for placing case units on the multiple storage shelf levels **130LS1-130LS4** (e.g. accessible from a common rail **1200S**) as described above. Here the transfer arm drive **250** (which may be substantially similar to one or more of drive **250A**, **250B**) includes only sufficient Z-travel for lifting the case units from the case unit support plane CUSP of the single level of storage shelves, for transferring the case units to and from the payload area **110PL** and for transferring the case units between the fingers **273** of the transfer arm **110PA** and the payload bed **110PB**. Suitable examples of bots **110'** can be found in, for example, U.S. patent application Ser. No. 13/326,993 filed on Dec. 15, 2011, the disclosure of which is incorporated herein by reference in its entirety.

In one aspect of the disclosed embodiment, referring also to FIG. **2A**, the rack shelves **1210** (inclusive of the rack shelf formed by rail **1200**) are sectioned SECA, SECB longitudinally (e.g. along the length of the picking aisle **130A** in the X direction, with respect to a storage structure frame of reference REF2) to form ordered or otherwise matched rack shelf sections along each picking aisle **130A**. The aisle shelf sections SECA, SECB are ordered/matched to each other based on, for example, a pick sequence of a bot **110** traversing the aisle in a common pass picking case units destined for a common order fill (e.g. based on the order out sequence). In other words, a bot **110** makes a single pass (e.g. traversal in a single direction) down a single or common picking aisle while picking one or more case units from aisle shelf sections SECA, SECB on a common side of the picking aisle **130A** to build a pickface on the bot **110** where the pickface includes case units that are arranged on the bot according to the order fill/order out sequence as will be described in greater detail below. Each of the aisle rack sections SECA, SECB includes intermediate shelves in the manner described above. In other aspects some of the aisle shelves do not include intermediate shelves while others do include intermediate shelves.

In one aspect, the ordered aisle rack sections SECA, SECB include shelf pitches that are different between sections SECA, SECB. For example, aisle rack section SECA has shelves with one or more pitches while aisle rack section SECB has shelves with one or more different pitches (e.g.

14

different than the pitches of the shelves in section SECA). In accordance with the aspects of the disclosed embodiment, the pitch of at least one intermediate shelf of one aisle rack section SECA, SECB is related to the pitch of at least one intermediate shelf of another of the ordered aisle rack sections SECA, SECB of the common picking aisle **130A**. The different pitches of the intermediate shelves **1210** in the ordered aisle rack section SECA, SECB are selected so as to be related and to effect multiple (at least two) ordered picks (i.e. picks in an ordered sequence) with a bot **110**, in accordance with a mixed SKU load out sequence (e.g. palletizing to a common pallet load), from shelves of different pitches, from a common pass of a common picking aisle **130A**. As may be realized, the mixed load output from the storage and retrieval system **100** (e.g. to fill a truck loadport/pallet load) is sequenced in a predetermined order according to various load out picking aisles (e.g. aisles from which case units are picked for transfer to an outgoing pallet) and the shelf pitch in the ordered sections SECA, SECB facilitates a bot **110** pick of more than one case unit in ordered sequence according to an order of the load out sequence in a common picking aisle pass (e.g. more than one case unit is picked in a predetermined order from a common picking aisle in one pass of the common picking aisle). The different aisle shelf pitches of the ordered rack sections SECA, SECB are so related to increase the probability of such an ordered multi-pick (the picking of two or more case units from a single aisle with a single pass of the aisle as described above) so that the multi-pick is performed by each bot order fulfillment pass along each aisle, and so related such that more than a majority of cases picked in the storage and retrieval system **100** by the bots **110** and destined for a common load out (e.g. a common pallet load) are picked by a common bot **110** in an ordered sequence corresponding to the load out sequence during a single pass of a common picking aisle (e.g. the two or more cases picked by the bot **110** are picked from the same picking aisle in a single pass, e.g. the bot travels in a single direction once through the picking aisle). As may be realized, in one aspect of the disclosed embodiment both sides PAS1, PAS2 of the picking aisle **130A** have ordered aisle rack sections SECA, SECB where one ordered section may be matched with one or more sections on the same side PAS1, PAS2 of the common picking aisle **130A**. As may be realized, the matched aisle rack sections may be located adjacent one another or spaced apart from one another along the picking aisle **130A**.

Referring again to FIG. **2A** each transfer deck or storage level **130L** includes one or more lift pickface interface/handoff stations TS (referred to herein as interface stations TS) where case unit(s) (of single or combined case pick-faces) or totes are transferred between the lift load handling devices LHD and bots **110** on the transfer deck **130B**. The interface stations TS are located at a side of the transfer deck **130B** opposite the picking aisles **130A** and rack modules RM, so that the transfer deck **130B** is interposed between the picking aisles and each interface station TS. As noted above, each bot **110** on each picking level **130L** has access to each storage location **130S**, each picking aisle **130A** and each lift **150** on the respective storage level **130L**, as such each bot **110** also has access to each interface station TS on the respective level **130L**. In one aspect the interface stations are offset from high speed bot travel paths HSTP along the transfer deck **130B** so that bot **110** access to the interface stations TS is undeterministic to bot speed on the high speed travel paths HSTP. As such, each bot **110** can move a case unit(s) (or pickface, e.g. one or more cases, built by the bot)

15

from every interface station TS to every storage space 130S corresponding to the deck level and vice versa.

In one aspect the interface stations TS are configured for a passive transfer (e.g. handoff) of case units (and/or pick-faces) between the bot 110 and the load handling devices LHD of the lifts 150 (e.g. the interface stations TS have no moving parts for transporting the case units) which will be described in greater detail below. For example, also referring to FIG. 2B the interface stations TS and/or buffer stations BS include one or more stacked levels TL1, TL2 of transfer rack shelves RTS (e.g. so as to take advantage of the lifting ability of the bot 110 with respect to the stacked rack shelves RTS) which in one aspect are substantially similar to the storage shelves described above (e.g. each being formed by rails 1210, 1200 and slats 1210S) such that bot 110 handoff (e.g. pick and place) occurs in a passive manner substantially similar to that between the bot 110 and the storage spaces 130S (as described herein) where the case units or totes are transferred to and from the shelves. In one aspect the buffer stations BS on one or more of the stacked levels TL1, TL2 also serve as a handoff/interface station with respect to the load handling device LHD of the lift 150. In one aspect, where the bots, such as bots 110', are configured for the transfer of case units to a single level 130L of storage shelves, the interface stations TS and/or buffer stations BS also include a single level of transfer rack shelves (which are substantially similar to the storage rack shelves of the storage levels 130L described above with respect to, for example, FIG. 1D). As may be realized, operation of the storage and retrieval system with bots 110' serving the single level storage and transfer shelves is substantially similar to that described herein. As may also be realized, load handling device LHD handoff (e.g. pick and place) of case units (e.g. individual case units or pickfaces) and totes to the stacked rack shelves RTS (and/or the single level rack shelves) occurs in a passive manner substantially similar to that between the bot 110 and the storage spaces 130S (as described herein) where the case units or totes are transferred to and from the shelves. In other aspects the shelves may include transfer arms (substantially similar to the bot 110 transfer arm 110PA shown in FIG. 6, although Z direction movement may be omitted when the transfer arm is incorporated into the interface station TS shelves) for picking and placing case units or totes from one or more of the bot 110 and load handling device LHD of the lift 150. Suitable examples of an interface station with an active transfer arm are described in, for example, U.S. patent application Ser. No. 12/757,354 filed on Apr. 9, 2010, the disclosure of which is incorporated by reference herein in its entirety.

In one aspect, the location of the bot 110 relative to the interface stations TS occurs in a manner substantially similar to bot location relative to the storage spaces 130S. For example, in one aspect, location of the bot 110 relative to the storage spaces 130S and the interface stations TS occurs in a manner substantially similar to that described in U.S. patent application Ser. No. 13/327,035 filed on Dec. 15, 2011 (now U.S. Pat. No. 9,008,884) and Ser. No. 13/608,877 filed on Sep. 10, 2012 (now U.S. Pat. No. 8,954,188), the disclosures of which are incorporated herein by reference in their entireties. For example, referring to FIGS. 1 and 1D, the bot 110 includes one or more sensors 110S that detect the slats 1210S or a locating feature 130F (such as an aperture, reflective surface, RFID tag, etc.) disposed on/in the rail 1200. The Slats and/or locating features 130F are arranged so as to identify a location of the bot 110 within the storage and retrieval system, relative to e.g. the storage spaces

16

and/or interface stations TS. In one aspect the bot 110 includes a controller 110C that, for example, counts the slats 1210S to at least in part determine a location of the bot 110 within the storage and retrieval system 100. In other aspects the location features 130F may be arranged so as to form an absolute or incremental encoder which when detected by the bot 110 provides for a bot 110 location determination within the storage and retrieval system 100.

As may be realized, referring to FIG. 2B, the transfer rack shelves RTS at each interface/handoff station TS define multi-load stations (e.g. having one or more storage case unit holding locations for holding a corresponding number of case units or totes) on a common transfer rack shelf RS. As noted above, each load of the multi-load station is a single case unit/tote or a multi-case pickface (e.g. having multiple case units/totes that are moved as a single unit) that is picked and paced by either the bot or load handling device LHD. As may also be realized, the bot location described above allows for the bot 110 to position itself relative to the multi-load stations for picking and placing the case units/totes and pickfaces from a predetermined one of the holding locations of the multi-load station. The interface/handoff stations TS define multi-place buffers (e.g. buffers having one or more case holding location—see FIG. 4B—arranged along, for example, the X axis of the bot 110 as the bot 110 interfaces with the interface station TS) where inbound and/or outbound case units/totes and pickfaces are temporarily stored when being transferred between the bots 110 and the load handling devices LHD of the lifts 150.

In one aspect one or more peripheral buffer/handoff stations BS (substantially similar to the interface stations TS and referred to herein as buffer stations BS) are also located at the side of the transfer deck 130B opposite the picking aisles 130A and rack modules RM, so that the transfer deck 130B is interposed between the picking aisles and each buffer station BS. The peripheral buffer stations BS are interspersed between or, in one aspect as shown in FIGS. 2A and 2B, otherwise in line with the interface stations TS. In one aspect the peripheral buffer stations BS are formed by rails 1210, 1200 and slats 1210S and are a continuation of (but a separate section of) the interface stations TS (e.g. the interface stations and the peripheral buffer stations are formed by common rails 1210, 1200). As such, the peripheral buffer stations BS, in one aspect, also include one or more stacked levels TL1, TL2 of transfer rack shelves RTS as described above with respect to the interface stations TS while in other aspects the buffer stations include a single level of transfer rack shelves. The peripheral buffer stations BS define buffers where case units/totes and/or pickfaces are temporarily stored when being transferred from one bot 110 to another different bot 110 on the same storage level 130L as will be described in greater detail below. As may be realized, in one aspect the peripheral buffer stations are located at any suitable location of the storage and retrieval system including within the picking aisles 130A and anywhere along the transfer deck 130B.

Still referring to FIGS. 2A and 2B in one aspect the interface stations TS are arranged along the transfer deck 130B in a manner akin to parking spaces on the side of a road such that the bots 110 “parallel park” at a predetermined interface station TS for transferring case units to and from one or more shelves RTS at one or more levels TL1, TL2 of the interface station TS. In one aspect, a transfer orientation of the bots 110 (e.g. when parallel parked) at an interface station TS is the same orientation as when the bot 110 is travelling along the high speed bot transport path HSTP (e.g. the interface station is substantially parallel with

17

a bot travel direction of the transfer deck and/or a side of the transfer deck on which the lifts **150** are located). Bot **110** interface with the peripheral buffer stations BS also occurs by parallel parking so that a transfer orientation of the bots **110** (e.g. when parallel parked) at a peripheral buffer station BS is the same orientation as when the bot **110** is travelling along the high speed bot transport path HSTP.

In another aspect, referring to FIGS. **3A** and **3B**, at least the interface stations TS are located on an extension portion or pier **130BD** that extends from the transfer deck **130B**. In one aspect, the pier **130BD** is similar to the picking aisles where the bot **110** travels along rails **1200S** affixed to horizontal support members **1200** (in a manner substantially similar to that described above). In other aspects, the travel surface of the pier **130BD** may be substantially similar to that of the transfer deck **130B**. Each pier **130BD** is located at the side of the transfer deck **130B**, such as a side that is opposite the picking aisles **130A** and rack modules RM, so that the transfer deck **130B** is interposed between the picking aisles and each pier **130BD**. The pier(s) **130BD** extends from the transfer deck at a non-zero angle relative to at least a portion of the high speed bot transport path HSTP. In other aspects the pier(s) **130BD** extend from any suitable portion of the transfer deck **130B** including the ends **130BE1**, **130BE2** of the transfer deck **130BD**. As may be realized, peripheral buffer stations BSD (substantially similar to peripheral buffers stations BS described above) may also be located at least along a portion of the pier **130BD**.

Referring now to FIGS. **4A**, **4B** and **5**, as described above, in one aspect the interface stations TS are passive stations and as such the load transfer device LHD of the lifts **150A**, **150B** have active transfer arms or pick heads **4000A**, **4000B**. In one aspect the inbound lift modules **150A** and the outbound lift modules **150B** have different types of pick heads (as will be described below) while in other aspects the inbound lift modules **150A** and the outbound lift modules **150B** have the same type of pick head similar to one of the pick heads described below (e.g. both the lifts **150A**, **150B** have pick head **4000A** or both lifts **150A**, **150B** have pick head **4000B**). The pick heads of the lifts **150A**, **150B** may, at least in part, define the Y throughput axis as described herein. In one aspect, both the inbound and outbound lift modules **150A**, **150B** have a vertical mast **4002** along which a slide **4001** travels under the motive force of any suitable drive unit **4002D** (e.g. connected to, for example, control server **120**) configured to lift and lower the slide (and the pick head **4000A**, **4000B** mounted thereto). The inbound lift module(s) **150A** include a pick head **4000A** mounted to the slide **4001** so that as the slide moves vertically the pick head **4000A** moves vertically with the slide **4001**. In this aspect the pick head **4000A** includes one or more tines or fingers **4273** mounted to a base member **4272**. The base member **4272** is movably mounted to one or more rail **4360S** of frame **4200** which in turn is mounted to the slide **4001**. Any suitable drive unit **4005**, such as a belt drive, chain drive, screw drive, gear drive, etc. (which is substantially similar in form but may not be similar in capacity to drive **4002D** as the drive **4005** may be smaller than drive **4002D**) is mounted to the frame **4200** and coupled to the base member **4272** for driving the base member **4272** (with the finger(s)) in the direction of arrow **4050**.

The outbound lift module(s) **150B** also include a pick head **4000B** mounted to the slide **4001** so that as the slide moves vertically the pick head **4000B** moves vertically with the slide **4001**. In this aspect the pick head **4000B** includes one or more pick head portions or effectors (e.g. transfer arms) LHDA, LHDB each having one or more tines or

18

fingers **4273** mounted to a respective base member **4272A**. Each base member **4272A** is movably mounted to one or more rail **4360SA** of frame **4200A** which in turn is mounted to the slide **4001**. Any suitable drive unit(s) **4005A**, such as a belt drive, chain drive, screw drive, gear drive, etc. is mounted to the frame **4200A** and coupled to a respective base member **4272A** for driving the respective base member **4272A** (with the finger(s)) in the direction of arrow **4050** (each effector has a respective drive unit so that each effector is independently movable in the direction of arrow **4050**). While two effectors LHDA, LHDB are illustrated on pick head **4000B** the pick head **4000B** includes any suitable number of effectors that correspond to a number of case unit/pickface holding locations of, for example, the interface stations TS so that case units/pickfaces are individually picked from the interface stations TS as described in greater detail below.

As may be realized, the lift modules **150A**, **150B** are under the control of any suitable controller, such as control server **120**, such that when picking and placing case unit(s) the pick head is raised and/or lowered to a predetermined height corresponding to an interface station TS at a predetermined storage level **130L**. As may be realized, the lift modules **150A**, **150B** provide the Z throughput axis (relative to both the bot frame of reference REF and the rack frame of reference REF2) of the storage and retrieval system where the output lift modules **150B** sort case units on the fly for delivery to the output stations **160US** as will be described below. At the interface stations TS the pick head **4000A**, **4000B** or individual portion thereof (e.g. effector LHDA, LHDB), corresponding to one or more case unit holding location(s) of the interface station TS from which one or more case unit(s) are being picked, is extended so that the fingers **4273** are interdigitated between the slats **1210S** (as illustrated in FIG. **4B**) underneath the case unit(s) being picked. The lift **150A**, **150B** raises the pick head **4000A**, **4000B** to lift the case unit(s) from the slats **1210S** and retracts the pick head **4000A**, **4000B** for transport of the case unit(s) to another level of the storage and retrieval system, such as for transporting the case unit(s) to output station **160UT**. Similarly, to place one or more case unit(s) the pick head **4000A**, **4000B** or individual portion thereof (e.g. effector LHDA, LHDB), corresponding to one or more case unit holding location(s) of the interface station TS from which one or more case unit(s) are being placed, is extended so that the fingers **4273** are above the slats. The lift **150A**, **150B** lowers the pick head **4000A**, **4000B** to place the case unit(s) on the slats **1210S** and so that the fingers **4273** are interdigitated between the slats **1210S** underneath the case unit(s) being picked.

Referring now to FIG. **6**, as noted above, the bot **110** includes a transfer arm **110PA** that effects the picking and placement of case units from the stacked storage spaces **130S**, interface stations TS and peripheral buffer stations BS, BSD defined at least in part, in the Z direction) by one or more of the rails **1210A-1210C**, **1200** (e.g. where the storage spaces, interface stations and/or peripheral buffer stations may be further defined in the X and Y directions, relative to either of the rack frame of reference REF2 or the bot frame of reference REF, through the dynamic allocation of the case units as described above). As may be realized, the bot defines the X throughput axis and, at least in part, the Y throughput axis (e.g. relative to the bot frame of reference REF) as will be described further below. The bots **110**, as noted above, transport case units between each lift module **150** and each storage space **130S** on a respective storage level **130L**. The bots **110** include a frame **110F** having a

19

drive section **110DR** and a payload section **110PL**. The drive section **110DR** includes one or more drive wheel motors each connected to a respective drive wheel(s) **202** for propelling the bot **110** along the X direction (relative to the bot frame of reference REF so as to define the X throughput axis). As may be realized, the X axis of bot travel is coincident with the storage locations when the bot **110** travels through the picking aisles **130A**. In this aspect the bot **110** includes two drive wheels **202** located on opposite sides of the bot **110** at end **110E1** (e.g. first longitudinal end) of the bot **110** for supporting the bot **110** on a suitable drive surface however, in other aspects any suitable number of drive wheels are provided on the bot **110**. In one aspect each drive wheel **202** is independently controlled so that the bot **110** may be steered through a differential rotation of the drive wheels **202** while in other aspects the rotation of the drive wheels **202** may be coupled so as to rotate at substantially the same speed. Any suitable wheels **201** are mounted to the frame on opposite sides of the bot **110** at end **110E2** (e.g. second longitudinal end) of the bot **110** for supporting the bot **110** on the drive surface. In one aspect the wheels **201** are caster wheels that freely rotate allowing the bot **110** to pivot through differential rotation of the drive wheels **202** for changing a travel direction of the bot **110**. In other aspects the wheels **201** are steerable wheels that turn under control of, for example, a bot controller **110C** (which is configured to effect control of the bot **110** as described herein) for changing a travel direction of the bot **110**. In one aspect the bot **110** includes one or more guide wheels **110GW** located at, for example, one or more corners of the frame **110F**. The guide wheels **110GW** may interface with the storage structure **130**, such as guide rails (not shown) within the picking aisles **130A**, on the transfer deck **130B** and/or at interface or transfer stations for interfacing with the lift modules **150** for guiding the bot **110** and/or positioning the bot **110** a predetermined distance from a location to/from which one or more case units are placed and/or picked up as described in, for example, U.S. patent application Ser. No. 13/326,423 filed on Dec. 15, 2011 the disclosure of which is incorporated herein by reference in its entirety. As noted above, the bots **110** may enter the picking aisles **130A** having different facing directions for accessing storage spaces **130S** located on both sides of the picking aisles **130A**. For example, the bot **110** may enter a picking aisle **130A** with end **110E2** leading the direction of travel or the bot may enter the picking aisle **130A** with end **110E1** leading the direction of travel.

The payload section **110PL** of the bot **110** includes a payload bed **110PB**, a fence or datum member **110PF**, a transfer arm **110PA** and a pusher bar or member **110PR**. In one aspect the payload bed **110PB** includes one or more rollers **110RL** that are transversely mounted (e.g. relative to a longitudinal axis LX of the bot **110**) to the frame **110F** so that one or more case units carried within the payload section **110PL** can be longitudinally moved (e.g. justified with respect to a predetermined location of the frame/payload section and/or a datum reference of one or more case units) along the longitudinal axis of the bot, e.g., to position the case unit at a predetermined position within the payload section **110PL** and/or relative to other case units within the payload section **110PL** (e.g. longitudinal forward/aft justification of case units). In one aspect the rollers **110RL** may be driven (e.g. rotated about their respective axes) by any suitable motor for moving the case units within the payload section **110PL**. In other aspects the bot **110** includes one or more longitudinally movable pusher bar (not shown) for pushing the case units over the rollers **110RL** for

20

moving the case unit(s) to the predetermined position within the payload section **110PL**. The longitudinally movable pusher bar may be substantially similar to that described in, for example, U.S. patent application Ser. No. 13/326,952 filed on Dec. 15, 2011, the disclosure of which was previously incorporated by reference herein in its entirety. The pusher bar **110PR** is movable in the Y direction, relative to the bot **110** reference frame REF to effect, along with the fence **110PF** and or pick head **270** of the transfer arm **110PA**, a lateral justification of case unit(s) within the payload area **110PL** in the manner described in U.S. Provisional Patent Application No. 62/107,135 filed on Jan. 23, 2015, previously incorporated herein by reference in their entireties.

Still referring to FIG. 6, the case units are placed on the payload bed **110PB** and removed from the payload bed **110PB** with the transfer arm **110PA** along the Y throughput axis. The transfer arm **110PA** includes a lift mechanism or unit **200** located substantially within the payload section **110PL** as described in, for example, U.S. Provisional Patent Application No. 62/107,135 filed on Jan. 23, 2015, previously incorporated herein by reference in their entireties. The lift mechanism **200** provides both gross and fine positioning of pickfaces carried by the bot **110** which are to be lifted vertically into position in the storage structure **130** for picking and/or placing the pickfaces and/or individual case units to the storage spaces **130S** (e.g. on a respective storage level **130L** on which the bot **110** is located). For example, the lift mechanism **200** provides for picking and placing case units at the multiple elevated storage shelf levels **130LS1-130LS4**, **TL1**, **TL2** accessible from the common picking aisle or interface station deck **1200S** (see e.g. FIGS. 1B, 2B and 3B).

The lift mechanism **200** is configured so that combined robot axis moves are performed (e.g. combined substantially simultaneous movement of the pusher bar **110PR**, lift mechanism **200**, pick head extension and fore/aft justification mechanism(s) such as, e.g., the longitudinally movable pusher bar described above), so that different/multi-sku or multi-pick payloads are handled by the bot. In one aspect, the actuation of the lifting mechanism **200** is independent of actuation of the pusher bar **110PR** as will be described below. The decoupling of the lift mechanism **200** and pusher bar **110PR** axes provides for combined pick/place sequences effecting a decreased pick/place cycle time, increased storage and retrieval system throughput and/or increased storage density of the storage and retrieval system as described above. For example, the lift mechanism **200** provides for picking and placing case units at multiple elevated storage shelf levels accessible from a common picking aisle and/or interface station deck **1200S** as described above.

The lifting mechanism may be configured in any suitable manner so that a pick head **270** of the bot **110** bi-directionally moves along the Z axis (e.g. reciprocates in the Z direction—see FIG. 6). In one aspect, the lifting mechanism includes a mast **200M** and the pick head **270** is movably mounted to the mast **200M** in any suitable manner. The mast is movably mounted to the frame in any suitable manner so as to be movable along the lateral axis LT of the bot **110** (e.g. in the Y direction so as to define the Y throughput axis). In one aspect the frame includes guide rails **210A**, **210B** to which the mast **200** is slidably mounted. A transfer arm drive **250A**, **250B** may be mounted to the frame for effecting at least movement of the transfer arm **110PA** along the lateral axis LT (e.g. Y axis) and the Z axis. In one aspect the transfer arm drive **250A**, **250B** includes an extension motor **301** and a lift motor **302**. The extension motor **301** may be mounted to the frame **110F** and coupled to the mast **200M** in any

21

suitable manner such as by a belt and pulley transmission 260A, a screw drive transmission (not shown) and/or a gear drive transmission (not shown). The lift motor 302 may be mounted to the mast 200M and coupled to pick head 270 by any suitable transmission, such as by a belt and pulley transmission 271, a screw drive transmission (not shown) and/or a gear drive transmission (not shown). As an example, the mast 200M includes guides, such as guide rails 280A, 280B, along which the pick head 270 is mounted for guided movement in the Z direction along the guide rails 280A, 280B. In other aspects the pick head is mounted to the mast in any suitable manner for guided movement in the Z direction. With respect to the transmissions 271, a belt 271B of the belt and pulley transmission 271 is fixedly coupled to the pick head 270 so that as the belt 271 moves (e.g. is driven by the motor 302) the pick head 270 moves with the belt 271 and is bi-directionally driven along the guide rails 280A, 280B in the Z direction. As may be realized, where a screw drive is employed to drive the pick head 270 in the Z direction, a nut may be mounted to the pick head 270 so that as a screw is turned by the motor 302 engagement between the nut and screw causes movement of the pick head 270. Similarly, where a gear drive transmission is employed a rack and pinion or any other suitable gear drive may drive the pick head 270 in the Z direction. In other aspects any suitable linear actuators are used to move the pick head in the Z direction. The transmission 260A for the extension motor 301 is substantially similar to that described herein with respect to transmission 271.

Still referring to FIG. 6 the pick head 270 of the bot 110 transfers case units between the bot 110 and a case unit pick/place location such as, for example, the storage spaces 130S, peripheral buffer stations BS, BSD and/or interface stations TS (see FIGS. 2A-3B) and in other aspects substantially directly between the bot 110 and a lift module(s) 150. In one aspect, the pick head 270 includes a base member 272, one or more tines or fingers 273A-273E and one or more actuators 274A, 274B. The base member 272 is mounted to the mast 200M, as described above, so as to ride along the guide rails 280A, 280B. The one or more tines 273A-273E are mounted to the base member 272 at a proximate end of the tines 273A-273E so that a distal end of the tines 273A-273E (e.g. a free end) is cantilevered from the base member 272. Referring again to FIG. 1D, the tines 273A-273E are configured for insertion between slats 1210S that form the case unit support plane CUSP of the storage shelves.

One or more of the tines 273A-273E is movably mounted to the base member 272 (such as on a slide/guide rail similar to that described above) so as to be movable in the Z direction. In one aspect any number of tines are mounted to the base member 272 while in the aspect illustrated in the figures there are, for example, five tines 273A-273E mounted to the base member 272. Any number of the tines 273A-273E are movably mounted to the base member 272 while in the aspect illustrated in the figures, for example, the outermost (with respect to a centerline CL of the pick head 270) tines 273A, 273E are movably mounted to the base member 272 while the remaining tines 273B-273D are immovable relative to the base member 272.

In this aspect the pick head 270 employs as few as three tines 273B-273D to transfer smaller sized case units (and/or groups of case units) to and from the bot 110 and as many as five tines 273A-273E to transfer larger sized case units (and/or groups of case units) to and from the bot 110. In other aspects, less than three tines are employed (e.g. such as where more than two tines are movably mounted to the

22

base member 272) to transfer smaller sized case units. For example, in one aspect all but one tine 273A-273E is movably mounted to the base member so that the smallest case unit being transferred to and from the bot 110 without disturbing other case units on, for example, the storage shelves has a width of about the distance X1 between slats 1210S (see FIG. 1D).

The immovable tines 273B-273D define a picking plane SP of the pick head 270 and are used when transferring all sizes of case units (and/or pickfaces) while the movable tines 273A, 273E are selectively raised and lowered (e.g. in the Z direction with the actuators 274A, 274B) relative to the immovable tines 273B-273D to transfer larger case units (and/or pickfaces). Still referring to FIG. 6 an example is shown where all of the tines 273A-273E are positioned so that a case unit support surface SF of each tine 273A-273E is coincident with the picking plane SP of the pick head 270 however, as may be realized, the two end tines 273A, 273E are movable so as to be positioned lower (e.g. in the Z direction) relative to the other tines 273B-273D so that the case unit support surface SF of tines 273A, 273E is offset from (e.g. below) the picking plane SP so that the tines 273A, 273E do not contact the one or more case units carried by the pick head 270 and do not interfere with any unpicked case units positioned in storage spaces 130S on the storage shelves or any other suitable case unit holding location.

The movement of the tines 273A-273E in the Z direction is effected by the one or more actuators 274A, 274B mounted at any suitable location of the transfer arm 110PA. In one aspect, the one or more actuators 274A, 274B are mounted to the base member 272 of the pick head 270. The one or more actuators are any suitable actuators, such as linear actuators, capable of moving one or more tines 273A-273E in the Z direction. In the aspect illustrated in, for example, FIG. 6 there is one actuator 274A, 274B for each of the movable tines 273A, 273E so that each moveable tine is independently movable in the Z direction. In other aspects one actuator may be coupled to more than one movable tine so that the more than one movable tine move as a unit in the Z direction.

As may be realized, movably mounting one or more tines 273A-273E on the base member 272 of the pick head 270 provides for full support of large case units and/or pickfaces on the pick head 270 while also providing the ability to pick and place small case units without interfering with other case units positioned on, for example, the storage shelves, interface stations and/or peripheral buffer stations. The ability to pick and place variably sized case units without interfering with other case units on the storage shelves, interface stations and/or peripheral buffer stations reduces a size of a gap GP (see FIG. 1B) between case units on the storage shelves. As may be realized, because the tines 273B-273D are fixed to the base member 272 there is no duplicative motion when picking/placing case units as the lifting and lowering of case units and/or pickfaces to and from the case unit holding location is effected solely by the lift motor 301, 301A.

Referring again to FIG. 6, it is again noted that the pusher bar 110PR is movable independent of the transfer arm 110PA. The pusher bar 110PR is movably mounted to the frame in any suitable manner such as by, for example, a guide rod and slide arrangement and is actuated along the Y direction (e.g. in a direction substantially parallel to the extension/retraction direction of the transfer arm 110PA). In one aspect at least one guide rod 360 is mounted within the payload section 110PL so as to extend transversely relative to the longitudinal axis LX of the frame 110F. The pusher

bar **110PR** may include at least one slide member **360S** configured to engage and slide along a respective guide rod **360**. In one aspect, at least the guide rod/slide arrangement holds the pusher bar **110PR** captive within the payload section **110PL**. The pusher bar **110PR** is actuated by any suitable motor and transmission, such as by motor **303** and transmission **303T**. In one aspect the motor **303** is a rotary motor and the transmission **303T** is a belt and pulley transmission. In other aspects the pusher bar **110PR** may be actuated by a linear actuator having substantially no rotary components.

The pusher bar **110PR** is arranged within the payload section **110PL** so as to be substantially perpendicular to the rollers **110RL** and so that the pusher bar **110PR** does not interfere with the pick head **270**. As can be seen in FIG. **10B**, the bot **110** is in a transport configuration where at least one case unit would be supported on the rollers **110RL** (e.g. the rollers collectively form the payload bed). In the transport configuration the tines **273A-273E** of the pick head **270** are interdigitated with the rollers **110RL** and are located below (along the Z direction) a case unit support plane RSP (see FIG. **10**) of the rollers **110RL**. The pusher bar **110PR** is configured with slots **351** (FIG. **10C**) into which the tines **273A-273E** pass where sufficient clearance is provided within the slots **351** to allow the tines to move below the case unit support plane RSP and to allow free movement of the pusher bar **110PR** without interference from the tines **273A-273E**. The pusher bar **110PR** also includes one or more apertures through which the rollers **110RL** pass where the apertures are sized to allow free rotation of the rollers about their respective axes. As may be realized, the independently operable pusher bar **110PR** does not interfere with the rollers **110PR**, extension of the transfer arm **110PA** in the transverse direction (e.g. Y direction) and the lifting/lowering of the pick head **270**.

As noted above, because the pusher bar **110PR** is a separate, standalone axis of the bot **110** that operates free of interference from the pick head **270** extension and lift axes, the pusher bar **110PR** can be operated substantially simultaneously with the lifting and/or extension of the transfer arm **110PA**. The combined axis moves (e.g. the simultaneous movement of the pusher bar **110PR** with the transfer arm **110PA** extension and/or lift axes) provides for increased payload handling throughput in along the Y throughput axis and effects the ordered (e.g. according to the predetermined load out sequence) multi-pick of two or more case units from a common picking aisle, in one common pass of the picking aisle. For example, referring to FIGS. **10-10A** during a transfer arm **110PA** multi-pick/place sequence the pusher bar **110PR** is prepositioned (as the case unit(s) and/or pickface are being picked and transferred into the payload section **110PL**) to a location that is a predetermined distance **X2** away from the contact depth **X3** (e.g. the depth of the tines occupied by the case unit(s) and/or pickface CU when being picked/placed from a storage space or other case unit holding location) (FIG. **14**, Block **1100**). The distance **X2** is a minimized distance that only allows sufficient clearance between pusher bar **110PR** and the case unit(s) to allow the case unit(s) to be seated on the rollers **110RL**. As the case unit(s) CU are lowered onto the rollers **110RL** (FIG. **14**, Block **1110**) the distance travelled by the pusher bar **110PR** to contact the case unit(s) CU is a shorter distance **X2** when compared to moving from a back side **402** (relative to the lateral direction and an access side **401** of the payload section **110PL**) of the payload section **110PL** a distance **X4** as with conventional transport vehicles. When the case unit(s) CU are lowered by the transfer arm **110PA** and

transferred to the rollers **110RL** so as to be solely supported by the rollers **110RL**, the pusher bar **110PR** is actuated to forward (relative to the lateral direction and an access side **401** of the payload section **110PL**) justify the case unit(s) CU (FIG. **14**, Block **1120**). For example, the pusher bar **110PR** may push the case unit(s) CU laterally in the Y direction so that the case unit(s) contact the fence **110PF** (which is located at the access side **401** of the payload section **110PL** so that a case unit reference datum may be formed through contact between the case unit(s) CU and the fence **110PF**. In one aspect the pusher bar **110PR** may engage or otherwise grip the case unit(s) CU during transport of the case units (e.g. so as to hold the case unit(s) against the fence **110PF**) for maintaining the case unit(s) CU in a predetermined spatial relationship with each other and a reference frame REF (FIG. **6**) of the bot **110** (FIG. **14**, Block **1130**). When placing the case unit(s) the pusher bar **110PR**, after justifying the case unit(s) CU against the fence **110PF**, is withdrawn (e.g. in the Y direction) from contact with the case unit(s) CU (FIG. **14**, Block **1140**). Substantially immediately after the pusher bar **110PR** disengages the case unit(s) CU one or more of the lift axis (e.g. in the Z direction) and extension axis (e.g. in the Y direction) of the transfer arm **110PA** are actuated substantially simultaneously with the withdrawing movement of the pusher bar **110PR** (FIG. **14**, Block **1150**). In one aspect both the lift and extension axes are actuated when the pusher bar is withdrawn from contact with the case unit(s) CU while in other aspect one of the lift and extension axes is actuated. As may be realized, the simultaneous movement of the transfer arm **110PA** lift axis and/or extension axis with the withdrawal of the pusher bar **110PR** as well as the decreased distance the pusher moves to justify the case unit(s) CU decreases the time needed to transfer case unit(s) CU to and from the bot **110** and increases throughput of the storage and retrieval system **100**.

As described herein, referring to FIGS. **2A**, **2B** and **12**, each bot **110** is configured to transport pickfaces between the picking aisles **130A** and the transfer/handoff stations TS and buffer stations BS. In one aspect, the control server **120** is configured to command the bot **110**, and effect with the bot **110** outbound flow (which may also be referred to as order fulfillment stream, outbound stream(s) or order fulfillment) sortation of case order(s) independent of the pick order of cases from the storage area by the bot **110** forming a pickface. In one aspect, the bot controller **110C** is configured to command the bot **110**, and effect with the bot **110** outbound flow sortation of case order(s) independent of the pick order of cases from the storage area by the bot **110** forming a pickface. In still other aspects, the control server **120** and the bot controller **110C** are both configured to command the bot **110**, and effect with the bot **110** outbound flow sortation of case order(s) independent of the pick order of cases from the storage area by the bot **110** forming a pickface. Thus, the control server **120** and/or the bot controller **110C** is/are configured to set the outbound case flow, at least in part with bot **110** sortation of the cases carried in common by the both **110** and decoupled from the pick order of the cases by the bot **110** from storage. As may be realized, in one aspect, each bot **110** is configured to transport pickfaces between a first pickface interface station (e.g. transfer/handoff station TS and/or buffer station BS) and a second pickface interface station (e.g. transfer/handoff station TS and/or buffer station BS) that is spaced apart from the first pickface interface station) where, as described herein, the bot **110** picks a first pickface from the first interface station, traverses the transfer deck **130B** and places/buffers the first pickface, or at least a portion thereon, at the second

pickface interface location so that the second pickface interface station has multiple pickfaces buffered on a common support/surface CS in an ordered sequence of pickfaces according to a predetermined case out order sequence of mixed case pickfaces. As will be described below, the bot **110** is configured to transfer a first pickface PCF1 having any suitable number of case units therein from the picking aisles **130A** (or a transfer station TS or a buffer station BS) and place second pickface PCF2, that is different than the first pickface PCF1, onto a common surface CS (such as of a rack shelf RTS) of the transfer/handoff station TS (or buffer station BS) that is common to both the bot **110** and the lift **150B**. This may be referred to for description purposes as outbound flow sortation with the bot at transfer stations (and/or at buffer stations). As will also be described below, the first and second pickfaces, in one aspect, have at least one case unit that is common to both the first and second pickfaces. In one aspect, as described herein, the bot **110** is configured to build the first pickface (e.g. at least one of the multiple pickfaces) on the fly, e.g. during traverse (e.g. while the bot is moving) from a first pick location in the picking aisles **130A** to placement of the second pickface at the transfer/handoff station TS (or buffer station BS), in a multi-pick/place sequence. In another aspect, the bot **110** is configured to build the first pickface (e.g. at least one of the multiple pickfaces placed on the common surface CS) on the fly, e.g. during traverse (e.g. while the bot is stationary at the second pickface interface station or the second pickface interface station buffer) from the first pick location to the transfer/handoff station TS (or buffer station BS), in a multi-pick/place sequence. As may be realized, where the pickfaces are picked, e.g. by the bot **110**, from a first pickface interface station, such as a transfer station TS or buffer station BS and placed at the second pickface interface station, such as another transfer station TS or buffer station BS the pickface bypasses storage (e.g. is not placed in a storage space **130S** before delivery to the second pickface interface station). In other aspects, at least a portion of the pickface picked from the first pickface interface station is placed in a storage space **130S** (e.g. in the storage rack array RMA) by the bot **110** before transport to the second pickface interface station. In one aspect, the pickface picked from an inbound transfer station TS (or buffer station BS) may be the same pickface that is placed at an outbound transfer station TS (or buffer station BS) (i.e. the pickface is not broken up during transport from the inbound transfer station TS/buffer station BS and the outbound transfer station TS/buffer station BS where the transport between the inbound and outbound stations may or may not include placement of the pickface in storage).

The controller **110C** of the bot **110** is configured to effect the on the fly building of the first pickface (or any other pickface picked by the bot **110**). In one aspect the bot **110** is configured, as described herein, to build the pickface onboard the bot **110** such as, for example, in the payload section **110PL** where case units/pickfaces are picked by the bot and arranged in the payload section in a predetermined order or sequence. In one aspect, the bot **110** is also configured to pick/build a pickface PCF3 that is different than the first pickface PCF1 and place the different pickface PCF3 on a shelf (such as another rack shelf RTS stacked above or below the rack shelf forming the common surface CS) of the transfer/handoff station TS (or buffer station BS). The bot **110** includes case manipulation, as described herein. The bot has picked the first pickface PCF1 and is configured to further pick the second pickface PCF2 from one or more case units (forming the different pickface PCF3) from the

rack shelf RTS (or other location such as a storage shelf in the picking aisles) and place the different pickface PCF3 on the common surface CS. As may be realized, the lift **150B**, in one aspect is configured to pick the second pickface PCF2 from the transfer/handoff station TS. In other aspects, the lift **150** is configured, as described herein, to pick a third pickface PCF4 from the common surface CS (such as the rack transfer shelf RTS) of the transfer/handoff station TS (or buffer station BS) where the third pickface PCF4 is different than the first and second pickfaces PCF1, PCF2 and the common case is common to the first, second and third pickfaces PCF1, PCF2, PCF4. As may be realized, the second interface station (such as the transfer station TS or buffer station BS) forms a common pickface transfer interface for the lift **150** so that the commonly supported pickfaces are picked in common with the lift **150**. It is noted that the ability of the lift **150** to pick individual pickfaces, as noted above, from different deck levels effects sorting of the pickfaces in the Z throughput axis.

In one aspect of the disclosed embodiment, as may be realized, in the multi-pick/place sequence multiple case units are substantially simultaneously carried and manipulated (e.g. so as to form one or more pickfaces) within the payload section **110PL** to further increase throughput of the storage and retrieval system **100** and to effect the multi-pick/place sequence in accordance with a predetermined order out sequence. Referring also to FIG. 1, the bot receives pick and place commands from, for example, control server **120** (and/or warehouse management system **2500**) and the bot controller **110C** executes those commands for forming the ordered multi-pick. Here the bot **110** enters the common aisle **130A1** from, for example, the transfer deck **130B** for making a single or common pass through the picking aisle **130A1** during which the bot **110** picks two or more case units according to the predetermined order out sequence (FIG. 15, Block **1201A**). In one aspect the manipulation of the case units CU is a sorting of the case units (in other words picking and placing of case units according to the predetermined load out sequence) where the cases are positioned on the transfer arm **110PA** for picking/placement of the case units and/or positioned so that the case units are not transferred and remain on the transfer arm **110PA** while other case units are transferred to and from the transfer arm **110PA**. Here, the bot **110** travels through the common picking aisle **130A1** in the direction of arrow XC and stops at a predetermined storage space **130S1**, according to the predetermined order out sequence, where the bot **110** picks one or more case units from the predetermined storage space **130S1** with a common transfer arm **110PA** where placement of the case units on the common transfer arm **110PA** corresponds to the predetermined order out sequence as will be described in greater detail below (e.g. the case units are sorted on-the-fly, e.g. during transport, with the bot **110**).

As an example of case manipulation on the bot **110**, referring also to FIGS. **10B-10E**, case unit(s) CUA may be picked from a case unit holding location (e.g. such as storage spaces **130S** in a common picking aisle for effecting the ordered multi-pick, and in other aspects from a lift interface station TS, and/or a case unit buffer station BS located in a picking aisle or on the transfer deck) and transferred into the payload section **110PL** (FIG. 15, Block **1201B**). As the case unit(s) CUA is being transferred into the payload section **110PL** the pusher bar **110PR** may be pre-positioned (FIG. 15, Block **1204**) adjacent the fence **110PF** so that the pusher bar **110PR** is positioned between the case unit(s) CUA and the fence **110PF** when the case unit(s) CUA is lowered for transfer to the rollers **110RL** (FIG. 15, Block **1205**). The

pusher bar **110PR** is actuated to push the case unit(s) CUA (resting on the rollers **110RL**) in the Y direction towards the back (e.g. rear) **402** of the payload section **110PL** so that the case unit(s) CUA contacts a justification surface **273JS** (FIG. **10**) of the tines **273A-273E** and is justified to the back **402** of the payload section **110PL** (FIG. **15**, Block **1210**).

In one aspect, the bot **110** continues to traverse the common picking aisle **130A1** in the same direction XC (e.g. so that all of the case units in the ordered multi-pick are picked in the common pass of the picking aisle with the bot **110** travelling in a single direction) and stops at another predetermined storage space **130S** according to the predetermined order out sequence. As noted above, the pusher bar **110PR** remains in contact with (e.g. grips) the case unit(s) CUA during transport of the case unit(s) between case unit holding locations so that the case unit(s) CUA remains in a predetermined location at the back **402** of the payload section **110PL** (and/or at a predetermined location longitudinally) relative to the reference frame REF of the bot **110** (FIG. **15**, Block **1215**). To pick subsequent case units, from for example, the other storage space **130S2** of the common picking aisle **130A1** the pusher bar **110PR** is moved in the Y direction to disengage the case unit(s) CUA and the lift and extension axes of the transfer arm **110PA** are actuated to retrieve another case unit(s) CUB from the other storage space **130S2** (or in other aspects from e.g. a lift/handoff interface station TS and/or a buffer/handoff station BS as noted above) (FIG. **15**, Block **1220**). While the case unit(s) CUB are being picked the pusher bar **110PR** is positioned in the Y direction adjacent the back **402** of the payload section **110PL** so as to be located between the case units CUA and the justification surface **273JS** of the tines **273A-273E** (FIG. **15**, Block **1225**). The case unit(s) CUB are transferred into the payload section and lowered/placed on the rollers **110RL** (FIG. **15**, Block **1230**) so that the case units CUA, CUB are arranged relative to each other along the Y axis. The pusher bar **110PR** is actuated in the Y direction to push the case units CUA, CUB towards the fence **110PF** to forward justify the case units CUA, CUB (FIG. **15**, Block **1234**) and grip/hold the case units CUA, CUB for transport (FIG. **15**, Block **1235**). As may be realized, in one aspect the case units CUA, CUB are placed at a case unit holding location together as a unit while in other aspects the case units CUA, CUB are sorted, e.g. transported to and placed at separate positions of a common case unit holding location or at different case unit holding locations (FIG. **15**, Block **1240**) as will be described in greater detail below. For example, referring also to FIGS. **7-9**, the bot **110** carrying the ordered multi-pick payload transfers the case units of the ordered multi-pick to one or more interface stations TS (which include buffer shelves **7000A-7000L**) corresponding to output lifts **150B1**, **150B2**.

As may be realized, in one aspect where the bots **110** “parallel park” into an interface station TS (FIG. **7**) or turn into a pier **130BD** (FIG. **8**) the spacing between bots travelling on the high speed bot travel path HSTP of the transfer deck **130B** (FIG. **2A**) is such that the bot interfacing with the interface station TS is able to slow down and turn into the interface station TS substantially without interference from and/or interference with another bot **110** travelling along the transfer deck **130B**. In other aspects, the bots travelling on the transfer deck may drive around the bots turning into the interface stations as the transfer deck(s) **130B** is substantially open and configured for the undetermined traversal of bots **110** across and along the transfer deck(s) **130B** as described above. Where the case units of the multi-pick are placed at different positions of, for example,

a common buffer shelf of interface/handoff station **7000A-7000L** of the lifts **150B1**, **150B2** the bot **110** places a first one of the case units CUB (corresponding to, for exemplary purposes pickface **7** in FIG. **9** which in this example includes a single case unit) in a first position of the buffer shelf **7000B** and places the second one of the case units CUA (corresponding to, for exemplary purposes pickface **5** in FIG. **9** which in this example includes a single case unit) in a second position of the buffer shelf **7000B**. Where the case units of the multi-pick are placed at a common case unit holding location the bot **110** places both case units CUA, CUB as a unit (e.g. a pickface) at for example, a common position of buffer shelf **7000A** (corresponding to, for exemplary purposes pickface **9** in FIG. **9** which in this example, includes two case units).

Where the case units CUA, CUB are sorted (FIG. **15**, Block **1250**) for placement at separate positions of a common case holding location or at different case holding locations, the case units CUA, CUB are separated from each other in the payload section **110PL**. For example, the pick head **270** of the transfer arm **110PA** may be moved in the Z direction to lift the case units CUA, CUB from the rollers **110RL** by an amount sufficient to allow the pusher bar **110PR** to pass beneath the case unit(s) (FIG. **16**, Block **1250A**). As the case units CUA, CUB are lifted the pusher bar **110PR** is positioned along the Y direction so as to be located between the case units CUA, CUB (see FIG. **10E**) (FIG. **16**, Block **1250B**). The pick head **270** is lowered so that the case units CUA, CUB are transferred to the rollers **110RL** and so that the pusher bar is inserted between the case units CUA, CUB (FIG. **16**, Block **1250C**). The pusher bar **110PR** is moved in the Y direction (e.g. to separate the case unit(s)) to move case unit(s) CUA towards the back **402** of the payload section **110PL** (e.g. against the justification surface **273JS** of the tines **273A-273E** or any other suitable position) while the case unit(s) CUB remain at the front of the payload section **110PL** adjacent the fence **110PF** (e.g. as shown in FIG. **10C**) (FIG. **16**, Block **1250D**). As may be realized, where the case units are held against the justification surface **273JS** of the tines during transport, the pusher bar is moved in the Y direction (e.g. to separate the case unit(s)) to move case unit(s) CUB towards the front **401** of the payload section **110PL** (e.g. against the fence **110PF** or any other suitable position) while the case unit(s) CUA remain at the back of the payload section **110PL** adjacent the justification surface **273JS**. The pusher bar **110PR** may also be moved in the Y direction to re-justify the case unit(s) CUB against the fence **110PF** to position the case unit(s) on the tines **273A-273E** for placement at a case unit holding location (FIG. **16**, Block **1250E**). As may be realized, with the case unit(s) CUA being positioned substantially against the justification surface **273JS** of the tines **273A-273E** (e.g. of the pick head **270**) the case unit(s) CUB can be placed at a case unit holding location substantially without interference from the case unit(s) CUA (FIG. **16**, Block **1250F**), e.g. the case unit CUA is free from contacting case units disposed at the case unit holding location. The case unit(s) CUA is lowered/transferred back into the payload section **110PL** (e.g. by retracting and lowering the transfer arm **110PA**) (FIG. **16**, Block **1250G**). The pusher bar **110PR**, which is pre-positioned between the justification surface **273JS** and the case unit(s) CUA, pushes the case unit(s) CUA, which is disposed on the rollers **110RL**, against the fence **110PF** to forward justify the case unit(s) CUA for placement at another case unit holding location (e.g. different than the holding location that case unit(s) CUB were placed) (FIG. **16**, Block **1250H**). The pusher bar **110PR**

remains against the case unit(s) CUA for gripping (e.g. with the fence) the case unit(s) during transport to the other case unit holding location (FIG. 16, Block 1250I). The pusher bar 110PR moves away from the case unit(s) CUA and the transfer arm is actuated to lift and extend the pick head 270 for placing the case unit(s) CUA at the other case unit holding location (FIG. 16, Block 1250J).

An example of a bot 110 case unit(s) transfer transaction including a case unit(s) multi-pick and place operation with on the fly sortation of the case units for creating a mixed pallet load MPL (as shown in FIG. 1F) and/or to fill the predetermined order sequence of picked items in one or more bag(s), tote(s) or other container(s) TOT at an operator station or cell 160EP (as shown in FIG. 26, such as to e.g., fill a customer order) according to a predetermined order out sequence will be described with respect to FIGS. 9 and 11-13 in accordance with an aspects of the disclosed embodiment. For example, referring to FIG. 11 a customer order may require case unit(s) 7 to be delivered to output lift 150B1 and case units 5 to also be delivered to output lift 150B1 (in other aspects, it is noted that customer orders may require case units carried by a common bot 110 to be delivered to different output lifts 150B1, 150B2 (FIG. 9) such that the transfer of the case units carried by the common bot 110 to different output lifts occurs in a manner substantially similar to that described herein). In the aspects of the disclosed embodiment described herein the output lift 150B1 (e.g. each of the output lifts 150B1, 150B2 of the storage and retrieval system/order fulfillment system) defines a fulfillment course or pathway (also referred to as a stream) of mixed case pickfaces outbound from the storage array to a load fill where the mixed case pickfaces enter and exit the fulfillment course in substantially the same order. As may be realized, while the input and output lifts 150A, 150B are described as vertically reciprocating lifts it should be understood that in other aspects the input and output lifts 150A, 150B are any suitable transport modules for transporting case pickfaces to and from the storage structure 130 (e.g. between a respective pickface interface station, such as transfer station TS or buffer station BS, and a respective one of an input station 160IN, e.g. an input cell, and an output station 160UT, e.g. a load fill section/cell). For example, in other aspects the lift modules 150A, 150B are one or more of vertically reciprocating lifts, any suitable automated material handling systems, conveyors, bots, turntables, roller beds, multilevel vertical conveyor (e.g. paternoster conveyor) that operate synchronously or asynchronously. To efficiently use each bot 110 in the storage and retrieval system 100 the controller, such as control server 120, determines which picking aisle(s) case units 5, 7 are located. The controller also determines which inbound case unit(s) ICU are to be stored in the picking aisle(s) from which case units 5, 7 (e.g. the outbound case units) are to be picked. The controller sends commands to a bot 110 on a level where case units 5, 7 are located to pick one or more inbound case units ICU from an interface station TS of one or more lift modules 150A in a manner similar to that described above (FIG. 17, Block 1400A). The bot 110 grips the case unit(s) ICU (FIG. 17, Block 1420) and transports the case unit(s) to one or more storage space 130 within one or more picking aisle 130A2 (FIG. 17, Block 1421) where at least one of the picking aisles in which the inbound case units are placed includes one of the outbound case units 5, 7. As may be realized, where the inbound case units are placed at different storage locations 130S the inbound case units are sorted (FIG. 17, Block 1425) as described above where one or more case unit(s) are transferred to one case unit holding location,

such as a storage space 130S or buffer, (FIG. 17, Block 1430) while case units that are not transferred are returned to the payload section of the bot 110 for transfer to another case unit holding location (FIG. 17, Block 1435).

As may be realized, the outbound case units 5, 7 are located in the same or different picking aisles and are retrieved by one bot 110 or different bots 110 depending on a proximity of the outbound case units and the predetermined storage position(s) of the inbound case unit(s). For example, referring to FIG. 11, the bot 110 picks an inbound case unit ICU from interface station TS of lift module 150A for placement in picking aisle 130A2 (in a manner substantially similar to that described above), which is the aisle case unit 5 is located. Case unit 7 in this example is located in picking aisle 130A1. After placement of the inbound case unit ICU the bot continues to travel along picking aisle 130A2 in a common pass (e.g. a single traversal of the picking aisle in a single direction) to pick the outbound case unit 5 (FIG. 17, Block 1400). Where it is more efficient to have a single bot 110 pick multiple case units, the outbound case unit 5 is justified on the bot 110 as described above (FIG. 17, Block 1405) and the bot travels to the location of another case unit, such as outbound case unit 7 in aisle 130A1 (it is noted that where a second outbound case is located in a common aisle with the first outbound case both outbound case units are picked in a common pass of the picking aisle with the common transfer arm 110PA (FIG. 6) of the bot 110). The second outbound case unit(s) 7 is picked with the common transfer arm 110PA (FIG. 17, Block 1410) and both case units 5, 7 are transferred and placed at one or more of peripheral buffer station BS and interface station TS of a pickface transport system such as lift module 150B (FIG. 17, Blocks 1420-1435) in a manner substantially similar to that described above with respect to the placement of the inbound case unit(s). Where it is more efficient to have a two different bots 110 pick a respective one of case units 5, 7 after picking the respective outbound case (FIG. 17, Block 1400) the case unit is gripped (FIG. 17, Block 1420) and transferred to and placed at one of the peripheral buffer station BS or the interface station TS of outbound lift 150B (FIG. 17, Block 1421-1435) as described herein. In one aspect, where an outbound case unit, such as case unit 5 is placed at a peripheral buffer station BS a different bot 110, than the bot that placed the case unit 5 at the peripheral buffer station BS, transfers the case unit 5 to the interface station TS while in other aspects the same bot 110 returns to the peripheral buffer station BS to transfer case unit 5 to the interface station TS. In the aspects of the disclosed embodiment described herein, the buffer stations BS and/or the transfer stations TS (e.g. at least one pickface handoff station) commonly supports more than one of the mixed case pickfaces defining a portion of the mixed case pickfaces outbound from the storage array/structure 130 entering the fulfillment course in an ordered sequence of pickfaces based on a predetermined sequence of the load fill. In one or more of the aspects of the disclosed embodiment described herein, the buffer station BS and/or transfer stations TS forms a common pickface transfer interface for the outbound lift(s) 150B1, so that the commonly supported pickfaces are picked in common with the outbound lift(s) 150B1. In one or more of the aspects of the disclosed embodiment described herein, each of the buffer stations BS and/or transfer stations TS commonly supports more than one of the mixed case pickfaces defining a portion of the mixed case pickfaces outbound from the storage array (see for exemplary purposes only pickfaces 1-4 in FIG. 9) in an ordered sequence of pickfaces based on the predetermined sequence

31

of the load fill. In one or more aspects of the disclosed embodiment described herein, the mixed case pickfaces defining the portion of the mixed case pickfaces outbound from the storage array/structure **130** in the ordered sequence and commonly supported on the buffer station BS and/or transfer station TS is based on an ordered sequence of pickfaces on another buffer station BS and/or transfer station TS of another fulfillment course (see e.g. the mixed cases outbound from the outbound lift **150B2**). In one or more aspects of the disclosed embodiment, any suitable controller, such as controller **120** is in communication with the bot(s) **110** and is configured to effect placement of pickfaces on the buffer station BS and or transfer station TS based on the ordered sequence of pickfaces.

In one aspect the outbound case units are picked and transferred as a unit (e.g. a pickface) by a common transfer arm **110PA** (FIG. 6) of bot **110**. Referring now to FIG. 12 again a customer order may require case unit(s) **7** to be delivered to output lift **150B1** and case units **5** to also be delivered to output lift **150B1** (in other aspects, it is noted that customer orders may require case units carried by a common bot **110** to be delivered to different output lifts **150B1**, **150B2** (FIG. 9) such that the transfer of the case units carried by the common bot **110** to different output lifts occurs in a manner substantially similar to that described herein). As described above, the controller determines which inbound case unit(s) ICU are to be stored in the picking aisle(s) from which case units **5**, **7** (e.g. the outbound case units) are to be picked. The controller sends commands to a bot **110** on a level where case units **5**, **7** are located to pick one or more inbound case units ICU as a unit (e.g. pickface) from an interface station TS of a lift module **150A** in a manner similar to that described above (FIG. 17, Block **1400A**). The bot **110** grips the pickface PF1 (FIG. 17, Block **1420**), transports the pickface PF1 to a storage space **130** within the picking aisle **130A2** (FIG. 17, Block **1421**) where the outbound case units **5**, **7** are located and places the pickface PF1 into a storage space **130S** (FIG. 17, Block **1430**). It is noted that since the whole pickface is transferred to a common storage space and no case units are left on the bot that the flow, in this example, does not proceed to block **1435** of FIG. 17.

After placing the inbound pickface PF1, the bot **110** continues to travel through aisle **130A2** in a common pass (e.g. a single traversal of the picking aisle in a single direction) to the storage space holding outbound case units **5**, **7** (which are arranged on the storage shelves adjacent one another so as to be picked simultaneously as outbound pickface PF2). The bot **110** picks pickface PF2 with the common transfer arm **110PA** (FIG. 6) (FIG. 17, Block **1415**), grips the pickface PF2 (FIG. 17, Block **1420**) and transports the pickface PF2 (FIG. 17, Block **1421**) to the outbound lift **150B1**. In one aspect the case units **5**, **7** of the pickface PF2 are placed at one of the peripheral buffer station BS or the interface station TS as a unit (FIG. 17, Block **1430**). In another aspect the case units **5**, **7** of the pickface are separated and justified (in a manner similar to that described above) for placement in different locations (FIG. 17, Block **1425**). For example, bot **110** places case unit **7** at the peripheral buffer station BS (FIG. 17, Block **1430**), returns case unit **5** to the payload area of the bot **110** (FIG. 17, Block **1435**), grips the case unit **5** (FIG. 17, Block **1420**), transports the case unit **5** to the interface station TS (FIG. 17, Block **1421**) and transfers the case unit **5** to the interface station (FIG. 17, Block **1430**).

In another aspect, referring to FIG. 13, the outbound case units **5**, **7** are picked from different storage locations within

32

a common aisle **130A2** with the common transfer arm **110PA** (FIG. 6 of the bot **110**. Here, the bot **110** transfers one or more inbound case units ICU to one or more storage locations in the manner described above where at least one of the inbound case units ICU is located in a common picking aisle **130A2** with the outbound case units **5**, **7**. After placing at least one inbound case unit at a predetermined storage location **130S** of aisle **130A2** the bot **110** continues to travel through picking aisle **130A1**, in a common pass of the picking aisle **130A2**, and picks case unit **5** from storage space **130S1** in the manner described above (FIG. 17, Block **1400**). The case unit(s) **5** is justified on the bot **110** towards the rear of the payload section **110PL** as described above (FIG. 17, Block **1405**). The bot **110** continues to travel through the picking aisle **130A1** in a common pass of the picking aisle and picks case unit **7** from a different storage space **130S2** with the common transfer arm **110PA** so that both case unit(s) **7**, **5** are located adjacent one another on the common transfer arm **110PA** (FIG. 17, Block **1410**). As may be realized, in one aspect, the controller **110C** is configured to effect picking of the case unit(s) in any suitable order such as, for example, an order that is opposite an order in which the case unit(s) are placed.

In this multi-pick example, the case unit holding location(s) correspond to storage spaces **130S** of the picking aisles **130** but in other aspects the case unit holding location(s) include input lift modules **150A1**, **150A2** (where a direct transfer between bots and the lift occurs), interface or peripheral buffer stations TS, BS for interfacing with the input lift modules **150A1**, **150A2**, (where an indirect transfer between the lift modules and the bots occurs) and storage spaces **130S** (picking from the interface stations TS and the input lift modules **150A** with the bot **110** is noted where case units are needed for a predetermined order out sequence are not located in the storage spaces **130S** but are being input into the storage rack array in a just in time manner to be delivered substantially directly to the output lift(s) **150B1**, **150B2**).

The bot **110** grips both case units **7**, **5** within the payload section **110PL** in the manner described above and exits the picking aisle **130A1** (FIG. 17, Block **1420**). The bot travels along the transfer deck **130B** and interfaces with output lift **150B1** (FIG. 17, Block **1421**). The bot separates the case units **7**, **5** within the payload section **110PL**, as described above, so that case unit(s) in any suitable manner such as, for example, so that case unit(s) **7** is justified towards the front of the payload section **110PL** and case unit(s) **5** is justified towards the back of the payload section **110PL** (FIG. 17, Block **1425**). The case unit **7** is transferred to the peripheral buffer station BS (FIG. 17, Block **1430**). The bot retracts the transfer arm **110PA** to return the case unit(s) **5** to the payload section **110PL** (FIG. 17, Block **1435**) and grips the case unit **5** (FIG. 17, Block **1420**). The case unit(s) **5** is transported to the interface station TS of output lift **150B1** (FIG. 17, Block **1421**), justified toward the front of the payload section **110PL** (FIG. 17, Block **1425**), as described above, and transferred to transfer station TS, as described above (FIG. 17, Block **1430**). In other aspects, depending on the predetermined case unit output sequence, the bot **110** places both case unit(s) **7**, **5** at a common location/position, such as at one of output lifts **150B1**, **150B2**. For example, pickface **20** on shelf **7000H** (FIG. 9) may include both case units **7**, **5** such that the bot **110** places both case units as a multi-case unit pickface at a single position of shelf **7000H**. As may be realized, the case unit(s) placed at the buffer station BS are, in one aspect, transferred to the interface station TS by a bot **110** or, in other aspects, by any suitable conveyor that

connects the buffer station BS to the interface station TS. In one aspect, where the case unit(s) are transferred from the buffer station BS to the interface station TS by a bot **110** that transfer is an opportunistic transfer such that a bot **110** travelling along the transfer deck, for example, in route for another task (e.g. transferring pickface(s) to storage, sorting pickfaces, transferring pickface(s) from storage, etc.) travelling by the buffer station BS stops to pick the pickface from the buffer station BS and transfer the pickface to the interface station TS while in the process of performing the other task.

An example of a bot **110** case unit(s) transfer transaction including a case unit(s) multi-pick and place operation with on the fly sortation of the case units for creating a mixed pallet load MPL (as shown in FIG. 1F) according to a predetermined order out sequence will be described with respect to FIGS. 9 and 24 in accordance with an aspects of the disclosed embodiment. The transfer of pickfaces with respect to FIGS. 9 and 24 is substantially similar to that described above with respect to FIGS. 11-13, however, in this aspect the storage of the pickfaces is bypassed such that pickfaces are transferred substantially directly between the inbound and outbound lifts **150A**, **150B1**. In one aspect, a bot picks a first pickface **5** from a first shelf of a first pickface handoff station such as transfer station TS of inbound lift **150A** where the inbound lift **150A** transfers one or more pickfaces/cases on the pickface handoff station (FIG. 23, Block **2300**). The bot **110** traverses the transfer deck **130B** and buffers the first pickface **5** (or a portion thereof) on a second shelf of a second pickface handoff station such as, for example, buffer station BS of the transfer station TS at the outbound lift **150B1** (FIG. 23, Block **2310**). In other aspects, the first pickface **5** (or a portion thereof) is buffered at the transfer station TS of the outbound lift **150B1** rather than at the buffer station BS. The bot **110** forms a second pickface **5, 7** at the second shelf, the second pickface being different than the first pickface **5** and comprising more than one case in ordered sequence corresponding to a predetermined case out order sequence of mixed cases where the first pickface **5** and the second pickface **5, 7** have at least one case in common (FIG. 23, Block **2320**). In one aspect the lift **150B1** picks the second pickface **5, 7** from the second shelf, such as the buffer station BS or transfer station TS (FIG. 23, Block **2330**). In one aspect, the bot **110** forms the second pickface **5, 7** at the second shelf (e.g. the buffer station BS or transfer station TS) on the fly during transport of the first pickface **5** between the first shelf and the second shelf. In one aspect the bot forms the second pickface **5, 7** onboard of the autonomous transport vehicle. In one aspect the bot **110** forms the second pickface **5, 7** at the second shelf or at a buffer portion of the second shelf. In one aspect the bot **110** places at least a portion of the first pickface (such as where pickface **5** includes more than one case) picked from the first shelf on a storage rack of a storage array (such as in storage spaces **130S**) before transporting at least the portion of the first pickface to the second shelf. In one aspect the second shelf forms a common pickface transfer interface for the vertically reciprocating lift, the method further comprising picking in common, with the vertically reciprocating lift, the commonly supported pickfaces.

While the bot **110** in FIG. 24 is illustrated picking one case/pickface **5** from the transfer station TS of the inbound lift **150A**, in other aspects the bot **110** picks two (or more than two) inbound pickfaces such as cases/pickfaces **5, 7**. Here, in one aspect, the bot **110** places one pickface **5** on the outbound buffer station BS (or outbound transfer station TS), then moves to another shelf location (such as another

outbound buffer or transfer station BS, TS or an adjacent location on a common buffer or transfer station BS, TS shelf) to place the second pickface **7**. In one aspect the lift **150B1** removes the pickface(s) **5, 7** from the buffer or transfer station BS, TS as described herein. In another aspect, the bot **110** places both pickfaces **5, 7** on the outbound buffer or transfer station BS, TS shelf. Here the lift **150B1** picks one of the pickfaces **5, 7** and transports the pickface **5, 7** to the output station **160UT**. The lift **150B1** returns to the buffer or transfer station BS, TS shelf and picks the other pickface **5, 7** for transfer to the output station **160UT**. In still another aspect, the bot **110** places both pickfaces **5, 7** at the outbound buffer or transfer station BS, TS shelf where the lift **150B1** picks both pickfaces **5, 7** for transfer to the output station **160UT**. Here the pickface **5, 7** are singulated or handled together in any suitable for building a mixed pallet as illustrated in FIG. 1F.

In the examples described herein the transfer of case units between the bots **110** and the lifts **150** occurs passively through the interface stations TS as described above. As an example of the transfer, referring to FIG. 18, the autonomous transport vehicle is positioned relative to the interface station TS in a manner similar to that described above with respect to the slats **1210S** and/or a locating features **130F** (FIG. 18, Block **1800**). The transfer arm **110PA** (e.g. end effector) of the bot **110** extends to transfer a pickface to the interface station TS where the fingers **273A-273E** of the transfer arm **110PA** interface with, for example, the slats **1210S** of the interface station TS (FIG. 18, Block **1801**). As may be realized, and as noted above, multiple pickfaces may be placed on the interface station TS (e.g. multiple individual pickfaces are simultaneously held on the interface station) for simultaneous or independent transfer to the lift **150**. The lift **150** is moved to position the load handling device LHD, LHDA, LHDB adjacent the interface station TS (FIG. 18, Block **1802**). The load handling device LHD, LHDA, LHDB is extended to lift the pickface from the interface station and transfer the pickface to the lift **150** where the fingers **4273** of the load handling device LHD, LHDA, LHDB interface with the slats **1210S** of the interface station TS in the manner described above with respect to, for example, FIG. 4B (FIG. 18, Block **1803**). As may be realized, the interface station TS has no moving parts and the transfer of the pickface(s) between the bots **110** and the lifts **150** through the interface station TS is a passive transfer. As may also be realized, transfer of pickfaces from the lifts **150** to the bots **110** may occur in a manner substantially opposite that described above with respect to FIG. 18.

In one aspect, the pickface, built by the bot **110** (e.g. in the manner described above), that is transferred to (e.g. placed to), for example, the interface station TS (and/or buffer station BS) is not the same pickface that is picked from the interface station TS (and/or buffer station BS) by the vertical lift **150**. For example, referring to FIG. 9, the bot **110** builds a first pickface from the storage spaces **130S** within the rack modules RM (e.g. FIG. 2A) that includes individual pickfaces **7** and **5** (FIG. 19, Block **1900**). The bot **110** transfers the first pickface to and places the first pickface on, for example, shelf **7000B** of interface station TS for transfer to a vertical lift **150** (FIG. 19, Block **1910**). As may be realized, while in this example, the individual pickfaces **5, 7** (e.g. forming the first pickface) are placed on a common shelf **7000B** for exemplary purposes only, in other aspects the individual pickfaces **5, 7** are placed on different shelves **7000A-7000F** so that the pickface placed on the shelves by the bot **110** is different than the first pickface but includes at least one case unit in common with the first pickface. For

example, the first pickface is broken up such that a different pickface including individual pickface **5** is placed on shelf **7000B** while another different pickface including individual pickface **7** is placed on, for example, shelf **7000H**. A vertical lift, such as lift **150B1** picks a second pickface from one or more shelves **7000A-7000F** (e.g. common to both the bot **110** and the vertical lift **150B1**) of the transfer stations TS (FIG. **19**, Block **1920**). Here the second pickface is different than the first pickface but includes at least one of the individual pickfaces **5**, **7** so that at least one case unit is common between the first pickface and the second pickface.

Similarly, in one aspect, the pickface that is transferred to (e.g. placed to), for example, the interface station TS (and/or buffer station BS) by the inbound vertical lift **150** (see vertical lift **150A** in FIG. **1**) is not the same pickface that is picked from the interface station TS (and/or buffer station BS) by the bot **110**. In one aspect, the control server **120** is configured to command the bot **110**, and effect with the bot **110** inbound flow (which may also be referred to as warehouse replenishment or inbound stream(s)) case sortation at a handoff station TS (and/or buffer station BS) with the bot **110** forming a pickface, independent of the pick order of cases from an input station by the lift **150**. In one aspect, the bot controller **110C** is configured to command the bot **110**, and effect with the bot **110** inbound flow case sortation at the handoff station TS (and/or buffer station BS) with the bot **110** forming the pickface, independent of the pick order of cases from an input station by the lift **150**. In still other aspects, the control server **120** and the bot controller **110C** are both configured to command the bot **110**, and effect with the bot **110** inbound flow case sortation at the handoff stations TS (and/or buffer station BS) with the bot **110** forming the pickface, independent of the pick order of cases from an input station by the lift **150**. Thus, the control server **120** and/or the bot controller **110C** is/are configured to set the inbound case flow, at least in part with bot **110** sortation of the cases carried in common by the both **110** and decoupled from the pick order of the cases by the lift **150**. This may be referred to for description purposes as inbound flow case sortation with the bot **110** at the handoff station TS (and/or buffer station BS). For example, referring to FIG. **9A**, first pickfaces are transferred to one or more vertical lifts **150A1**, **150A2** from the input station(s) **160IN** by the inbound conveyors **160CB** (FIG. **20**, Block **2000**). In this example, one of the first pickfaces includes a combination of individual pickfaces **5**, **7** while the other first pickface includes a combination of individual pickfaces **20**, **22**. The vertical lift **150A1** places the respective first pickface **5**, **7** to shelf **7000B** of interface station TS while vertical lift **150A2** places the other respective first pickface **20**, **22** to shelf **7000H** of another interface station TS on the same storage level **130L** (FIG. **20**, Block **2010**). The bot **110** builds or otherwise picks a second pickface(s) from the interface station(s) TS so that the first pickface(s) placed on the shelf(s) **7000B**, **7000H** (e.g. common to both the bot **110** and a respective vertical lift **150A**) by the vertical lifts **150A1**, **150A2** is/are different than the second pickface but the second pickface includes at least one case unit in common with the first pickface (FIG. **20**, Block **2020**). For example, the first pickface **5**, **7** is broken up such that a different pickface including individual pickface **5** (or individual pickface **7**) is picked by the bot **110** and/or the other first pickface **20**, **22** is broken up such that a different pickface including individual pickface **20** (or individual pickface **22**) is picked by the bot **110**. Here the second pickface is different than the first pickface but includes at least one of the individual pickfaces of the first pickface so

that at least one case unit is common between the first pickface and the second pickface. As may be realized, the second pickface may be broken up by the bot so that a pickface placed on at least one storage shelf with the **110** is different than the second pickface and where at least one case unit is common between the second pickface and the pickface placed on the at least one storage shelf.

The output lifts **150B1**, **150B2** transfer the ordered multi-pick(s) placed on the shelves **7000A-7000L** by the bots **110** to the output station **160UT** also in accordance with the predetermined order out sequence. For example, referring again to FIG. **9**, the pickfaces **1-22** are picked by the lifts **150B1**, **150B2** in sequenced order so that the pickfaces **1-22** are delivered to the output station **160UT** in the predetermined order (indicated by, for example, the number associated with each case unit/pickface illustrated in FIG. **9**) needed to form the mixed pallet load MPL (FIG. **1F**) and/or to fill the predetermined order sequence of picked items in one or more bag(s), tote(s) or other container(s) TOT at an operator station **160EP** (such as to fill e.g., a customer order). As such, each of the interface stations TS of each lift **150B1**, **150B2** forms a buffer that holds one or more case unit(s) until the case unit(s) are needed and picked by the respective lift **150B1**, **150B2** for forming the mixed pallet load.

In one aspect, the storage and retrieval system **100** described herein is effected by providing a storage array RMA with rack storage spaces **130S** arrayed on racks along aisles **130A** (FIG. **25**, Block **2500**). At least one transfer deck **130B** communicably connected with each of the aisles **130A** is also provided (FIG. **25**, Block **2505**). At least one autonomous transport vehicle or bot **110** is provided and is configured for holding at least one pickface and traversing the at least one transfer deck **130B** and aisles **130A**, and having an extendable effector or transfer arm **110PA** for picking and placing the at least one pickface to and from one of the rack storage spaces **130S** (FIG. **25**, Block **2510**). Pickface transport axes X, Y of the storage array are defined with the aisles **130A**, the at least one transfer deck **130B**, the at least one autonomous transport vehicle **110**, traversing thereon, and the extendable effector **110PA**, such that pickfaces are transported along the pickface transport axes X, Y between an inbound section of the automated storage and retrieval system **160IN**, where pickfaces inbound to the storage array are generated, and a load fill section of the automated storage and retrieval system **160UT**, where outbound pickfaces from the storage array are arranged to fill a load in accordance with a predetermined load fill order sequence. On the fly sortation of mixed case pickfaces is effected (FIG. **25**, Block **2520**) coincident with transport on at least one of the pickface transport axes X, Y with the storage racks and the autonomous transport vehicle **110** in combination, so that two or more of the at least one pickface are picked from one or more of the rack storage spaces **130S** and placed at one or more pickface holding locations (such as, for example, transfer or buffer stations TS, BS), different than the one or more of the rack storage spaces **130S**, according to the predetermined load fill order sequence. In one aspect the controller **120** (which is operably connected to the at least one autonomous transport vehicle as described above) manages the pickface transport axes X, Y, Z wherein the pickface transport axes includes a plurality of transport axes. As described above, the plurality of pickface transport axes X, Y, Z are oriented in at least two directions angled relative to each other. As also described above, one of the plurality of pickface transport axes Y is defined by extension of the extendable effector **110PA** and is in a different direction

angled relative to another of the plurality of pickface transport axes X defined by the autonomous transport vehicle **110** traverse along the picking aisle **130A**. In one aspect, as described above, on the fly sortation is effected, with the racks and the at least one autonomous transport vehicle in combination, coincident with transport on at least one of each of the plurality of pickface transport axes. In one aspect the lifts **150** define another pickface transport axis Z of the storage array. As described herein on the fly sortation of mixed case pickfaces is effected by the lifts **150** coincident with transport on the other pickface transport axis so that two or more of the pickfaces are picked from one or more deck levels and transported to the load fill section according to the predetermined load fill order sequence.

Referring now to FIGS. **21**, **22A** and **22B**, in one aspect the transfer of pickfaces from the input station **160IN** to the output station **160UT** occurs without pickface transfer by the bots **110**. For example, referring to FIG. **21** the conveyors **160CA**, **160CB** of the input and output stations **160IN**, **160UT** are arranged so that each lift **150** serves both the input and output stations **160IN**, **160UT**. For example, conveyor **160CA1** of input station **160IN1** and conveyor **160CB1** of output station **160UT1** are both served by lifts **150A1**, **150B1**. As may be realized, each conveyor **160CA1**, **160CB1** are located at different levels (in a manner similar to that described above with respect to the shelves of the buffer and transfer stations BS, TS) of a common lift **150A1**, **150B1** so that pickfaces can be picked from one conveyor **160CA1**, **160CB1** by the common lift **150A1**, **150B1** at one level and transferred to the other conveyor **160CA1**, **160CB1** at another level. Here pickfaces are substantially directly transferred, by the lifts **150A1**, **150B1**, from one conveyor **160CA1**, **160CB1** (e.g. from input station **160IN1** to output station **160UT1**) while bypassing the bots **110** and storage structure **130**. As may be realized, in one aspect the pickfaces from input station **160IN1** are placed on the shelves buffer or transfer stations BS, TS by the common lift **150A1**, **150B1** for sorting the pickfaces, as described above, before transferring the pickfaces to the output conveyor **160CB1**. Referring to FIGS. **22A** and **22B**, in one aspect pickfaces are transferred from the input station **160IN** to the output station **160UT**, while bypassing the bots **110** and storage structure **130**, through buffer lanes BL that communicably connect the input conveyors **160CA** to the output conveyors **160CB**.

Referring to FIG. **27**, in accordance with aspects of the disclosed embodiment, storage spaces arrayed on racks along picking aisles are provided (FIG. **27**, Block **1600**). Multiple level decks are also provided (FIG. **27**, Block **1610**), where at least one deck level of the multiple level decks communicates with each aisle, where the multiple level decks and aisles define a rolling surface for an autonomous transport vehicle at each level of the multiple level decks. Racks at multiple rack levels are accessed from a respective rolling surface that is common to the multiple rack levels (FIG. **27**, Block **1620**), where the racks are disposed along at least one aisle at each level of the multiple level decks. In one aspect, a vertical pitch between rack levels varies for a portion of a respective aisle. In one aspect, the vertical pitch between at least two rack levels of the portion of the respective aisle is related to another vertical pitch between at least two other rack levels of another aisle portion of the respective aisle so that the autonomous transport vehicle effects multiple picks in an ordered sequence in a common aisle pass. In one aspect, the vertical pitch between at least two rack levels of the portion of the respective aisle is related to another vertical pitch between

at least two other rack levels of another aisle portion of the respective aisle so that the vertical pitch and the other vertical pitch effects substantially filling a vertical space between the multiple deck levels with stored items.

In accordance with one or more aspects of the disclosed embodiment, an automated storage and retrieval system is provided. The automated storage and retrieval system including at least one autonomous transport vehicle, a transfer deck that defines a transport surface for the at least one autonomous transport vehicle, at least one reciprocating lift, a first pickface interface station and a second pickface interface station connected to the transfer deck and spaced apart from each other, each pickface interface station forming a pickface transfer interfacing between the at least one autonomous transport vehicle on the transfer deck and the at least one reciprocating lift at each pickface interface station so that a pickface is transferred between the at least one reciprocating lift and the at least one autonomous transport vehicle at each pickface interface station, wherein the at least one autonomous transport vehicle is configured to pick a first pickface at the first pickface interface station, traverse the transfer deck and buffer the first pickface, or at least a portion thereof, at the second pickface interface station so that the at least a portion of the first pickface is buffered at the second pickface interface station for transport with the outbound pickface transport system in an order sequence of pickfaces according to a predetermined case out order sequence of mixed case pickfaces.

In accordance with one or more aspects of the disclosed embodiment, the at least one autonomous transport vehicle is configured to buffer the first pickface, or at least a portion thereof, at the second pickface interface station so that the second pickface interface station has multiple pickfaces buffered on a common support.

In accordance with one or more aspects of the disclosed embodiment, at least one of the multiple pickfaces at the second pickface interface station is different from the first pickface and includes a case that is from the first pickface.

In accordance with one or more aspects of the disclosed embodiment, the autonomous transport vehicle builds at least one of the multiple pickfaces at the second pickface interface station on the fly during transport of the first pickface between the first pickface interface station and the second pickface interface station.

In accordance with one or more aspects of the disclosed embodiment, the autonomous transport vehicle builds the at least one of the multiple pickfaces onboard of the autonomous transport vehicle.

In accordance with one or more aspects of the disclosed embodiment, the autonomous transport vehicle builds the at least one of the multiple pickfaces at the second pickface interface station or at a buffer portion of the common support of the second pickface interface station buffer.

In accordance with one or more aspects of the disclosed embodiment, the first pickface is at least one of the multiple pickfaces at the second pickface interface station.

In accordance with one or more aspects of the disclosed embodiment, the automated storage and retrieval system comprises autonomous transport vehicle access aisles connected to the deck, and a storage array having storage racks arranged in multilevel shelves and distributed along the autonomous transport vehicle access aisles.

In accordance with one or more aspects of the disclosed embodiment, the autonomous transport vehicle is arranged so that at least another portion of the first pickface picked

from the first pickface interface station is placed on a storage rack of the storage array before transport to the second pickface interface station.

In accordance with one or more aspects of the disclosed embodiment, the second pickface interface station forms a common pickface transfer interface for the at least one reciprocating lift so that the commonly supported pickfaces are picked in common with the at least one reciprocating lift.

In accordance with one or more aspects of the disclosed embodiment, the transfer deck is undeterministic and has multiple travel lanes.

In accordance with one or more aspects of the disclosed embodiment, an automated storage and retrieval system is provided. The automated storage and retrieval system including at least one autonomous transport vehicle, a transfer deck that defines a transport surface for the at least one autonomous transport vehicle, at least one inbound pickface transport system disposed between an unload cell and a load fill section, at least one outbound pickface transport system disposed between the unload cell and the load fill section, a first pickface interface station, and a second pickface interface station connected to the transfer deck and spaced apart from each other, each pickface interface station forming a pickface transfer interfacing between the at least one autonomous transport vehicle on the transfer deck and a respective one of the inbound pickface transport system and the outbound pickface transport system at each pickface interface station so that a pickface is transferred between the respective one of the inbound pickface transport system and the outbound pickface transport system and the at least one autonomous transport vehicle at each pickface interface station, wherein the at least one autonomous transport vehicle is configured to pick a first pickface at the first pickface interface station, traverse the deck and buffer the first pickface, or at least a portion thereof, at the second pickface interface station so that the second pickface interface station has multiple pickfaces buffered on a common support in an order sequence of pickfaces according to a predetermined case out order sequence of mixed case pickfaces.

In accordance with one or more aspects of the disclosed embodiment, at least one of the multiple pickfaces at the second pickface interface station is different from the first pickface and includes a case that is from the first pickface.

In accordance with one or more aspects of the disclosed embodiment, the autonomous transport vehicle builds at least one of the multiple pickfaces at the second pickface interface station on the fly during transport of the first pickface between the first pickface interface station and the second pickface interface station.

In accordance with one or more aspects of the disclosed embodiment, the autonomous transport vehicle builds the at least one of the multiple pickfaces onboard of the autonomous transport vehicle.

In accordance with one or more aspects of the disclosed embodiment, the autonomous transport vehicle builds the at least one of the multiple pickfaces at the second pickface interface station or at a buffer portion of the common support of the second pickface interface station buffer.

In accordance with one or more aspects of the disclosed embodiment, the first pickface is at least one of the multiple pickfaces at the second pickface interface station.

In accordance with one or more aspects of the disclosed embodiment, the automated storage and retrieval system comprises autonomous transport vehicle access aisles connected to the deck, and a storage array having storage racks

arranged in multilevel shelves and distributed along the autonomous transport vehicle access aisles.

In accordance with one or more aspects of the disclosed embodiment, the autonomous transport vehicle is arranged so that at least another portion of the first pickface picked from the first pickface interface station is placed on a storage rack of the storage array before transport to the second pickface interface station.

In accordance with one or more aspects of the disclosed embodiment, the second pickface interface station forms a common pickface transfer interface for the respective one of the inbound pickface transport system and the outbound pickface transport system so that the commonly supported pickfaces are picked in common with the respective one of the inbound pickface transport system and the outbound pickface transport system.

In accordance with one or more aspects of the disclosed embodiment, the transfer deck is undeterministic and has multiple travel lanes.

In accordance with one or more aspects of the disclosed embodiment, a method for automated storage and retrieval is provided. The method including picking, with an autonomous transport vehicle, a first pickface from a first shelf of a first pickface handoff station, buffering, with the autonomous transport vehicle, the first pickface on a second shelf of a second pickface handoff station, forming a second pickface at the second shelf, the second pickface being different than the first pickface and comprising more than one case in ordered sequence corresponding to a predetermined case out order sequence of mixed cases where the first pickface and the second pickface have at least one case in common, and picking, with a reciprocating lift, the second pickface from the second shelf.

In accordance with one or more aspects of the disclosed embodiment, the method comprises forming, with the autonomous transport vehicle, the second pickface at the second shelf on the fly during transport of the first pickface between the first shelf and the second shelf.

In accordance with one or more aspects of the disclosed embodiment, the method comprises forming, with the autonomous transport vehicle, the second pickface onboard of the autonomous transport vehicle.

In accordance with one or more aspects of the disclosed embodiment, the method comprises forming, with the autonomous transport vehicle, the second pickface at the second shelf or at a buffer portion of the second shelf.

In accordance with one or more aspects of the disclosed embodiment, the method comprises placing, with the autonomous transport vehicle, at least a portion of the first pickface picked from the first shelf on a storage rack of a storage array before transporting at least the portion of the first pickface to the second shelf.

In accordance with one or more aspects of the disclosed embodiment, the second shelf forms a common pickface transfer interface for the reciprocating lift, the method further comprising picking in common, with the reciprocating lift, the commonly supported pickfaces.

In accordance with one or more aspects of the disclosed embodiment, an automated storage and retrieval system is provided. The automated storage and retrieval system including a storage array with rack storage spaces arrayed on racks along aisles, at least one transfer deck communicably connected with each of the aisles, at least one autonomous transport vehicle configured for holding at least one pickface and traversing the at least one transfer deck and aisles, and having an extendable effector for picking and placing the at least one pickface to and from one of the rack storage spaces,

wherein the aisles, the at least one transfer deck, the at least one autonomous transport vehicle, traversing thereon, and the extendable effector define pickface transport axes of the storage array along which pickfaces are transported between an inbound section of the automated storage and retrieval system, where pickfaces inbound to the storage array are generated, and a load fill section of the automated storage and retrieval system, where outbound pickfaces from the storage array are arranged to fill a load in accordance with a predetermined load fill order sequence, and wherein the racks and the autonomous transport vehicle are arranged so that in combination the racks and the autonomous transport vehicle effect on the fly sortation of mixed case pickfaces coincident with transport on at least one of the pickface transport axes so that two or more of the at least one pickface are picked from one or more of the rack storage spaces and placed at one or more pickface holding locations, different than the one or more of the rack storage spaces, according to the predetermined load fill order sequence.

In accordance with one or more aspects of the disclosed embodiment, the automated storage and retrieval system comprises a controller operably connected to the at least one autonomous transport vehicle and arranged to manage the pickface transport axes wherein the pickface transport axes comprises a plurality of transport axes.

In accordance with one or more aspects of the disclosed embodiment, the plurality of pickface transport axes are oriented in at least two directions angled relative to each other.

In accordance with one or more aspects of the disclosed embodiment, one of the plurality of pickface transport axes defined by extension of the extendable effector is in a different direction angled relative to another of the plurality of pickface transport axes defined by the autonomous transport vehicle traverse along the aisle.

In accordance with one or more aspects of the disclosed embodiment, the racks and the at least one autonomous transport vehicle in combination effect on the fly sortation coincident with transport on at least one of each of the plurality of pickface transport axes.

In accordance with one or more aspects of the disclosed embodiment, the at least one transfer deck comprises more than one transfer deck arranged at different deck levels.

In accordance with one or more aspects of the disclosed embodiment, the automated storage and retrieval system comprises a lift communicably connected to each of the decks at the different deck levels, the lift being arranged to transport the pickfaces between the different deck levels and defining another pickface transport axis of the storage array.

In accordance with one or more aspects of the disclosed embodiment, the lift is arranged to effect on the fly sortation of mixed case pickfaces coincident with transport on the other pickface transport axis so that two or more of the pickfaces are picked from one or more deck levels and transported to the load fill section according to the predetermined load fill order sequence.

In accordance with one or more aspects of the disclosed embodiment, on the fly sortation is effected coincident with transport on at least one of each of the plurality of pickface transport axes and each of the other transport axis of the lift.

In accordance with one or more aspects of the disclosed embodiment, an automated storage and retrieval system is provided. The automated storage and retrieval system including a storage array with rack storage spaces arrayed on racks along aisles, at least one transfer deck communicably connected with each of the aisles, at least one autonomous transport vehicle configured for holding at least one pickface

and traversing the at least one transfer deck and aisles, and having an extendable effector for picking and placing the at least one pickface to and from one of the rack storage spaces, at least one lift communicably connected to each transfer deck, the lift being arranged to transport pickfaces to and from the at least one transfer deck, and wherein the aisles, the at least one transfer deck, the at least one autonomous transport vehicle, traversing thereon, the extendable effector and the at least one lift define pickface transport axes of the storage array along which pickfaces are transported between an inbound section of the automated storage and retrieval system, where pickfaces inbound to the storage array are generated, and a load fill section of the automated storage and retrieval system, where outbound pickfaces from the storage array are arranged to fill a load in accordance with a predetermined load fill order sequence, the racks and the autonomous transport vehicle are arranged so that in combination the racks and the autonomous transport vehicle effect on the fly sortation of mixed case pickfaces on at least one of the pickface transport axes so that two or more of the at least one pickface are picked from one or more of the rack storage spaces and placed at one or more pickface holding locations, different than the one or more of the rack storage spaces, according to the predetermined load fill order sequence, and the at least one lift is arranged to effect on the fly sortation of the mixed case pickfaces on another of the at least one pickface transport axes so that two or more of the pickfaces are picked from different ones of the at least one transfer deck and transported to the load fill section according to the predetermined load fill order sequence where on the fly sortation is effected coincident with transport on at least one of each of the pickface transport axes.

In accordance with one or more aspects of the disclosed embodiment, the automated storage and retrieval system comprises a controller operably connected to the at least one autonomous transport vehicle and the at least one lift and arranged to manage the pickface transport axes.

In accordance with one or more aspects of the disclosed embodiment, the pickface transport axes are oriented in at least two directions angled relative to each other.

In accordance with one or more aspects of the disclosed embodiment, one of the pickface transport axes defined by extension of the extendable effector is in a different direction angled relative to another of the pickface transport axes defined by the autonomous transport vehicle traverse along the aisle.

In accordance with one or more aspects of the disclosed embodiment, the racks and the at least one autonomous transport vehicle in combination effect on the fly sortation coincident with transport on at least one of the pickface transport axes.

In accordance with one or more aspects of the disclosed embodiment, the at least one transfer deck comprises more than one transfer deck arranged at different deck levels and the at least one lift is configured to transport pickfaces between the different deck levels.

In accordance with one or more aspects of the disclosed embodiment, a method for automated storage and retrieval is provided. The method including providing a storage array with rack storage spaces arrayed on racks along aisles, providing at least one transfer deck communicably connected with each of the aisles, providing at least one autonomous transport vehicle configured for holding at least one pickface and traversing the at least one transfer deck and aisles, and having an extendable effector for picking and placing the at least one pickface to and from one of the rack storage spaces, defining, with the aisles, the at least one

transfer deck, the at least one autonomous transport vehicle, traversing thereon, and the extendable effector, pickface transport axes of the storage array, such that pickfaces are transported along the pickface transport axes between an inbound section of the automated storage and retrieval system, where pickfaces inbound to the storage array are generated, and a load fill section of the automated storage and retrieval system, where outbound pickfaces from the storage array are arranged to fill a load in accordance with a predetermined load fill order sequence, and effecting on the fly sortation of mixed case pickfaces coincident with transport on at least one of the pickface transport axes, with the racks and the autonomous transport vehicle in combination, so that two or more of the at least one pickface are picked from one or more of the rack storage spaces and placed at one or more pickface holding locations, different than the one or more of the rack storage spaces, according to the predetermined load fill order sequence.

In accordance with one or more aspects of the disclosed embodiment, the method comprises managing, with a controller operably connected to the at least one autonomous transport vehicle, the pickface transport axes wherein the pickface transport axes includes a plurality of transport axes.

In accordance with one or more aspects of the disclosed embodiment, the plurality of pickface transport axes are oriented in at least two directions angled relative to each other.

In accordance with one or more aspects of the disclosed embodiment, one of the plurality of pickface transport axes defined by extension of the extendable effector is in a different direction angled relative to another of the plurality of pickface transport axes defined by the autonomous transport vehicle traverse along the aisle.

In accordance with one or more aspects of the disclosed embodiment, the method comprises effecting, with the racks and the at least one autonomous transport vehicle in combination, on the fly sortation coincident with transport on at least one of each of the plurality of pickface transport axes.

In accordance with one or more aspects of the disclosed embodiment, the method comprises defining another pickface transport axis of the storage array with a lift that is communicably connected to each of the at least one transfer deck disposed at different deck levels and transports the pickfaces between the different deck levels.

In accordance with one or more aspects of the disclosed embodiment, the method comprises effecting, with the lift, on the fly sortation of mixed case pickfaces coincident with transport on the other pickface transport axis so that two or more of the pickfaces are picked from one or more deck levels and transported to the load fill section according to the predetermined load fill order sequence.

In accordance with one or more aspects of the disclosed embodiment, the method comprises effecting on the fly sortation coincident with transport on at least one of each of the plurality of pickface transport axes and each of the other transport axis of the lift.

It should be understood that the foregoing description is only illustrative of the aspects of the disclosed embodiment. Various alternatives and modifications can be devised by those skilled in the art without departing from the aspects of the disclosed embodiment. Accordingly, the aspects of the disclosed embodiment are intended to embrace all such alternatives, modifications and variances that fall within the scope of the appended claims. Further, the mere fact that different features are recited in mutually different dependent or independent claims does not indicate that a combination

of these features cannot be advantageously used, such a combination remaining within the scope of the aspects of the invention.

What is claimed is:

1. An automated storage and retrieval system comprising: at least one autonomous transport vehicle;

an elevated transfer deck that defines a transport surface for the at least one autonomous transport vehicle at an elevated level from an output conveyor at a lower level; and

more than one goods interface station connected to the transfer deck and spaced apart from each other, each goods interface station forming a goods transfer interfacing between the at least one autonomous transport vehicle on the transfer deck and a lowering transfer at each of the more than one goods interface station so that a goods unit is transferred via the lowering transfer from the at least one autonomous transport vehicle at each of the more than one goods interface station;

wherein the at least one autonomous transport vehicle is configured to traverse the transfer deck transporting a first goods unit and transfer via the lowering transfer the first goods unit at one of the more than one goods interface station so that the one of the more than one goods interface station has multiple goods units transferred down in an order sequence of goods units that defines a successive sequence, of ordered multiple goods units, dependent from and embodying a predetermined goods unit out order sequence of mixed goods units having a predetermined successive sequence of ordered mixed goods units.

2. The automated storage and retrieval system of claim 1, wherein the autonomous transport vehicle effects ordering of the at least one of the multiple goods units with the lowering transfer at the more than one goods interface station so as to form the predetermined goods unit out order sequence with the predetermined successive sequence at the output conveyor.

3. The automated storage and retrieval system of claim 1, wherein the first goods unit is at least one of the multiple goods units of mixed orders at the more than one goods interface station.

4. The automated storage and retrieval system of claim 1, further comprising:

autonomous transport vehicle access aisles connected to the deck; and

a storage array having storage racks arranged in multi-level shelves and distributed along the autonomous transport vehicle access aisles.

5. The automated storage and retrieval system of claim 4, wherein the autonomous transport vehicle is arranged so that at least another portion of the first goods unit engaged at the one of the more than one goods interface station is placed on a storage rack of the storage array before transport to another of the more than one goods interface station.

6. The automated storage and retrieval system of claim 1, wherein the lowering transfer is a reciprocating lift and the more than one goods interface station forms a common goods transfer interface for the at least one reciprocating lift so that commonly supported goods units at the more than one goods interface station are transferred in common with the at least one reciprocating lift.

7. The automated storage and retrieval system of claim 1, wherein the transfer deck is undeterministic and has multiple travel lanes.

45

8. An automated storage and retrieval system comprising:
 at least one autonomous transport vehicle;
 an elevated transfer deck that defines a transport surface
 for the at least one autonomous transport vehicle;
 at least one inbound goods transport system disposed 5
 between an unload cell and a load fill section;
 at least one outbound goods transport system disposed
 between the unload cell and the load fill section,
 wherein the elevated transfer deck is at an elevated
 level relative to the outbound goods transport system; 10
 more than one goods interface station connected to the
 transfer deck and spaced apart from each other, each
 goods interface station forming a goods transfer inter-
 facing between the at least one autonomous transport
 vehicle on the transfer deck and a lowering transfer at 15
 each of the more than one goods interface station so
 that a goods unit is transferred via the lowering transfer
 from the at least one autonomous transport vehicle at
 each of the more than one goods interface station;
 wherein the at least one autonomous transport vehicle is 20
 configured to traverse the deck transporting a first
 goods unit and transfer via the lowering transfer the
 first goods unit at one of the more than one goods
 interface station so that so that the at least a portion of
 the first goods unit is transferred down in an order 25
 sequence of goods units that defines a successive
 sequence, of ordered multiple goods units, dependent
 from and embodying a predetermined goods unit out
 order sequence of mixed goods units having a prede-
 termined successive sequence of ordered mixed goods 30
 units.
9. The automated storage and retrieval system of claim 8,
 wherein the autonomous transport vehicle effects ordering of
 the at least one of the multiple goods units with the lowering
 transfer at the more than one goods interface station so as to 35
 form the predetermined goods unit out order sequence with
 the predetermined successive sequence at the outbound
 goods transport system.
10. The automated storage and retrieval system of claim
 8, wherein the first goods unit is at least one of the multiple 40
 goods units of mixed loads at the more than one goods
 interface station.
11. The automated storage and retrieval system of claim
 8, further comprising:
 autonomous transport vehicle access aisles connected to 45
 the deck; and
 a storage array having storage racks arranged in multi-
 level shelves and distributed along the autonomous
 transport vehicle access aisles.
12. The automated storage and retrieval system of claim 50
 11, wherein the autonomous transport vehicle is arranged so

46

that at least another portion of the first goods unit engaged
 at the one of the more than one goods interface station is
 placed on a storage rack of the storage array before transport
 to another of the more than one goods interface station.

13. The automated storage and retrieval system of claim
 8, wherein the lowering transfer is a reciprocating lift and
 the more than one goods interface station forms a common
 goods transfer interface for a respective one of the inbound
 goods transport system and the outbound goods transport
 system so that commonly supported goods units at the more
 than one goods interface station are transferred in common
 with the respective one of the inbound goods transport
 system and the outbound goods transport system.

14. The automated storage and retrieval system of claim
 8, wherein the transfer deck is undeterministic and has
 multiple travel lanes.

15. A method comprising:

traversing, with an autonomous transport vehicle, a trans-
 fer deck with a first goods unit;

transferring, with a lowering transfer, the first goods unit
 on one of more than one goods handoff station so that
 the first goods unit is transferred down in ordered
 sequence of goods units that defines a successive
 sequence, of ordered multiple goods units, dependent
 from and embodying a predetermined goods unit out
 order sequence of mixed goods having a predetermined
 successive sequence of ordered mixed goods where the
 first goods unit and the second goods unit have at least
 one good in common.

16. The method of claim 15, further comprising effecting
 ordering, with the autonomous transport vehicle, of the first
 goods unit at the more than one goods handoff station so as
 to form the predetermined goods unit out order sequence
 with the predetermined successive sequence at an output
 conveyor.

17. The method of claim 15, further comprising placing,
 with the autonomous transport vehicle, at least a portion of
 the first goods unit engaged at the more than one goods
 handoff station on a storage rack of a storage array before
 transporting at least the portion of the first goods unit to
 another of the more than one goods handoff station.

18. The method of claim 15, wherein the lowering transfer
 is a reciprocating lift and the more than one goods handoff
 station forms a common goods transfer interface for the
 reciprocating lift, the method further comprising transferring
 in common, with the reciprocating lift, a commonly sup-
 ported goods units at the more than one goods handoff
 station.

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